

CLASSIFICATION THEORY AND THE CONSTRUCTION OF PAC FIELDS

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ABSTRACT. We analyze a construction of Cherlin, van den Dries, and Macintyre to code graphs in PAC fields. We show that, in many cases, model-theoretic properties of the graph are preserved in the passage from the graph to the field. As a corollary, we show that the SOP_n hierarchy is strict in the class of fields. The main ingredient is a detailed treatment of the model theory of inverse systems of certain profinite groups that code graphs and can be realized as the absolute Galois groups of PAC fields.

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One important thread within model theory focuses on whether the various regions of the model-theoretic map of the universe have (theories of) fields in them and, if so, what those fields might look like. Some old and central conjectures, like the stable fields conjecture (predicting that all stable fields are separably closed) or the supersimple fields conjecture (predicting that all simple fields are pseudo-algebraically closed) concern this issue, as well as new ones, regarding dp-finite, NIP, or NIP_k fields. The general picture presented by these conjectures is that model-theoretic tameness notions tend to place very strong constraints on what a field can look like; the fact that many of them remain open suggests that finding new fields with prescribed model-theoretic properties is quite challenging. With groups, the situation is rather different. Although model-theoretic tameness notions also place strong constraints on what groups can look like, the problem of constructing a group with a specific degree of tameness and complexity is often tractable. Coding constructions of Ershov and of Mekler allow one to encode arbitrary graphs into nilpotent groups of class 2 and exponent p [Erš74, Mek81]. Mekler's analysis of this construction showed that the type-counting function of the group is equal to that of the graph, and so properties like stability and NIP are preserved. Subsequent

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research has elaborated this idea, showing that many additional dividing lines are preserved by this graph-coding construction [BPT24, Ahn19, CH19, Bau02].

This paper is concerned with analyzing a graph-coding construction of Cherlin, van den Dries, and Macintyre [CvdDM80] with the aim of demonstrating that it can be regarded as a kind of ‘Mekler construction’ for fields. Recall that a field K is called *pseudo-algebraically closed* (PAC) if every absolutely irreducible variety defined over K has a K -rational point. These fields entered model theory through Ax’s celebrated characterization of pseudo-finite fields as exactly the perfect PAC fields with Galois group $\hat{\mathbb{Z}}$ [Ax68]. In [CvdDM80], the authors give a systematic account of the model theory of PAC fields, showing that many model-theoretic questions about such fields reduce to questions about the absolute Galois group. In connection with issues around decidability, they show that an arbitrary graph can be coded into a profinite group that can be realized as the absolute Galois group of a PAC field in such a way that the graph can be interpreted in the field (and this coding was recently revisited, again in connection with decidability, in [Tyr24]). This construction yields a method for producing complicated PAC fields. To see it as a method for producing PAC fields with a prescribed degree of complexity and tameness, one would need additionally to know that the complexity of the field produced by this construction does not overshoot that of the graph.

Unfortunately, the complexity of the field produced by CDM graph coding often *will* exceed that of the graph. When the input is an infinite graph, the resulting PAC field has unbounded Galois group and therefore has both TP_2 and IP_k for all k [Cha99, Hem16]. This might have once seemed to model theorists to be the end of the story, as structures with these properties were formerly thought to inhabit an impossibly complicated region of the model-theoretic map. But current work in neostability has found a wide array of tameness properties that are compatible with these properties. And, indeed, work of Chatzidakis [Cha02] has shown that for certain unbounded PAC fields—namely the ω -free PAC fields and, more broadly, the Frobenius fields—there is a nice theory of independence and amalgamation that can be leveraged to treat core model-theoretic problems concerning them. It was this family of examples, in fact, that motivated the development of Kim-independence for NSOP_1 theories in its early stages. Thus, provided we are willing to take the leap of regarding theories with TP_2 and higher arity analogues of the independence property as potentially tame, it makes sense to ask whether the complexity of the fields produced by the CDM procedure can be determined as a function of the input graph.

The construction proceeds in two steps. In the first step, one takes a graph Γ and produces a projective profinite group $\widetilde{G}_\Gamma$ whose inverse system $S(\widetilde{G}_\Gamma)$ interprets Γ . In the second step, one realizes $\widetilde{G}_\Gamma$ as the absolute Galois group of a PAC field. This latter step is possible thanks to a theorem of van den Dries and Lubotzky, showing that the class of projective profinite groups is exactly the class of absolute Galois groups of PAC fields [LvD81]. The question of whether some model-theoretic property of the graph is preserved in the resulting field, then, breaks up into two questions: one about preservation in the passage from the graph to the group, the second about preservation from the group to the field. A specific test question in this vein comes from another theorem of Chatzidakis. She showed that the theory of a PAC field has Shelah’s property SOP_n if and only if the theory of the inverse system of its absolute Galois group has SOP_n , for $n \geq 3$. It was left open whether there

are examples showing that the SOP_n hierarchy is strict in fields, that is, showing that, for all $n \geq 3$, a field can have SOP_n and NSOP_{n+1} theory. We address this by showing that the inverse system $S(\widetilde{G}_\Gamma)$ of the profinite group produced by CDM graph coding has SOP_n theory if and only if the input graph Γ has SOP_n theory. Realizing these groups as absolute Galois groups of PAC fields, then, provides the desired family of examples. In a sequel, we use this construction again to produce PAC fields with prescribed amounts of independent amalgamation. These examples give a proof of concept, but it is our hope that this construction will later serve as an engine for producing new algebraic examples across the model-theoretic map.

The paper is structured as follows. In Section 1, we describe the basic framework for viewing the inverse system of profinite groups as model-theoretic structures. We also survey the mostly elementary facts about profinite groups we will need, especially concerning projective profinite groups and Frattini covers. In Section 2, we explain the CDM graph coding construction. Several subsections deal with specific algebraic issues involving the groups produced by this construction. The real model theory begins in Section 3, which describes all structures elementarily equivalent to the inverse system of a group produced by the CDM construction applied to a graph Γ . The simplest imaginable situation would have been for $S \equiv S(G_\Gamma)$ to imply that $S \cong S(G_{\Gamma'})$ for some graph $\Gamma' \equiv \Gamma$; in other words, S being elementarily equivalent to an inverse system produced by the CDM construction entails that S is also produced by the CDM construction applied to an elementarily equivalent graph. This, it turns out, is not the case, but it is almost correct. We show that, for an infinite graph Γ , if $S \equiv S(\widetilde{G}_\Gamma)$, then $S \cong S(\widetilde{G_{\Gamma'} \times C_2^I})$, where C_2 denotes the group $\{\pm 1\}$, Γ' is a graph elementarily equivalent to Γ , I is some index set, and the tilde denotes the universal Frattini cover. This means that an arbitrary $S \equiv S(\widetilde{G}_\Gamma)$ is determined by a choice of $\Gamma' \equiv \Gamma$ and the dimension of an $\mathbb{F}_2$ -vector space (namely, the continuous dual of $\mathbb{F}_2^I$). Section 4 leverages this representation to give a detailed model-theoretic analysis of algebraic closure and types in $\text{Th}(S(\widetilde{G}_\Gamma))$ and proves several preservation results, showing that properties like NSOP_n are preserved in the move from Γ to $S(\widetilde{G}_\Gamma)$. We show that the model-theoretic properties of $S(\widetilde{G}_\Gamma)$ are determined by the structure induced on the subsystem corresponding to the maximal pro-2 quotient of $\widetilde{G}_\Gamma$, which is a free pro-2 group. At this stage in the analysis, we were able to leverage Chatzidakis's work on inverse systems of free pro- p groups [Cha98] and results on independence and amalgamation in H -structures, due to Berenstein, Carmona, and Vassiliev [BCV17]. The arguments in this section draw inspiration from related work on preservation theorems for generic expansions [KTW22, CdH⁺25]. Section 5 finally applies these preservation results to obtain PAC fields with prescribed model-theoretic properties.

We have tried to make this paper as user-friendly as possible. A reader who is primarily interested in applying these results to produce examples of fields with certain properties should probably skip straight to Section 4, which gives the model-theoretic analysis, and then refer back to the earlier sections as needed.

This paper is long overdue. The core ideas were worked out during a week-long visit to Zoé Chatzidakis at ENS in 2017 and we talked about it, and related problems, for a long while after. Virtually every aspect of this paper owes a great debt both to her mathematical contributions to the study of PAC fields and to her personal mentorship and guidance.

1. PROFINITE GROUPS AND THEIR LOGIC

1.1. The inverse system language. Let G be a profinite group. To G , we will associate a first-order structure $S(G)$ that encodes the inverse system of G . The language L_{IS} of the inverse system consists of sorts X_n for all natural numbers $n \geq 1$, binary relations $\leq$ and C , and a ternary relation P .

The underlying set of $S(G)$ consists of all cosets of open normal subgroups of G . That is, taking $\mathcal{N}(G)$ to be the set of open normal subgroups of G , we have

$$S(G) = \{gN : N \in \mathcal{N}(G)\}.$$

A coset gN belongs to sort X_i if and only if $[G : N] \leq i$. We will often write $S(G)_i$ for the elements of $S(G)$ of sort i .

We interpret C so that $C(gN, hM)$ if and only if $N \subseteq M$ and $gM = hM$; in other words, $C(gN, hM)$ holds when $N \subseteq M$ and $\pi(gN) = hM$ where $\pi : G/N \rightarrow G/M$ is the canonical quotient map. Finally, P is interpreted so that $P(g_1N_1, g_2N_2, g_3N_3)$ holds if and only if $N_1 = N_2 = N_3$ and $g_1g_2N_1 = g_3N_1$.

We define $gN \leq hM$ if and only if $N \subseteq M$. We write $\sim$ for the equivalence relation defined by $gN \sim hM$ if and only if $gN \leq hM$ and $hM \leq gN$, or equivalently, if and only if $M = N$. We write $[gN]$ for the equivalence class of gN under $\sim$. The equivalence class $[gN] = [N]$ consists of the elements of the quotient G/N and has a group law given by P . For $N, M \in \mathcal{N}(G)$, we define $[N] \vee [M] = [NM]$ and $[N] \wedge [M] = [N \cap M]$. Then $[N] \vee [M]$ is the least upper bound and $[N] \wedge [M]$ is the greatest lower bound of $[N]$ and $[M]$. This turns $S(G)/\sim$ into a modular lattice.

An unusual feature here is that the sorts are not disjoint and the relations $\leq$, P , and C are not described with specified sorts. This follows the presentation of [Cha02, Appendix 1] and is a convention adopted for convenience. But, technically, one should have different relation symbols for each of the relevant tuples of sorts and, rather than allowing the sorts to intersect, one should have disjoint sorts and maps between the sorts that determine how elements of one sort get identified with elements of another. This more formally standard approach tends to obscure more than it illuminates, so we will stick with Chatzidakis's conventions.

There is an L_{IS} -theory T_{IS} which axiomatizes structures of the form $S(G)$ for a profinite group G . We refer to L_{IS} -structures satisfying this theory as *complete systems*. Conversely, if $S \models T_{\text{IS}}$, then one may form a profinite group G by setting

$$G(S) = \varprojlim_{\alpha \in S} [\alpha]$$

where the inverse limit is taken with respect to the maps $\pi_{\alpha, \beta} : [\alpha] \rightarrow [\beta]$ for $\alpha \leq \beta$ and where the graph of the $\pi_{\alpha, \beta}$ is given by $C \cap ([\alpha] \times [\beta])$. These operations are dual in the sense that G and $G(S(G))$ can be naturally identified for any profinite group G and, likewise, S and $S(G(S))$ may be naturally identified for any $S \models T_{\text{IS}}$. Epimorphisms of profinite groups $G \rightarrow H$ give rise to embeddings of L_{IS} -structures $S(H) \rightarrow S(G)$.

For elements $\alpha, \beta \in S$, we will write $\alpha \vee \beta$ for the identity element of $[\alpha] \vee [\beta]$ and likewise $\alpha \wedge \beta$ for the identity element of $[\alpha] \wedge [\beta]$. With this convention, $\vee$ and $\wedge$ pick out canonical (identity-coset) representatives of the join and meet on $S/\sim$. Note that $\vee$ and $\wedge$ are definable operations on S . We will often make use of the fact that, if $\alpha \vee \beta = \delta$, then

$$[\alpha \wedge \beta] \cong [\alpha] \times_{[\delta]} [\beta] = \{(a, b) \in [\alpha] \times [\beta] : \pi_{\alpha, \delta}(a) = \pi_{\beta, \delta}(b)\}.$$

Another important fact is the *modular law* [Cha98, Lemma 1.9], which says that for $\alpha \leq \beta$:

$$\alpha \vee (\beta \wedge \epsilon) = \beta \wedge (\alpha \vee \epsilon).$$

It is also worth noting that for any finite group A with $|A| = k$, the set $\{\alpha \in X_k(S(G)) : [\alpha] \cong A\}$ is a definable set (defined by asserting that the k elements in $[\alpha]$ have the same multiplication table as A). If A is a finite group and $[\alpha] \cong A$, we refer to α as an *A-element*.

Definition 1.1. Suppose $S \models T_{\text{IS}}$. We say $S_0 \subseteq S$ is a *subsystem* of S if it satisfies the following:

- (Upwards closed) If $\alpha \leq \beta$ and $\alpha \in S_0$, then $\beta \in S_0$.
- (Downwards directed) If $\alpha, \beta \in S_0$, then there is some $\gamma \in S_0$ with $\gamma \leq \alpha$ and $\gamma \leq \beta$.

If A is an arbitrary subset of S , we denote by $\langle A \rangle$ the smallest subsystem of S containing A . It is obtained by closing under $\wedge$ and closing upwards.

If $S_0, S_1 \subseteq S$ are subsystems, we can write $S_0 \vee S_1$ and $S_0 \wedge S_1$ for the subsystems generated by $\{\alpha \vee \beta : \alpha \in S_0, \beta \in S_1\}$ and by $\{\alpha \wedge \beta : \alpha \in S_0, \beta \in S_1\}$, respectively. By passing to the inverse limit, the observation before the above definition entails

$$G(S_0 \wedge S_1) \cong G(S_0) \times_{G(S_0 \vee S_1)} G(S_1).$$

The modular law likewise holds for subsystems, for $S_0 \subseteq S_1$: $S_0 \vee (S_1 \wedge S_2) = S_1 \wedge (S_0 \vee S_2)$. If $\alpha \in S$ and $S_0 \subseteq S$ is a subsystem, we will write $S_0 \wedge \alpha$ for $S_0 \wedge \langle \alpha \rangle$ and likewise for $S_0 \vee \alpha$.

1.2. Projective profinite groups.

Definition 1.2. We say G is *projective* if, whenever $\varphi : G \rightarrow A$ and $\psi : B \rightarrow A$ are continuous epimorphisms with A and B finite groups, there is a continuous homomorphism $\gamma : G \rightarrow B$ so that $\psi \circ \gamma = \varphi$ as in the following diagram:

$$\begin{array}{ccc} & G & \\ \swarrow \text{---} & \downarrow & \\ B & \twoheadrightarrow & A \end{array}$$

We say that G has the *embedding property* if, whenever A and B are finite groups, B is a quotient of G , and $\varphi : G \rightarrow A$, $\psi : B \rightarrow A$ are continuous epimorphisms, there is a continuous epimorphism $\gamma : G \rightarrow B$ making the above diagram commute. A group G is called *superprojective* if it is projective and has the embedding property.

Examples of projective profinite groups include the free profinite groups $\hat{F}_n$ and free pro- p groups $\hat{F}_n(p)$, though there are many others.

Definition 1.3. Let G be a profinite group.

- (1) The *Frattini subgroup* $\Phi(G) \leq G$ is the intersection of all maximal open subgroups of G .
- (2) ([FJ08, Definition 22.5.1]) A homomorphism $\varphi : H \rightarrow G$ is called a *Frattini cover* if one of the following equivalent conditions holds:
 - (a) φ is surjective and $\ker(\varphi) \leq \Phi(H)$.
 - (b) A closed subgroup $H_0 \leq H$ is equal to H if and only if $\varphi(H_0) = G$.
 - (c) A subset $S \subseteq H$ generates H if and only if $\varphi(S)$ generates G .

- (3) A Frattini cover $\alpha : H \rightarrow G$ is called *universal* if whenever $\beta : H' \rightarrow G$ is another Frattini cover of G , there exists an epimorphism $\gamma : H \rightarrow H'$ such that $\beta \circ \gamma = \alpha$, as in the following diagram:

$$\begin{array}{ccc} & H & \\ & \downarrow & \\ H' & \twoheadrightarrow & G \end{array}$$

Fact 1.4. Let G be a profinite group.

- (1) A universal Frattini cover of G exists. It is unique up to isomorphism over G [FJ08, Proposition 22.6.1], so will be referred to as *the* universal Frattini cover of G and will be denoted $\tilde{G}$. We will usually suppress mention of the map $\tilde{\varphi} : \tilde{G} \rightarrow G$.
- (2) The universal Frattini cover $\tilde{G}$ of G is projective and is, moreover, the smallest projective cover of G , in the sense that if H is a projective group, $\alpha : H \rightarrow G$ is an epimorphism and $\tilde{\varphi} : \tilde{G} \rightarrow G$ is the universal Frattini cover of G , then there is an epimorphism $\beta : H \rightarrow \tilde{G}$ making the following diagram commute [FJ08, Proposition 22.6.1]:

$$\begin{array}{ccc} & H & \\ & \downarrow \alpha & \\ \tilde{G} & \xrightarrow{\tilde{\varphi}} & G \end{array}$$

- (3) Let $\tilde{\varphi} : \tilde{G} \rightarrow G$ be the universal Frattini cover of a profinite group G . Then a profinite group H is a quotient of $\tilde{G}$ if and only if H is a Frattini cover of a quotient of G [FJ08, Lemma 22.6.3].
- (4) Suppose G is a profinite group, F is a free profinite group, and $\varphi : F \rightarrow G$ is an epimorphism. If $H \leq F$ is a minimal closed subgroup such that $\varphi|_H$ is an epimorphism onto G , then $\varphi|_H$ is a universal Frattini cover. In particular, the universal Frattini cover of G has the same rank as G . [FJ08, Corollary 22.5.3]

Fact 1.4(3) has the following consequence, which is worth spelling out explicitly:

Fact 1.5. Let $\tilde{\varphi} : \tilde{G} \rightarrow G$ be the universal Frattini cover of a profinite group G . Then a finite group B is a quotient of $\tilde{G}$ if and only if $B/\Phi(B)$ is a quotient of G .

Proof. Clearly B is a Frattini cover of $B/\Phi(B)$, so if $B/\Phi(B)$ is a quotient of G , then B is a quotient of $\tilde{G}$ by Fact 1.4(3). For the other direction, suppose $\psi : B \rightarrow A$ is a Frattini cover of A , for some quotient A of G . Then $A \cong B/\ker(\psi)$ and $\ker(\psi) \subseteq \Phi(B)$ so $B/\Phi(B)$ is a quotient of A and thus of G . $\square$

Note that, by Fact 1.4(2), if $\rho : G \rightarrow H$ is an epimorphism of profinite groups and $\varphi_G : \tilde{G} \rightarrow G$ and $\varphi_H : \tilde{H} \rightarrow H$ are the universal Frattini covers of G and H respectively, then $\rho \circ \varphi_G : \tilde{G} \rightarrow H$ is an epimorphism from a projective group to H and, since $\tilde{H}$ is the smallest projective group with an epimorphism to H , there must be some epimorphism $\tilde{\rho} : \tilde{G} \rightarrow \tilde{H}$ such that $\rho \circ \varphi_G = \varphi_H \circ \tilde{\rho}$. We will refer to $\tilde{\rho}$ as a *lift* of the epimorphism $\rho : G \rightarrow H$.

Fact 1.6. Suppose G is a profinite group and $N \trianglelefteq G$ is a closed normal subgroup. Assume that, for any open normal subgroup $M \trianglelefteq G$, the quotient map $G/M \rightarrow G/NM$ is a Frattini cover. Then the quotient map $G \rightarrow G/N$ is a Frattini cover.

Proof. By hypothesis, $G/M \rightarrow G/NM$ is a Frattini cover for every open normal $M \trianglelefteq G$, so $NM/M \leq \Phi(G/M)$. Taking the inverse limit over M , we have

$$N = \varprojlim_{M \trianglelefteq G} NM/M \leq \varprojlim_{M \trianglelefteq G} \Phi(G/M) = \Phi(G)$$

by [RZ10, Proposition 2.8.2]. This shows that $G \rightarrow G/N$ is a Frattini cover. $\square$

Finally, we will need the following fact concerning the model theory of projective profinite groups:

Fact 1.7. [Fro11, Proposition 2.18] The class of complete inverse systems of projective profinite groups is L_{IS} -axiomatizable.

2. CDM GRAPH CODING

2.1. The basic set-up. In this subsection, we will explain a construction introduced by Cherlin, van den Dries, and Macintyre for coding graphs into profinite groups [CvdDM80]. We will follow the presentation from [FJ08, Chapter 28].

For a natural number n , we write C_n for the cyclic group of order n . We fix, once and for all, distinct odd primes p and q . The dihedral group D_p is defined by

$$D_p = \langle \beta, \gamma : \beta^2 = 1, \gamma^p = 1, \beta\gamma\beta^{-1} = \gamma^{-1} \rangle.$$

Let $\tau : D_p \rightarrow C_2 = \{\pm 1\}$ denote the epimorphism obtained by quotienting D_p by the normal subgroup generated by γ . Then we may define an action of $D_p \times D_p$ on C_q by setting, for $(x, y) \in D_p \times D_p$ and $g \in C_q$,

$$(x, y) \cdot g = g^{\tau(x)\tau(y)}.$$

In other words, if $\tau(x)$ and $\tau(y)$ agree, then (x, y) acts trivially on g and if $\tau(x) \neq \tau(y)$, then (x, y) acts by inverting g . We let $W = C_q \rtimes (D_p \times D_p)$ be the associated semi-direct product and let $\lambda : W \rightarrow D_p \times D_p$ be the quotient map. When doing calculations involving W , it is often useful to represent elements as triples (a, b, c) where $a \in C_q$, and $(b, c) \in D_p \times D_p$, so then λ becomes the projection to the second and third coordinates.

Definition 2.1. (1) Given a graph $\Gamma = (V, R)$ with vertex set V and edge relation $R \neq \emptyset$, we define the group G_Γ by

$$G_\Gamma = \{((a_v)_{v \in V}, (b_r)_{r \in R}) \in D_p^V \times W^R : r = (v, v') \in R \implies \lambda(b_r) = (a_v, a_{v'})\}.$$

The group G_Γ is a closed subgroup of the profinite group $D_p^V \times W^R$ and is therefore profinite. When Γ has no edges, we set $G_\Gamma = D_p^V$.

(2) Given a profinite group G , we define a graph $\Gamma(G) = (V_G, R_G)$ where the vertex set V_G is given by

$$V_G = \{N \in \mathcal{N}(G) : G/N \cong D_p\}$$

and edge set R given by

$$R_G = \{(N, N') \in V_G^2 : N \neq N' \wedge (\exists M \in \mathcal{N}(G))(M \subseteq N \cap N' \text{ and } G/M \cong W)\}.$$

Remark 2.2. The construction of the graph $\Gamma(G)$ from the group G can be viewed as an interpretation of the graph, viewed as a first-order structure, in the structure $S(G)$. The vertex set is defined in $S(G)$ by

$$V_G = \{\alpha \in X_{2p}(S(G)) : P(\alpha, \alpha, \alpha) \text{ and } [\alpha] \cong D_p\}$$

and the edge set is defined by

$$R_G = \{(\alpha, \alpha') \in V_G^2 : \alpha \neq \alpha' \wedge (\exists \beta \in X_{4p^2q}(S(G))) (\beta \leq \alpha \wedge \alpha' \text{ and } [\beta] \cong W)\}.$$

When S is an inverse system, we will abuse notation by writing $\Gamma(S)$ instead of $\Gamma(G(S))$, and will refer to $\Gamma(S)$ as the graph interpreted in S .

The key fact about these constructions is that the second construction undoes the first:

Fact 2.3. [FJ08, Proposition 28.8.3] Suppose $\Gamma = (V, R)$ is a graph. Then the graph Γ is isomorphic to $\Gamma(G_\Gamma)$ via the map $v \mapsto \ker \pi_v$.

Underlying Fact 2.3 above is the following:

Fact 2.4. [FJ08, Example 28.7.4] If $\pi : D_p^I \rightarrow D_p$ is an epimorphism, then for some $i \in I$, $\ker(\pi_i) = \ker(\pi)$ where π_i is the projection onto the i th coordinate.

We will need to upgrade this slightly:

Lemma 2.5. *Suppose V , V' , and I are disjoint finite sets.*

- (1) *If $\pi : D_p^V \times C_2^I \rightarrow D_p$ is an epimorphism with $N = \ker(\pi)$, then there is some $v \in V$ such that $N = \ker(\pi_v)$.*
- (2) *Suppose $|V| \geq |V'|$ and $\pi : D_p^V \times C_2^I \rightarrow D_p^{V'}$ is an epimorphism. Then there is $V_0 \subseteq V$ with $|V_0| = |V'|$ such that $\ker(\pi) = \bigcap_{v \in V_0} \ker(\pi_v)$.*

Proof. (1) Note that there must be some $v \in V$ such that $(C_p)_v \cap N = 1$, since otherwise, the map π factors through $C_2^V \times C_2^I$, which is impossible. If there is some $i \in I$ such that $N \cap (C_2)_i$ is trivial, then it follows that $(D_p^V \times (C_2)^I)/N$ must contain $D_p \times C_2$ and hence $C_2 \times C_2$, which D_p does not. Hence for all i , $(C_2)_i \subseteq N$ and hence the result follows by Fact 2.4.

(2) By (1), we know that, for each $w \in V'$, there is some $j(w) \in V$ such that $\ker(\pi_w \circ \pi) = \ker(\pi_{j(w)})$. Let $V_0 = \{j(w) : w \in V'\}$. Since an element g of $D_p^V \times C_2^I$ is in the kernel of π if and only if $\pi(g)$ is the identity in each coordinate of $D_p^{V'}$, it follows that

$$\ker(\pi) = \bigcap_{v \in V_0} \ker(\pi_v).$$

Moreover, the map $w \mapsto j(w)$ is injective, since if $w \neq w'$ and $j(w) = j(w')$, then it follows that all elements in the image of π agree in both the w and w' coordinate, which is impossible since π is surjective. This implies $|V_0| = |V'|$. $\square$

For reasons that will soon become clear, we will consider groups of the form $G_\Gamma \times C_2^I$ for a set I . For a vertex $v \in V$, we will write $\pi_v : G_\Gamma \times C_2^I \rightarrow D_p$ for the projection to the v coordinate and likewise for $\pi_r : G_\Gamma \times C_2^I \rightarrow W$.

Lemma 2.5 has the following consequence:

Corollary 2.6. *Suppose $\Gamma = (V, R)$ is a graph and I is a set. Then the graph Γ is isomorphic to $\Gamma(G_\Gamma \times C_2^I)$.*

Proof. It follows from Lemma 2.5 and Fact 2.3 that $\Gamma \cong \Gamma(G_\Gamma \times C_2^I)$ via the map $v \mapsto \ker(\pi_v)$. Then the conclusion follows from [FJ08, Lemma 28.6.1], which shows that $\Gamma(H) \cong \Gamma(\tilde{H})$ for any profinite group H , where $\tilde{H}$ denotes the universal Frattini cover of H . $\square$

Lemma 2.7. *Suppose $\Gamma = (V, R)$ is a graph. Then $\Phi(G_\Gamma) = 1$.*

Proof. We use the description of $G_\Gamma = C_q^R \rtimes D_p^V$. For each vertex $v \in V$, consider the pair of open subgroups

$$\begin{aligned} A_v &= C_q^R \rtimes (D_p^{V \setminus \{v\}} \times \langle \beta \rangle) \\ A'_v &= C_q^R \rtimes (D_p^{V \setminus \{v\}} \times \langle \gamma\beta \rangle), \end{aligned}$$

which are $\langle \beta \rangle$ and $\langle \gamma\beta \rangle$ in the v coordinate, respectively. Now β and $\gamma\beta$ are both reflections in D_p so A_v and A'_v both have index p in G_Γ and are therefore maximal subgroups. For each $r \in R$, we can define

$$B_r = (C_q^{R \setminus \{r\}} \times 1) \rtimes D_p^V,$$

which is an open subgroup of G_Γ of index q and therefore also maximal. We have

$$\Phi(G_\Gamma) \subseteq \bigcap_{v \in V} (A_v \cap A'_v) \cap \bigcap_{r \in R} B_r = 1,$$

so the Frattini subgroup of G_Γ is trivial. $\square$

2.2. A bounding lemma. For a vertex $v \in V$, there is a function $D_p \rightarrow G_\Gamma \times C_2^I$ denoted $g \mapsto (g)_v$, which is defined so that $(g)_v$ is the identity in all coordinates except in the v coordinate, where it is g , and in the coordinates $r = (v, v')$, where it is equal to $(1, g, 1)$ in W . Similarly, for an edge $r = (v, v')$, there is a map $W \rightarrow G_\Gamma \times C_2^I$ denoted $g \mapsto (g)_r$ defined so that if $g = (a, b, c) \in W$, then $(g)_r$ is the identity in all coordinates except the v coordinate, where it is a , the v' coordinate, where it is b , and the r coordinate, where it is g . We will write $(C_p)_v$ and $(D_p)_v$ for the images of C_p and D_p under the map $g \mapsto (g)_v$; we similarly write $(C_q)_r$ and $(W)_r$ for the corresponding elements, for an edge $r \in R$.

The group G_Γ can be alternatively described as $C_q^R \rtimes D_p^V$ where $(D_p)_v$ acts trivially on $(C_q)_r$ for all r not incident to v , and $(D_p)_v \times (D_p)_{v'}$ acts on $(C_q)_r$ for $r = (v, v')$ as in W [FJ08, Lemma 28.8.2].

Lemma 2.8. *Suppose $\Gamma = (V, R)$ is a graph and I is a set.*

- (1) *Suppose $N \trianglelefteq D_p^V$ is an open normal subgroup. If $\pi_v(N) \supseteq C_p$ for some $v \in V$, then $(C_p)_v \subseteq N$.*
- (2) *Suppose $N \trianglelefteq G_\Gamma \times C_2^I$ is an open normal subgroup. If $\pi_v(N) \supseteq C_p$ for some $v \in V$, then $(C_p)_v \subseteq N$.*
- (3) *Suppose $N \trianglelefteq G_\Gamma \times C_2^I$ is an open normal subgroup. If $C_q \subseteq \pi_r(N)$ for some $r \in R$, then $(C_q)_r \subseteq N$.*
- (4) *Suppose $N \trianglelefteq G_\Gamma \times C_2^I$ and $N \cap C_q^R = 1$. Then we have*

$$N \subseteq \{(\bar{a}, \bar{b}, \bar{c}) \in G_\Gamma \times C_2^I : r = (v, v') \in R \implies \tau(a_v) = \tau(a_{v'}) \text{ and } b_r = (1, a_v, a_{v'})\}.$$

Proof. (1) We will use the description of $D_p = \langle \beta, \gamma | \beta^2 = 1, \gamma^p = 1, \gamma\beta = \gamma^{-1} \rangle$. Pick $g \in N$ such that $\pi_v(g) = \gamma$ and let h be the element of D_p^V which is 1 in the v coordinate and β in all other coordinates. Then $h = h^{-1}$ and we have $(\gamma^2)_v = hghg \in N$ and $(\gamma^2)_v$ generates $(C_p)_v$, since p is odd.

(2) Let $\pi : G_\Gamma \times C_2^I \rightarrow D_p^V$ be the projection to the V coordinates. By (1), $(C_p)_v \subseteq \pi(N)$, so we can find some $g = (\bar{a}, \bar{b}, \bar{c}) \in N$ such that a_v generates $(C_p)_v$ and $a_{v'} = 1$ for all $v' \neq v$ (here $\bar{a} \in D_p^V$, $\bar{b} \in W^R$, and $\bar{c} \in C_2^I$). By the definition of G_Γ , for an edge $r \in R$, if v is not incident to r , then $b_r = (x, 1, 1)$ for some $x \in C_q$, so $b_r^q = (1, 1, 1) \in W$. If, on the other hand, $r = (v, v')$, then $b_r = (x, \beta, 1)$ for some $x \in C_q$ and $b_r^q = (1, \beta^q, 1) \in W$. Also, every element of C_2 has order 2. Thus $g^{2q} \in N$ and generates $(C_p)_v$.

(3) Suppose $r = (v, v')$ and let δ denote the generator of C_q . Pick $g \in N$ with $\pi_r(g) = \delta$. After replacing g with g^{2p} , we may assume $g \in N \cap C_q^R$, that is, it is the identity in all the V and I coordinates, since δ^{2p} is also a generator of C_q . Let $r' \in R$ be any edge with $r' \neq r$ (if there is no such, we are already done) and let $w \in V \setminus \{v, v'\}$ be a vertex incident to the edge r' (which exists since $r' \neq r$). Then $(\gamma)_w = (\gamma)_w^{-1}$ and we have $g' = (\gamma)_w g (\gamma)_w \in N$. Note that since w is not incident to r , we have $\pi_r(g') = \delta^2$. If $\pi_{r'}(g) = (\delta^i, 1, 1)$, then we have

$$\pi_{r'}(g') = (1, \gamma, 1)(\delta^i, 1, 1)(1, \gamma, 1)(\delta^i, 1, 1) = (\delta^{-i}, 1, 1)(\delta^i, 1, 1) = (1, 1, 1).$$

Hence some power of g' will be in N and agree with $(\delta)_r$ in both the r and r' coordinate. Iterating, we can find an element of N which agrees with $(\delta)_r$ on any finitely many coordinates. Since N is closed, get $(\delta)_r \in N$.

(4) First, we prove the corresponding statement for W : if $N \trianglelefteq W$ and $N \cap C_q = 1$, then $N \subseteq \{(1, b, c) : \tau(b) = \tau(c)\}$. First, suppose towards contradiction that there is some $g = (a, b, c) \in N$ with $\tau(b) \neq \tau(c)$. Again letting δ denote the generator of C_q , we have that the commutator $[(\delta, 1, 1), g] \in N$ and we calculate

$$[(\delta, 1, 1), g] = (\delta^{-1}, 1, 1)(a, b, c)(\delta, 1, 1)(a, b^{-1}, c^{-1}) = (\delta^{-2}, 1, 1),$$

and since δ^{-2} also generates C_q , we have $C_q \subseteq N$. This shows that if $(a, b, c) \in N$, then $\tau(b) = \tau(c)$. Next, suppose there is some $g = (a, b, c) \in N$ with $a \neq 1$. Since $\tau(b) = \tau(c)$, we have $g^{2p} = (a^{2p}, b^{2p}, c^{2p}) = (a^{2p}, 1, 1)$ and since the order of a is prime to $2p$, we get that $C_q \subseteq N$. This contradiction completes the proof.

Now the statement follows. If $N \trianglelefteq G_\Gamma \times C_2^I$ is an open normal subgroup and $C_q^R \cap N = 1$, then, for each $r \in R$, $\pi_r(N)$ is a normal subgroup of W with $C_q \cap \pi_r(N) = 1$. Thus if $(\bar{a}, \bar{b}, \bar{c}) \in N$, it follows that, for all $r = (v, v') \in R$, $b_r = (1, a_v, a_{v'})$ with $\tau(a_v) = \tau(a_{v'})$. $\square$

Lemma 2.9. *Suppose $\Gamma = (V, R)$ is a graph. Let $S = C_q^R C_p^V$ be the (normal) product of the p - and q -Sylow subgroups of G_Γ . If $N \trianglelefteq G_\Gamma$ is a proper normal subgroup, then NS is also a proper normal subgroup.*

Proof. Suppose towards contradiction that $NS = G_\Gamma$. Fix some $v \in V$. Then there are $a \in S$ and $b \in N$ such that $ab = (\beta)_v$, where $(\beta)_v$ denotes the element which is an involution in the v coordinate and the identity in every other coordinate. We will show that $(\beta)_v \in N$.

Let $c = b^{pq} \in N$. Then $\pi_v(c) = \pi_v(b^p) = \beta$ and $\pi_{v'}(c) = \pi_{v'}(b^p) = 1$ for all $v' \neq v$ in V . Thus if r is an edge that is not incident to v , we have $\pi_r(b^p) = (x_r, 1, 1) \in C_q \rtimes (D_p \times D_p) = W$ so $\pi_r(c) = (1, 1, 1)$. On the other hand, if $r = (v, v')$ is incident to v , then $\pi_r(c) = (x_r, \beta, 1)$. Let $d \in G_\Gamma$ be an element such that $\pi_r(d) = (x_r^{\frac{q-1}{2}}, 1, 1)$ for all r incident to v and the identity in every other

coordinate. Then since N is normal, $d^{-1}cd \in N$, and for each $r = (v, v')$, we have

$$\begin{aligned}\pi_r(d^{-1}cd) &= (x_r^{-\left(\frac{q-1}{2}\right)}, 1, 1) \cdot (x_r, \beta, 1) \cdot (x_r^{\frac{q-1}{2}}, 1, 1) \\ &= (x_r^{1+\left(\frac{q-1}{2}\right)}, \beta, 1) \cdot (x_r^{\frac{q-1}{2}}, 1, 1) \\ &= (x_r^q, 1, 1) \\ &= (1, 1, 1).\end{aligned}$$

Thus, we have $d^{-1}cd = (\beta)_v \in N$.

We know, then, that $\pi_v(N)$ is a normal subgroup of D_p containing β , so $\pi_v(N) = D_p$ which entails $(C_p)_v \subseteq N$ by Lemma 2.8(2). Since $(\beta)_v$ is also in N , we get $(D_p)_v \subseteq N$. Moreover, if $r = (v', v'') \in R$, since $(\beta)_{v'} \in N$, there is an element $g \in N$ such that, writing $\pi_r(g) = (a, b, c)$, we have $\tau(b) \neq \tau(c)$. It follows that $C_q \subseteq \pi_r(N)$, by (the proof of) Lemma 2.8(4). Then, by Lemma 2.8(3), we obtain $(C_q)_r \subseteq N$. Since r is arbitrary, we have shown $N = G_\Gamma$, a contradiction. $\square$

The following technical lemma will be a key ingredient in our analysis of $\text{Th}(S(\widetilde{G}_\Gamma))$ when Γ is an infinite graph:

Lemma 2.10. *Let $\Gamma = (V, R)$ be a graph and I a (possibly empty) set. Suppose $A = (G_\Gamma \times C_2^I)/N$ is a quotient of $G_\Gamma \times C_2^I$, where $N \trianglelefteq G_\Gamma \times C_2^I$ is an open normal subgroup. If $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ is the induced subgraph of Γ with vertices*

$V_0 = \{v \in V : \pi_v(N) \subsetneq D_p \text{ or there exists } r \in R \text{ incident to } v \text{ with } \pi_r(N) \cap C_q = 1\}$, *then A is a finite quotient of $G_{\Gamma_0} \times C_2^\ell$ for some ℓ . Moreover, we can bound ℓ and $|V_0|$ in terms of $|A|$.*

Proof. By definition of V_0 , if $v \notin V_0$, then $\pi_v(N) = D_p$. By Lemma 2.8(2), it follows that $(C_p)_v \subseteq N$ for all $v \notin V_0$. The subgroup generated by the groups $(C_p)_v$ for $v \in V \setminus V_0$ is dense in the subgroup $C_p^{V \setminus V_0}$ so, since N is closed, we must have $C_p^{(V \setminus V_0)} \subseteq N$. Similarly, if $r \notin R_0$, then $C_q \subseteq \pi_r(N)$, so, by Lemma 2.8(3), $(C_q)_r \subseteq N$ and since N is closed, we obtain $C_q^{R \setminus R_0} \subseteq N$. Thus, A is a quotient of $(G_\Gamma \times C_2^I)/(C_p^{(V \setminus V_0)} C_q^{(R \setminus R_0)}) \cong G_{\Gamma_0} \times C_2^{(V \setminus V_0)} \times C_2^I$. Since A is finite, we obtain that A is a quotient of $G_{\Gamma_0} \times C_2^\ell$ for some ℓ .

Now we explain how to get the bounds. Write $|A| = 2^k p^n q^m$. Since C_p is the only nontrivial normal subgroup of D_p , we may partition $V_0 = V_1 \cup V_2 \cup V_3$, where

$$\begin{aligned}V_1 &= \{v \in V_0 : \pi_v(N) = 1\} \\ V_2 &= \{v \in V_0 : \pi_v(N) = C_p\} \\ V_3 &= \{v \in V_0 : \pi_v(N) = D_p\}.\end{aligned}$$

We will bound the cardinality of each V_i as a function of k , n , and m .

Claim 1: $|V_1| \leq \min\{k, n\}$.

Proof of Claim: If we replace A by a quotient $A' := (G_\Gamma \times C_2^I)/(NC_q^R C_2^I)$, we still have $|A'| = p^n c$ for some c with $\gcd(c, p) = 1$. Additionally, by Lemma 2.8(2), we know $C_p^{V_2 \cup V_3} \subseteq N$. Hence, this A' may itself be viewed as a quotient of

$$(G_\Gamma \times C_2^I)/(C_p^{V_2 \cup V_3} C_q^R C_2^I) \cong D_p^{V_1} \times C_2^{V_2} \times C_2^{V_3}.$$

Thus we can write $A' = (D_p^{V_1} \times C_2^{V_2} \times C_2^{V_3})/\overline{N}$ for some normal subgroup $\overline{N}$. Quotienting further, we can replace A' by $A'' = (D_p^{V_1} \times C_2^{V_2} \times C_2^{V_3})/(NC_2^{V_2} C_2^{V_3})$

and we have that A'' may be viewed as a quotient of $D_p^{V_1}$ and $|A''| = p^n c$ for some c with $\gcd(p, c) = 1$. Write $A'' = D_p^{V_1}/N''$ and note that $\pi_v(N'') = 1$ for all $v \in V_1$ as well, by construction and the choice of V_1 . In other words, $N'' = 1$ and $A'' \cong D_p^{V_1}$ so we have $|A''| = (2p)^{|V_1|}$. Since $|A| \geq |A''|$, we obtain $|V_1| \leq \min\{k, n\}$. $\square$

Claim 2: $|V_2| \leq k$.

Proof of Claim: Again by Lemma 2.8(2), we have $(C_p)_v \subseteq N$ for all $v \in V_2$. We know, then, that $\bar{g} \in \bigcap_{v \in V_2} N \ker(\pi_v)$ if and only if $\bar{g}$ is in C_p in the v coordinate for every $v \in V_2$ (and can be arbitrary otherwise). From this it follows that

$$N \bigcap_{v \in V_2} \ker(\pi_v) = \bigcap_{v \in V_2} N \ker(\pi_v).$$

Therefore, we have

$$(G_\Gamma \times C_2^I) / \left(N \bigcap_{v \in V_2} \ker(\pi_v) \right) = (G_\Gamma \times C_2^I) / \bigcap_{v \in V_2} N \ker(\pi_v) \cong (C_2)^{V_2}.$$

This shows A has $C_2^{V_2}$ as a quotient. Since this group has cardinality $2^{|V_2|}$, we get $|V_2| \leq k$. $\square$

Claim 3: $|V_3| \leq 2m$.

Proof of Claim: Let $R' = \{r \in R : \pi_r(N) \cap C_q = 1\}$ and let V' be the vertices that are on some edge in R' . Each vertex $v \in V_3$ is on some edge in R' and each edge has only two vertices so we have $|R'| \geq \frac{|V_3|}{2}$. Let $N' = NC_2^I C_p^V$. Then A has a quotient

$$(G_\Gamma \times C_2^I) / N' \cong (C_q^R \rtimes C_2^V) / M$$

for some open normal $M \trianglelefteq C_q^R \rtimes C_2^V$. Recall that C_2^V acts on each coordinate $r = (v, v')$ of R by inverting if the v and v' coordinates of an element of C_2^V disagree and otherwise acts trivially. Thus $C_2^V = C_2^{V'} \times C_2^{(V \setminus V')}$ and the $C_2^{(V \setminus V')}$ factor acts trivially on all of the R' coordinates. Moreover, if $r \notin R'$, then $\pi_r(N) \supseteq C_q$ and hence $(C_q)_r \subseteq N$ by Lemma 2.8(3) (and hence $(C_q)_r \subseteq M$ for all $r \notin R'$). It follows, then, that

$$(C_q^R \rtimes C_2^V) / M \cong ((C_q^{R'} \rtimes C_2^{V'}) \times C_2^{(V \setminus V')}) / M''$$

for some open normal $M'' \trianglelefteq ((C_q^{R'} \rtimes C_2^{V'}) \times C_2^{(V \setminus V')})$. Then we have $M'' C_2^{(V \setminus V')}$ is normal in $((C_q^{R'} \rtimes C_2^{V'}) \times C_2^{(V \setminus V')})$ and we can write

$$((C_q^{R'} \rtimes C_2^{V'}) \times C_2^{(V \setminus V')}) / M'' C_2^{(V \setminus V')} \cong C_q^{R'} \rtimes C_2^{V'} / H$$

for some $H \trianglelefteq C_q^{R'} \rtimes C_2^{V'}$ with $H \cap C_q^{R'} = 1$. Note that we have obtained this group by further quotienting A by subgroups that contain no elements of order q . By Lemma 2.8(4) (applied to the graph (V', R')), we have that if $((a_r)_{r \in R'}, (b_v)_{v \in V'}) \in H$, then for all $r = (v, v') \in R'$, $a_r = 1$ and $b_v = b_{v'}$. It follows that every element of H has order 2. The quotient $(C_q^{R'} \rtimes C_2^{V'}) / H$ of A also has order $q^m c$ for some c with $\gcd(q, c) = 1$. Therefore, we have $|R'| = m$, so $|V_3| \leq 2m$. $\square$

Note that Claims 1, 2, and 3 give a bound for $|V_0|$. Note that if we have a bound for $|V_0|$, we also obtain a bound for $|G_{\Gamma_0}|$. To finish, we bound ℓ :

Claim 4: $\ell \leq \log_2(|A||G_{\Gamma_0}|)$.

Proof of Claim: By Lemma 2.8(2) and (3), we know that N contains $(C_q)_r$ for every edge r not in Γ_0 and contains $(C_p)_v$ for every vertex v not in Γ_0 . Therefore,

A is a quotient of the group

$$G_{\Gamma_0} \times C_2^{V \setminus V_0} \times C_2^I.$$

Call this group H and let $\overline{N}$ denote the image of N in H , so $A \cong H/\overline{N}$. Let $\pi_{\Gamma_0} : H \rightarrow G_{\Gamma_0}$ denote the projection to the G_{Γ_0} coordinates and let $\overline{N}' = \overline{N} \cap \ker(\pi_{\Gamma_0})$ and let $\tilde{A} = H/\overline{N}'$. Clearly A is a quotient of $\tilde{A}$ and if M is the projection of $\overline{N}'$ to the $(C_2^{V \setminus V_0} \times C_2^I)$ factor, we have

$$\tilde{A} = H/\overline{N}' \cong \left((C_2^{V \setminus V_0} \times C_2^I) / M \right) \times G_{\Gamma_0}.$$

Now $(C_2^{V \setminus V_0} \times C_2^I) / M \cong C_2^{\ell'}$ for some ℓ' so if we bound ℓ' in terms of $|A|$ we are done. We calculate

$$|\tilde{A}| = [H : \overline{N} \cap \ker(\pi_{\Gamma_0})] \leq [H : \overline{N}][H : \ker(\pi_{\Gamma_0})] = |A||G_{\Gamma_0}|.$$

Since $\ell' \leq \log_2(|\tilde{A}|) \leq \log_2(|A||G_{\Gamma_0}|)$ and we have already bounded $|V_0|$ (and thus $|G_{\Gamma_0}|$) in terms of $|A|$, we obtain the desired bound. $\square$

Remark 2.11. It would have been more natural to define

$$V_0 = \{v \in V : \pi_v(N) \not\leq D_p\},$$

but the statement of the lemma with this definition would be incorrect. Consider the graph Γ with vertices $a \neq b$ and an edge between them. Then $G_\Gamma \cong W$ and W has a normal subgroup $M = \{(1, g, h) : \tau(g) = \tau(h)\}$ with the quotient $W/M \cong C_q \rtimes C_2 \cong D_q$. But $\pi_a(M)$ and $\pi_b(M)$ are both equal to D_p .

2.3. Maps between the groups G_Γ . If $\Phi : S \rightarrow S'$ is an embedding of complete systems, then Φ induces an injective homomorphism of graphs, which we will denote $\Gamma(\Phi) : \Gamma(S) \rightarrow \Gamma(S')$. In general, this will not be an embedding of graphs: for example, one can consider the embedding $\Lambda : S(D_p \times D_p) \rightarrow S(W)$ dual to quotient map $\lambda : W \rightarrow D_p \times D_p$. The graph interpreted in $S(D_p \times D_p)$ consists of two vertices with no edge between them, while the graph interpreted in $S(W)$ also has two vertices but now there is an edge. When $\varphi : G' \rightarrow G$, we will sometimes write $\Gamma(\varphi)$ for the induced map $\Gamma(G) \rightarrow \Gamma(G')$ which is the same as $\Gamma(\Phi)$ for the dual map $\Phi : S(G) \rightarrow S(G')$, since the map of graphs is the same, regardless of whether we see it as coming from the map of groups or the map of inverse systems.

Lemma 2.12. *Suppose $\sigma_1, \sigma_2 \in \text{Aut}(D_p)$. Then there is $\nu \in \text{Aut}(W)$ so that the following diagram commutes:*

$$\begin{array}{ccc} W & \xrightarrow{\lambda} & D_p \times D_p \\ \downarrow \nu & & \downarrow \sigma_1 \times \sigma_2 \\ W & \xrightarrow{\lambda} & D_p \times D_p. \end{array}$$

Proof. Note that, because it is the only nontrivial normal subgroup, $\ker(\tau) = C_p$ is a characteristic subgroup of D_p . Because the image of τ is C_2 , it follows that $\tau(\sigma_i(g)) \neq \tau(g)$ if and only if exactly one of $\sigma_i(g)$ and g are in C_p , which is impossible. Therefore $\tau(\sigma_i(g)) = \tau(g)$ for $i = 1, 2$ and the bijection

$$C_q \times D_p \times D_p \xrightarrow{\text{id}_{C_q} \times \sigma_1 \times \sigma_2} \tilde{C}_q \times D_p \times D_p.$$

is compatible with the semi-direct product structure on $W = C_q \rtimes D_p \times D_p$ and therefore may be regarded as an automorphism $\nu \in \text{Aut}(W)$. The desired compatibilities are clear. $\square$

Note that if $f : \Gamma \rightarrow \Gamma$ is an automorphism of the graph Γ , we obtain a ‘coordinate-change’ automorphism φ_f of G_Γ defined by

$$\varphi_f((a_v)_{v \in V}, (b_r)_{r \in R}) = ((a_{f^{-1}(v)})_{v \in V}, (b_{f^{-1}(r)})_{r \in R}).$$

Note that the induced map of graphs $\Gamma(\varphi_f)$ maps $\ker(\pi_v) \mapsto \ker(\pi_v \circ \varphi_f)$. It is easy, then, to check that $\Gamma(\varphi_f) = f$.

On the other hand, if, for each $v \in V$, we have an automorphism $\sigma_v \in \text{Aut}(D_p)$, and then for each $r = (v, v') \in R$, we have an automorphism $\sigma_r \in \text{Aut}(W)$ which is compatible with σ_v and $\sigma_{v'}$ as in Lemma 2.12, then we get an automorphism $\bar{\sigma}$ of G_Γ defined by

$$\bar{\sigma}((a_v)_{v \in V}, (b_r)_{r \in R}) = ((\sigma_v(a_v))_{v \in V}, (\sigma_r(b_r))_{r \in R}).$$

Then we clearly have $\Gamma(\bar{\sigma}) = \text{id}_\Gamma$. The following lemma shows that we can factor any automorphism of G_Γ into a composition of two automorphisms, one of which is a coordinate-change automorphism and the other which is a tuple of automorphisms applied coordinate-wise.

Lemma 2.13. *Suppose $\varphi \in \text{Aut}(G_\Gamma)$ is an automorphism and $f = \Gamma(\varphi)$. Then there is a tuple of automorphisms $\bar{\sigma} = ((\sigma_v)_{v \in V}, (\sigma_r)_{r \in R}) \in \text{Aut}(D_p)^V \times \text{Aut}(W)^R$ such that*

$$\varphi = \bar{\sigma} \circ \varphi_f,$$

where φ_f is the coordinate-change automorphism associated to f .

Proof. Since $\Gamma(\varphi \circ \varphi_f^{-1}) = \text{id}_\Gamma$, $\varphi \circ \varphi_f^{-1}$ preserves $\ker(\pi_v)$ for each vertex v and hence induces an automorphism σ_v on the quotient $G_\Gamma / \ker(\pi_v) \cong D_p$. Likewise, for each $r = (v, v') \in R$, we obtain an automorphism σ_r on $G_\Gamma / \ker(\pi_r)$ which is compatible with $\sigma_v, \sigma_{v'}$ as in Lemma 2.12. Unravelling definitions, we have

$$(\varphi \circ \varphi_f^{-1})((a_v)_{v \in V}, (b_r)_{r \in R}) = ((\sigma_v(a_v))_{v \in V}, (\sigma_r(b_r))_{r \in R}) = \bar{\sigma}((a_v)_{v \in V}, (b_r)_{r \in R}),$$

which shows $\varphi = \bar{\sigma} \circ \varphi_f$. $\square$

Lemma 2.14. *Suppose $\Gamma_0 \subseteq \Gamma$ and $\Gamma'_0 \subseteq \Gamma'$ are graphs and we are given an isomorphism of groups $\varphi_0 : G_{\Gamma_0} \rightarrow G_{\Gamma'_0}$ and an isomorphism of graphs $f : \Gamma' \rightarrow \Gamma$ such that $f|_{\Gamma'_0} = \Gamma(\varphi_0)$. Then there is an isomorphism $\varphi : G_\Gamma \rightarrow G_{\Gamma'}$ such that $\Gamma(\varphi) = f$ and the following diagram commutes:*

$$\begin{array}{ccc} G_\Gamma & \xrightarrow{\varphi} & G_{\Gamma'} \\ \downarrow \pi_{\Gamma_0} & & \downarrow \pi_{\Gamma'_0} \\ G_{\Gamma_0} & \xrightarrow{\varphi_0} & G_{\Gamma'_0}. \end{array}$$

Proof. Let $\Gamma_0 = (V_0, R_0)$, $\Gamma = (V, R)$ and, similarly, $\Gamma'_0 = (V'_0, R'_0)$ and $\Gamma' = (V', R')$. Denote by f_0 the restriction $f|_{\Gamma'_0}$. Let $\varphi_f : G_\Gamma \rightarrow G_{\Gamma'}$ be the coordinate-change isomorphism associated to f . That is, φ_f is defined by

$$\varphi_f((a_v)_{v \in V}, (b_r)_{r \in R}) = ((a_{f^{-1}(v)})_{v \in V'}, (b_{f^{-1}(r)})_{r \in R'}),$$

and, likewise, let $\varphi_{f_0} : G_{\Gamma_0} \rightarrow G_{\Gamma'_0}$ be defined as the coordinate-change isomorphism associated to f_0 . Since the automorphism $(\varphi_0 \circ \varphi_{f_0}^{-1}) \in \text{Aut}(G_{\Gamma'_0})$ satisfies $\Gamma(\varphi_0 \circ \varphi_{f_0}^{-1}) = \text{id}_{\Gamma'_0}$, it follows by Lemma 2.13 that $\varphi_0 \circ \varphi_{f_0}^{-1} = \bar{\sigma}_0$ for some tuple of automorphisms $\bar{\sigma}_0 = ((\sigma_v)_{v \in V'_0}, (\sigma_r)_{r \in R'_0}) \in \text{Aut}(D_p)^{V'_0} \times \text{Aut}(W)^{R'_0}$. Hence $\varphi_0 = \bar{\sigma}_0 \circ \varphi_{f_0}$.

Define $\sigma_v = \text{id}_{D_p}$ for each $v \in V' \setminus V'_0$, and then, for each $r = (v, v') \in R' \setminus R'_0$, define $\sigma_r \in \text{Aut}(W)$ to be an automorphism compatible with σ_v and $\sigma_{v'}$, using Lemma 2.12. This defines a map $\bar{\sigma} = ((\sigma_v)_{v \in V'}, (\sigma_r)_{r \in R'})$ and $\bar{\sigma}_0$ is a subtuple which can be obtained from $\bar{\sigma}$ by restriction to the Γ'_0 coordinates.

Define $\varphi = \bar{\sigma} \circ \varphi_f : G_{\Gamma} \rightarrow G_{\Gamma'}$. Then we have $\Gamma(\varphi) = \Gamma(\varphi_f) = f$ and, by construction, we have $\pi_{\Gamma'_0} \circ \varphi = \varphi_0 \circ \pi_{\Gamma_0}$, which completes the proof. $\square$

Lemma 2.15. *Suppose $\Gamma = (V, R)$ is a graph and $G \cong G_{\Gamma}$. Suppose, moreover, that there is a bijection $V \rightarrow \{N \trianglelefteq G : G/N \cong D_p\}$, denoted by $v \mapsto N_v$, and a bijection $R \rightarrow \{M \trianglelefteq G : G/M \cong W\}$, denoted by $r \mapsto M_r$ satisfying the following conditions:*

- (1) *For each $v \in V$, there is an isomorphism $\varphi_v : G_{\Gamma}/\ker(\pi_v) \rightarrow G/N_v$.*
- (2) *For each $r = (v, v') \in R$, we have $M_r \subseteq N_v \cap N_{v'}$ and there is an isomorphism $\varphi_r : G_{\Gamma}/\ker(\pi_r) \rightarrow G/M_r$ such that the following diagram commutes:*

$$\begin{array}{ccc} G_{\Gamma}/\ker(\pi_r) & \xrightarrow{\varphi_r} & G/M_r \\ \downarrow & & \downarrow \lambda'_r \\ G_{\Gamma}/\ker(\pi_v) \times G_{\Gamma}/\ker(\pi_{v'}) & \xrightarrow{\varphi_v \times \varphi_{v'}} & G/N_v \times G/N_{v'}, \end{array}$$

where the vertical arrows denote the canonical projections.

Then there is a unique isomorphism $\varphi : G_{\Gamma} \rightarrow G$ which induces φ_v for each $v \in V$ and φ_r for each $r \in R$.

Proof. Since $G \cong G_{\Gamma}$, we know that the canonical map

$$G \rightarrow \{((a_v)_{v \in V}, (b_r)_{r \in R}) \in \prod_{v \in V} G/N_v \times \prod_{r \in R} G/M_r : r = (v, v') \in R \implies \lambda'_r(b_r) = (a_v, a_{v'})\}$$

is an isomorphism. Identifying G with the range of this isomorphism, we see that the conditions (1) and (2) imply that the map $G_{\Gamma} \rightarrow G$ defined by $g \mapsto ((\varphi_v(g))_{v \in V}, (\varphi_r(g))_{r \in R})$ is an isomorphism. Moreover, any two isomorphisms that induce the same φ_v and φ_r will clearly induce the same isomorphism so it is unique. $\square$

Corollary 2.16. *Suppose $\Gamma = (V, R)$ is a graph. Working in $S(G_{\Gamma})$, assume we are given, for each $v \in V$, some automorphism $\sigma_v : [\ker(\pi_v)] \rightarrow [\ker(\pi_v)]$, and, for each $r = (v, v') \in R$, some automorphism $\sigma_r : [\ker(\pi_r)] \rightarrow [\ker(\pi_r)]$ such that the diagram*

$$\begin{array}{ccc} [\ker(\pi_r)] & \xrightarrow{\sigma_r} & [\ker(\pi_r)] \\ \downarrow & & \downarrow \\ [\ker(\pi_v)] \times [\ker(\pi_{v'})] & \xrightarrow{(\sigma_v, \sigma_{v'})} & [\ker(\pi_v)] \times [\ker(\pi_{v'})], \end{array}$$

commutes, where the vertical arrows are the natural quotient maps (whose graph is given by the relation C). Then there is a unique $\sigma_* \in \text{Aut}(S(G_\Gamma))$ such that $\sigma_*|_{[\ker(\pi_v)]} = \sigma_v$ and $\sigma_*|_{[\ker(\pi_r)]} = \sigma_r$ for all $v \in V$ and $r \in R$.

Proof. Take $G = G_\Gamma$ and $v \mapsto \ker(\pi_v)$ in Lemma 2.15. $\square$

2.4. The universal ℓ -Frattini cover. In general, universal Frattini covers of a profinite group G can be analyzed as fiber products of Frattini covers of G whose kernels are pro- ℓ for different primes ℓ . Taking this point of view, we are able to explicitly describe the universal Frattini cover $\widetilde{G}_\Gamma$ of G_Γ as a certain semi-direct product of free pro- ℓ groups for $\ell \in \{2, p, q\}$. This digression will take us a little far afield from the model-theoretic issues of interest for us, but this level of granularity will turn out to be important for the description of algebraic closure and independence in the theories $\text{Th}(S(\widetilde{G}_\Gamma))$.

We use the notation F_n to refer to the free profinite group of rank n and $F_n(p)$ to denote the free pro- p group of rank n . This makes sense even when n is not a natural number but an infinite cardinal.

Definition 2.17. Let ℓ be a fixed prime.

- (1) We say that an epimorphism of profinite groups $\pi : G \rightarrow H$ is an ℓ -cover if $\ker(\pi)$ is a pro- ℓ group.
- (2) We say that an ℓ -cover $\pi : G \rightarrow H$ is ℓ -Frattini if $\ker(\pi) \subseteq \Phi(G)$, i.e. π is both an ℓ - and Frattini cover.

Fact 2.18. [FJ08, Construction 22.11.5] Let ℓ be a prime number. If G is a profinite group, then it has a universal ℓ -Frattini cover $\pi_\ell : {}_\ell\tilde{G} \rightarrow G$. That is, π_ℓ is an ℓ -Frattini cover and, if $\rho : H \rightarrow G$ is ℓ -Frattini, then there is an epimorphism $\chi : {}_\ell\tilde{G} \rightarrow H$ so that the following diagram commutes:

$$\begin{array}{ccc} & & {}_\ell\tilde{G} \\ & \swarrow \chi & \downarrow \pi_\ell \\ H & \xrightarrow{\rho} & G \end{array}$$

Moreover, the universal ℓ -Frattini cover may be constructed as follows. Suppose $\pi : \tilde{G} \rightarrow G$ is the universal Frattini cover. Then since $\ker(\pi)$ is a closed subgroup of $\Phi(\tilde{G})$ and $\Phi(\tilde{G})$ is pro-nilpotent [FJ08, Lemma 22.1.2], it follows that $\ker(\pi) = \prod_{p' \text{ prime}} K_{p'}$ where $K_{p'}$ is pro- p' [RZ10, Proposition 2.3.8]. Then, setting $K'_\ell = \prod_{p' \neq \ell} K_{p'}$, we let ${}_\ell\tilde{G} = \tilde{G}/K'_\ell$ and let $\pi_\ell : {}_\ell\tilde{G} \rightarrow G$ be the map induced by π . Then $\pi_\ell : {}_\ell\tilde{G} \rightarrow G$ is a universal ℓ -Frattini cover.

The previous construction applies equally well to a set of Π of primes. Extend the definition of ℓ -Frattini to Π -Frattini in the obvious way: a cover is Π -Frattini if the kernel is a pro- Π -group (i.e. the only primes that divide the order of any of its finite quotients lie in Π). One can define $K'_\Pi = \prod_{\ell \notin \Pi} K_\ell$ and define ${}_\Pi\tilde{G} = \tilde{G}/K'_\Pi$ and let $\pi_\Pi : {}_\Pi\tilde{G} \rightarrow G$ denote the induced map. Since π_Π is a Frattini cover and its ℓ -Sylows are free pro- ℓ for all $\ell \in \Pi$, an identical argument to [FJ08, Construction 22.11.5] establishes that this map is a universal Π -Frattini cover.

Specializing to the case of $G = G_\Gamma$, the relevant primes are 2, p , and q . We will denote by ${}_{pq}\tilde{G}_\Gamma$ the universal $\{p, q\}$ -Frattini cover (and refer to this as the universal

pq -Frattini cover of G_Γ); similarly ${}_{pq}\tilde{H}$ will denote the universal pq -Frattini cover of any profinite group H .

The following fact follows by an identical argument to [FJ08, Lemma 22.6.3]:

Fact 2.19. Let $\tilde{\varphi} : {}_{pq}\tilde{G} \rightarrow G$ be the universal pq -Frattini cover of a profinite group G . Then a profinite group H is a quotient of ${}_{pq}\tilde{G}$ if and only if H is a pq -Frattini cover of a quotient of G .

Fact 2.20. [Rib85, Theorem 3.4] Let $p_1, \dots, p_n$ be distinct prime numbers and let $t_1, \dots, t_n$ be natural numbers. Suppose G is a profinite group that admits a subnormal series

$$G = G_0 \triangleright G_1 \triangleright \dots \triangleright G_n = 1$$

such that G_{i-1}/G_i is a pro- p_i group of rank t_i . Assume $\tilde{G}$ is a profinite group with subnormal series

$$\tilde{G} = \tilde{G}_0 \triangleright \tilde{G}_1 \triangleright \dots \triangleright \tilde{G}_n = 1,$$

such that $\tilde{G}_{i-1}/\tilde{G}_i$ is the free pro- p_i group of rank t_i . Then any epimorphism $\tilde{\gamma} : \tilde{G} \rightarrow G$ is a universal Frattini cover of G .

We want to characterize the universal Frattini cover of the group G_Γ for a graph $\Gamma = (V, R)$, or more generally, of groups of the form $G_\Gamma \times C_2^I$ for an index set I (which we may suppose is disjoint from V). We use the description $G_\Gamma = C_q^R \rtimes D_p^V = (C_q^R) \rtimes (C_p^V \rtimes C_2^V)$, from which we also get a description $G_\Gamma \times C_2^I = (C_q^R) \rtimes (C_p^V \rtimes C_2^{V \cup I})$, where the C_2^I factor acts trivially. Let $(z_r)_{r \in R}$ be a set of generators converging to 1 for $F_{|R|}(q)$, and likewise generators $(y_v)_{v \in V}$ and $(x_v)_{v \in V \cup I}$ for $F_{|V|}(p)$ and $F_{|V \cup I|}(2)$ respectively.

Throughout this section, $\langle X \rangle$ denotes the closed subgroup *topologically* generated by X in a profinite group, not merely the abstractly generated subgroup. If $\Gamma_0 = (V_0, R_0)$ is a finite induced subgraph of Γ with $|V_0| = n$ and $I_0 \subseteq I$ is a subset with $|I_0| = m$, then there is a quotient $\langle y_v : v \in V_0 \rangle \cong F_n(p)$ of $F_{|V|}(p)$ obtained by mapping $y_v \mapsto y_v$ for $v \in V_0$ and $y_v \mapsto 1$ for all $v \in V \setminus V_0$. We can define an action of $C_2^{V_0 \cup I_0}$ on $\langle y_v : v \in V_0 \rangle$ by setting, for all $\bar{a} = (a_v)_{v \in V_0 \cup I_0} \in C_2^{V_0 \cup I_0}$, $\bar{a} \cdot y_v = y_v^{a_v}$ for all $v \in V_0$. Notice that these actions are compatible in the sense that if $\Gamma_0 = (V_0, R_0) \subseteq (V_1, R_1)$ are both finite induced subgraphs of Γ and $I_0 \subseteq I_1 \subseteq I$, then we have a diagram

$$\begin{array}{ccc} C_2^{V_1 \cup I_1} \times \langle y_v : v \in V_1 \rangle & \longrightarrow & \langle y_v : v \in V_1 \rangle \\ \downarrow & & \downarrow \\ C_2^{V_0 \cup I_0} \times \langle y_v : v \in V_0 \rangle & \longrightarrow & \langle y_v : v \in V_0 \rangle \end{array}$$

where the horizontal maps are the respective actions, the map $C_2^{V_1 \cup I_1} \rightarrow C_2^{V_0 \cup I_0}$ is projection to the $V_0 \cup I_0$ coordinates, and the map $\langle y_v : v \in V_1 \rangle \rightarrow \langle y_v : v \in V_0 \rangle$ maps $y_v \mapsto y_v$ if $v \in V_0$ and maps $y_v \mapsto 1$ otherwise. By passing to the inverse limit, we obtain a continuous action of $C_2^{V \cup I}$ on $F_{|V|}(p)$. This action has the property that if $\bar{a} = (a_v)_{v \in V \cup I} \in C_2^{V \cup I}$, then $\bar{a} \cdot y_v = y_v^{a_v}$. Let $F_{|V|}(p) \rtimes C_2^{V \cup I}$ be the associated semidirect product and let $\rho : F_{|V|}(p) \rtimes C_2^{V \cup I} \rightarrow C_2^{V \cup I}$ denote the quotient map.

Now we define an action of $C_2^{V \cup I}$ on $F_{|R|}(q)$. If $\Gamma_0 = (V_0, R_0)$ is a finite induced subgraph with $|R_0| = n$ and $I_0 \subseteq I$ is a finite subset, then there is an action $C_2^{V_0 \cup I_0}$ on $\langle z_r : r \in R_0 \rangle \cong F_n(q)$ defined so that, if $r = (v, v')$ and $\bar{a} = (a_v)_{v \in V_0 \cup I_0}$,

then $\bar{a} \cdot z_r = z_r^{a_v a_{v'}}$. As above, these actions cohere and we obtain a continuous action of $C_2^{V \cup I}$ on $F_{|R|}(q)$ by passage to the inverse limit, with the property that $\bar{a} = (a_v)_{v \in V \cup I} \in C_2^{V \cup I}$ and $r = (v, v')$, then $\bar{a} \cdot z_r = z_r^{a_v a_{v'}}$. We obtain, then, an action of $F_{|V|}(p) \rtimes C_2^{V \cup I}$ on $F_{|R|}(q)$ by $g \cdot h = \rho(g) \cdot h$ for all $g \in F_{|V|}(p) \rtimes C_2^{V \cup I}$ and $h \in F_{|R|}(q)$, where the action on the right-hand side of the equation is the action of $C_2^{V \cup I}$ we just defined. Let $\widetilde{{}_{pq}(G_\Gamma \times C_2^I)}$ be defined to be the semi-direct product:

$$\widetilde{{}_{pq}(G_\Gamma \times C_2^I)} = F_{|R|}(q) \rtimes (F_{|V|}(p) \rtimes C_2^{V \cup I}).$$

There is an epimorphism $\pi_{pq} : \widetilde{{}_{pq}(G_\Gamma \times C_2^I)} \rightarrow (G_\Gamma \times C_2^I)$ defined by mapping $y_v \mapsto (\gamma)_v$ for all $v \in V$, where γ denotes a generator of C_p , and mapping $z_r \mapsto (\delta)_r$ for all $r \in R$, where δ denotes a generator of C_q . Our construction ensures that the resulting maps $F_{|V|}(p) \rightarrow C_p^V$ and $F_{|R|}(q) \rightarrow C_q^R$ are compatible with the semi-direct product structure on $\widetilde{{}_{pq}(G_\Gamma \times C_2^I)}$ so we obtain a well-defined epimorphism.

Note that we have explicitly constructed $\widetilde{{}_{pq}(G_\Gamma \times C_2^I)}$ as the inverse limit of $\langle z_r : r \in R_0 \rangle \rtimes (\langle y_v : v \in V_0 \rangle \rtimes C_2^{V_0 \cup I_0})$ for finite induced subgraphs $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ and finite subsets $I_0 \subseteq I$. We may view each such group as a quotient of

$$\langle z_r : r \in R_0 \rangle \rtimes (\langle y_v : v \in V_0 \rangle \rtimes \langle x_v : v \in V_0 \cup I_0 \rangle)$$

where $\langle x_v : v \in V_0 \cup I_0 \rangle$ maps to $C_2^{V_0 \cup I_0}$ by sending $x_v \mapsto (-1)_v$ for all $v \in V_0 \cup I_0$. By taking the inverse limit over all finite induced subgraphs, we obtain a group

$$\widetilde{G_\Gamma \times C_2^I} = F_{|R|}(q) \rtimes (F_{|V|}(p) \rtimes F_{|V \cup I|}(2)).$$

We will denote by $\pi_2 : \widetilde{G_\Gamma \times C_2^I} \rightarrow \widetilde{{}_{pq}(G_\Gamma \times C_2^I)}$ the quotient map obtained by mapping each $x_v \mapsto (-1)_v$ for $v \in V \cup I$.

Lemma 2.21. *The map $\pi_{pq} \circ \pi_2 : \widetilde{G_\Gamma \times C_2^I} \rightarrow G_\Gamma \times C_2^I$ is a universal Frattini cover of $G_\Gamma \times C_2^I$.*

Proof. Note that if $\Gamma = (V, R)$ is finite, with $|V| = n$, $|R| = m$, and $|I| = k$, then we have that $G_\Gamma \times C_2^I$ has a subnormal series

$$G_\Gamma \times C_2^I = C_q^R \rtimes (C_p^V \rtimes C_2^{V \cup I}) \triangleright C_p^V \rtimes C_2^{V \cup I} \triangleright C_2^{V \cup I} \triangleright 1$$

with quotients being q -, p -, and 2-groups of rank m , n and $n + k$, respectively.

Then, by construction, $G_\Gamma \times C_2^I$ can be described as

$$\widetilde{G_\Gamma \times C_2^I} = F_m(q) \rtimes (F_n(p) \rtimes F_{n+k}(2)) \triangleright F_n(p) \rtimes F_{n+k}(2) \triangleright F_{n+k}(2) \triangleright 1,$$

so by Fact 2.20, the epimorphism $\pi_{pq} \circ \pi_2 : \widetilde{G_\Gamma \times C_2^I} \rightarrow G_\Gamma \times C_2^I$ is a universal Frattini cover.

For the general case, we note that the map $\pi_{pq} \circ \pi_2 : \widetilde{G_\Gamma \times C_2^I} \rightarrow G_\Gamma \times C_2^I$ is the inverse limit (in the obvious sense) of the maps $G_{\Gamma_0} \times C_2^{I_0} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ for $\Gamma_0 \subseteq \Gamma$ ranging over the finite induced subgraphs of Γ and $I_0 \subseteq I$ ranging over the finite subsets of I . Then $\pi_{pq} \circ \pi_2$ is a universal Frattini cover by the finite case and Fact 1.6. $\square$

Lemma 2.22. *Suppose $\Gamma = (V, R)$ is a graph, I is a set, and $\pi = \pi_{pq} \circ \pi_2 : \widetilde{G_\Gamma \times C_2^I} \rightarrow G_\Gamma \times C_2^I$ is the universal Frattini cover, with π_2 and π_{pq} defined as*

above. Then $\ker(\pi_2)$ is the unique closed normal subgroup M of $\widetilde{G_\Gamma \times C_2^I}$ contained in $\ker(\pi)$ such that the induced map $(\widetilde{G_\Gamma \times C_2^I})/M \rightarrow (\widetilde{G_\Gamma \times C_2^I})/\ker(\pi)$ is a universal pq -Frattini cover.

Proof. It is clear that $\ker(\pi_2)$ is at least one closed normal subgroup of $\widetilde{G_\Gamma \times C_2^I}$ with the desired property. Moreover, any such normal subgroup must be equal to the 2-Sylow subgroup of $\Phi(\widetilde{G_\Gamma \times C_2^I})$ since $\Phi(G_\Gamma \times C_2^I) = 1$ and is therefore uniquely determined. $\square$

Lemma 2.23. *Suppose $\Gamma = (V, R)$ is a graph, I a set, $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ is an induced subgraph and $I_0 \subseteq I$ a finite subset. Let $\pi_0 : G_\Gamma \times C_2^I \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ be the projection map. Let N be the kernel of the map $\pi_0 \circ \pi_{pq} : \widetilde{pq(G_\Gamma \times C_2^I)} \rightarrow \widetilde{G_{\Gamma_0} \times C_2^{I_0}}$. Then there is a unique closed normal subgroup $M \trianglelefteq \widetilde{pq(G_\Gamma \times C_2^I)}$ with $M \subseteq N$ and the natural map $\widetilde{pq(G_\Gamma \times C_2^I)}/M \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/N$ a universal pq -Frattini cover.*

Proof. There is clearly at least one such M , since $\widetilde{pq(G_\Gamma \times C_2^I)}/N \cong \widetilde{G_{\Gamma_0} \times C_2^{I_0}}$ and $\widetilde{pq(G_{\Gamma_0} \times C_2^{I_0})}$ is a quotient of $\widetilde{pq(G_\Gamma \times C_2^I)}$. More specifically, we recall that we may identify $\widetilde{pq(G_{\Gamma_0} \times C_2^{I_0})} = \langle z_r : r \in R_0 \rangle \rtimes (\langle y_v : v \in V_0 \rangle \rtimes C_2^{V_0 \cup I_0})$ as a quotient of $\widetilde{pq(G_\Gamma \times C_2^I)}$ via the map that sends $z_r \mapsto z_r$ if $r \in R_0$ and $z_r \mapsto 1$ otherwise, $y_v \mapsto y_v$ if $v \in V_0$ and $y_v \mapsto 1$ otherwise, and $C_2^{V \cup I}$ mapping to $C_2^{V_0 \cup I_0}$ by projection to the $V_0 \cup I_0$ coordinates. Let M_* denote the kernel of this mapping, so we have $\widetilde{pq(G_\Gamma \times C_2^I)}/M_* \cong \widetilde{pq(G_{\Gamma_0} \times C_2^{I_0})}$ and the natural map $\widetilde{pq(G_\Gamma \times C_2^I)}/M_* \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/N$ is a universal pq -Frattini cover.

Now suppose M is an arbitrary closed normal subgroup of $\widetilde{pq(G_\Gamma \times C_2^I)}$ contained in N such that the natural map $\pi_{MN} : \widetilde{pq(G_\Gamma \times C_2^I)}/M \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/N$ is a universal pq -Frattini cover. We will show $M = M_*$. Let $\pi : \widetilde{pq(G_\Gamma \times C_2^I)} \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/M$ denote the quotient map. First note that if $v \in V \cup I$ and $(\pi_{MN} \circ \pi)((-1)_v) = 1$ then $(-1)_v \in \ker(\pi)$ since $\ker(\pi_{MN})$ has no nontrivial elements of order 2, and thus $(-1)_v \in M$. It follows, then, that $(-1)_v \in M$ for all $v \in (V \cup I) \setminus (V_0 \cup I_0)$. Then, by definition of the action, since $(-1)_v \in M$, we have $y_v M = y_v^{-1} M$ for all $v \in V \setminus V_0$. Since $y_v M$ cannot have order 2, we get $y_v \in M$ for all $v \in V \setminus V_0$. Arguing similarly, if $r = (v, v') \in R \setminus R_0$, then, without loss of generality, $v \notin V_0$ so, since $(-1)_v \in M$, we have $z_r M = z_r^{-1} M$ and since $z_r M$ cannot have order two, we again get $z_r \in M$. By the choice of M_* , this yields that $M_* \subseteq M$ so we have natural quotient map $\widetilde{pq(G_\Gamma \times C_2^I)}/M_* \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/M$ and the composition with the quotient map $\widetilde{pq(G_\Gamma \times C_2^I)}/M \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/N$ is a universal pq -Frattini cover, so $\widetilde{pq(G_\Gamma \times C_2^I)}/M_* \rightarrow \widetilde{pq(G_\Gamma \times C_2^I)}/M$ is also a universal pq -Frattini cover, which must therefore be an isomorphism. This shows $M = M_*$. $\square$

Corollary 2.24. *Suppose $\Gamma = (V, R)$ is a graph, I is a set, $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ is an induced subgraph, and $I_0 \subseteq I$ is a finite set. Let $\pi : (G_\Gamma \times C_2^I) \rightarrow$*

$(G_\Gamma \times C_2^I)$ denote the universal Frattini cover. There is a unique closed normal subgroup $\widetilde{M} \trianglelefteq (G_\Gamma \times C_2^I)$ with $M \subseteq \ker(\pi_{\Gamma_0} \circ \pi)$ such that the induced map $\overline{\pi_{\Gamma_0} \circ \pi} : (G_\Gamma \times C_2^I)/\widetilde{M} \rightarrow (G_{\Gamma_0} \times C_2^{I_0})$ is the universal pq -Frattini cover.

Proof. Let S denote the 2-Sylow subgroup of $\ker(\pi)$. Then S is the unique closed normal subgroup of $G_\Gamma \times C_2^I$ contained in $\ker(\pi)$ with $G_\Gamma \times C_2^I/S \cong_{pq} (G_\Gamma \times C_2^I)$ and such that the induced map from $(G_\Gamma \times C_2^I)/S$ to $G_\Gamma \times C_2^I$ is the universal pq -Frattini cover. Let $\rho : (G_\Gamma \times C_2^I)/S \rightarrow (G_\Gamma \times C_2^I)$ be the map induced by π . Let $\pi_0 : G_\Gamma \times C_2^I \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ denote the projection (as in Lemma 2.23). Since $(G_\Gamma \times C_2^I)/S \cong_{pq} (G_\Gamma \times C_2^I)$, we may apply Lemma 2.23 to $(\pi_0 \circ \rho) : (G_\Gamma \times C_2^I)/S \rightarrow (G_{\Gamma_0} \times C_2^{I_0})$ and pulling back to $(G_\Gamma \times C_2^I)$, to obtain a unique closed normal $\overline{M} \trianglelefteq (G_\Gamma \times C_2^I)/S$ with $\overline{M} \subseteq \ker(\pi_0 \circ \rho)$ such that the induced map $((G_\Gamma \times C_2^I)/S)/\overline{M} \rightarrow (G_{\Gamma_0} \times C_2^{I_0})$ is the universal pq -Frattini cover. Such an $\overline{M}$ corresponds uniquely to a closed normal M with $S \subseteq M \trianglelefteq (G_\Gamma \times C_2^I)$ and with $M \subseteq \ker(\pi_0 \circ \pi)$, as desired. $\square$

Remark 2.25. Although Corollary 2.24 is stated for finite I_0 , it holds equally well when I_0 is an arbitrary (possibly infinite) subset of I . Indeed, write $I_0 = \varinjlim_F F$ as a directed union of its finite subsets $F \subseteq I_0$. For each finite F , Corollary 2.24 gives a unique M_F such that $(G_\Gamma \times C_2^I)/M_F \rightarrow G_{\Gamma_0} \times C_2^F$ is the universal pq -Frattini cover. By uniqueness, the M_F are compatible as F varies, and $M = \bigcap_F M_F$ is the unique closed normal subgroup of $(G_\Gamma \times C_2^I)$ such that $(G_\Gamma \times C_2^I)/M \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ is the universal pq -Frattini cover, using the fact that the universal pq -Frattini cover commutes with inverse limits of finite groups.

Lemma 2.26. *Let $\Gamma = (V, R)$ be a graph and I be a set disjoint from V , and fix a natural number $n \geq 1$. Let $N_1 \subseteq N_0$ be closed normal subgroups of $G_\Gamma \times C_2^I$ such that $(G_\Gamma \times C_2^I)/N_0 \cong G_{\Gamma_0} \times C_2^{I_0}$ and the quotient map $(G_\Gamma \times C_2^I)/N_1 \rightarrow (G_\Gamma \times C_2^I)/N_0$ is a universal pq -Frattini cover. Let $S_{pq} \trianglelefteq G_\Gamma \times C_2^I$ denote the (normal) product of the p - and q -Sylow subgroups of $G_\Gamma \times C_2^I$.*

Let M be an arbitrary closed normal subgroup of $(G_\Gamma \times C_2^I)$ with $M \subseteq N_1$. Set $A = (G_\Gamma \times C_2^I)/M$ and $B = (G_\Gamma \times C_2^I)/MS_{pq}$.

Identifying $F_{|V \cup I|}(2)$ with $(G_\Gamma \times C_2^I)/S_{pq}$ so that the maps $(G_\Gamma \times C_2^I) \rightarrow A$, $A \rightarrow B$, and $F_{|V \cup I|}(2) \rightarrow B$ are the quotient maps, we have the following:

- (1) $(G_\Gamma \times C_2^I) \cong A \times_B F_{|V \cup I|}(2)$.
- (2) Let P_n denote the group

$$(G_\Gamma \times C_2^I) \times_A (G_\Gamma \times C_2^I) \times_A \dots \times_A (G_\Gamma \times C_2^I)$$

which is the n fold fiber product of $(G_\Gamma \times C_2^I)$ over A and let H_n denote the group

$$F_{|V \cup I|}(2) \times_B F_{|V \cup I|}(2) \times_B \dots \times_B F_{|V \cup I|}(2)$$

the n -fold fiber product of $F_{|V \cup I|}(2)$ over B . Then we have an isomorphism

$$P_n \cong A \times_B H_n.$$

Proof. (1) Since $(\widetilde{G_\Gamma \times C_2^I})/N_1 \cong_{pq} (\widetilde{G_\Gamma \times C_2^I})$, the subgroup N_1 is necessarily a 2-group. Thus, since $M \subseteq N_1$ and the subgroup S_{pq} has only quotients whose order is divisible by p or q , we have $M \cap S_{pq} = 1$ so

$$\begin{aligned} (\widetilde{G_\Gamma \times C_2^I}) &= (\widetilde{G_\Gamma \times C_2^I})/(M \cap S_{pq}) \\ &\cong ((\widetilde{G_\Gamma \times C_2^I})/M) \times_{(\widetilde{G_\Gamma \times C_2^I})/MS_{pq}} ((\widetilde{G_\Gamma \times C_2^I})/S_{pq}) \\ &\cong A \times_B F_{|V \cup I|}(2), \end{aligned}$$

as desired.

(2) We will represent elements of P_n and H_n as n -tuples from $(\widetilde{G_\Gamma \times C_2^I})^n$ or $F_{|V \cup I|}(2)^n$ with the property that they have the same image in A or in B , respectively. By (1), there is an isomorphism $\psi : A \times_B F_{|V \cup I|}(2) \rightarrow (\widetilde{G_\Gamma \times C_2^I})$. Then we define a map $\bar{\psi} : A \times_B H_n \rightarrow P_n$ by

$$\bar{\psi}(g, \bar{h}) = (\psi(g, h_1), \psi(g, h_2), \dots, \psi(g, h_n)).$$

It is easy to see that this defines a map to P_n and, since ψ is an isomorphism, $\bar{\psi}$ is injective. To see that it is surjective, let $\bar{a} = (a_1, \dots, a_n) \in P_n$. It follows from (1) that we have a commutative diagram

$$\begin{array}{ccc} (\widetilde{G_\Gamma \times C_2^I}) & \longrightarrow & A \\ \downarrow & & \downarrow \\ F_{|V \cup I|}(2) & \longrightarrow & B, \end{array}$$

where all the maps are quotient maps (using the identification of $F_{|V \cup I|}(2)$ with $(\widetilde{G_\Gamma \times C_2^I})/S_{pq}$). Thus, if we let g be the (common) image of each a_i in A and let h_i be the image of a_i in $F_{|V \cup I|}(2)$ then g and h_i have the same image in B . It follows that, letting $\bar{h} = (h_1, \dots, h_n)$, we have $(g, \bar{h}) \in A \times_B H_n$ and $\bar{\psi}(g, \bar{h}) = \bar{a}$. $\square$

Remark 2.27. To see what the above lemma is saying, it can be helpful to consider the special case that $M = N_1$. Then we obtain

$$(1) \quad (\widetilde{G_\Gamma \times C_2^I}) \cong_{pq} (\widetilde{G_\Gamma \times C_2^I}) \times_{C_2^{V \cup I}} F_{|V \cup I|}(2).$$

(2) If given

$$P_n := (\widetilde{G_\Gamma \times C_2^I}) \times_{pq(\widetilde{G_\Gamma \times C_2^I})} (\widetilde{G_\Gamma \times C_2^I}) \times_{pq(\widetilde{G_\Gamma \times C_2^I})} \dots \times_{pq(\widetilde{G_\Gamma \times C_2^I})} (\widetilde{G_\Gamma \times C_2^I})$$

which is the n -fold fiber product of $(\widetilde{G_\Gamma \times C_2^I})$ over $_{pq}(\widetilde{G_\Gamma \times C_2^I})$ and

$$H_n := F_{|V \cup I|}(2) \times_{C_2^{V \cup I}} F_{|V \cup I|}(2) \times_{C_2^{V \cup I}} \dots \times_{C_2^{V \cup I}} F_{|V \cup I|}(2),$$

the n -fold fiber product of $F_{|V \cup I|}(2)$ over $C_2^{V \cup I}$, then we have an isomorphism

$$P_n \cong_{pq} (\widetilde{G_\Gamma \times C_2^I}) \times_{C_2^{V \cup I}} H_n.$$

Lemma 2.28. *Suppose $\Gamma = (V, R)$ is a graph and I is a set (disjoint from V). Let $I_0 \subseteq I$ be a subset and assume $|I| > |I_0| + |V| + \aleph_0$. Suppose we are given a profinite group A and epimorphisms $\tilde{\pi}_0 : (\widetilde{G_\Gamma \times C_2^I}) \rightarrow (\widetilde{G_\Gamma \times C_2^{I_0}})$, $\beta : (\widetilde{G_\Gamma \times C_2^{I_0}}) \rightarrow A$, and $\alpha : A \rightarrow {}_{pq}(G_\Gamma \times C_2^{I_0})$ such that $\tilde{\pi}_0$ is a lift of the projection $G_\Gamma \times C_2^I \rightarrow G_\Gamma \times C_2^{I_0}$ and $\alpha \circ \beta$ is a universal Frattini cover. Fix $n \geq 1$ and let P_n be the group defined by*

$$P_n := (\widetilde{G_\Gamma \times C_2^{I_0}}) \times_A (\widetilde{G_\Gamma \times C_2^{I_0}}) \times_A \dots \times_A (\widetilde{G_\Gamma \times C_2^{I_0}}),$$

the n -fold fiber product of $\widetilde{G_\Gamma \times C_2^{I_0}}$ over A , where the map from $\widetilde{G_\Gamma \times C_2^{I_0}}$ to A is given by β . Then there is an epimorphism $\gamma : (\widetilde{G_\Gamma \times C_2^I}) \rightarrow P_n$ such that the following diagram commutes:

$$\begin{array}{ccc} (\widetilde{G_\Gamma \times C_2^I}) & \xrightarrow{\gamma} & P_n \\ \downarrow \tilde{\pi}_0 & & \downarrow \bar{\beta} \\ (\widetilde{G_\Gamma \times C_2^{I_0}}) & \xrightarrow{\beta} & A. \end{array}$$

where $\bar{\beta} : P_n \rightarrow A$ is the map obtained by applying β to any one of the $(\widetilde{G_\Gamma \times C_2^{I_0}})$ factors.

Proof. As in the proof of Lemma 2.26, we let $S_{pq} \trianglelefteq (\widetilde{G_\Gamma \times C_2^I})$ denote the product of the p - and q -Sylow subgroups of $(\widetilde{G_\Gamma \times C_2^I})$. Let $D = (\widetilde{G_\Gamma \times C_2^I})/S_{pq}$ and let $C = (\widetilde{G_\Gamma \times C_2^{I_0}})/\tilde{\pi}_0(S_{pq})$. Set $\bar{\pi} : D \rightarrow C$ be the map induced by $\tilde{\pi}_0$. Note that we have $D \cong F_{|V \cup I|}(2)$ and $C \cong F_{|V \cup I_0|}(2)$.

Let $B = A/(\beta \circ \tilde{\pi}_0)(S_{pq})$ be the maximal pro-2 quotient of A and let $\delta : C \rightarrow B$ be the epimorphism induced by β . We define the group H_n by setting

$$H_n = C \times_B C \times_B \dots \times_B C,$$

the n -fold fiber product of C with itself over B , defined using the epimorphism δ . By Lemma 2.26, we may identify P_n with $A \times_B H_n$, where the map $A \rightarrow B$ is the quotient map.

Then by superprojectivity (using that D is a free pro-2 group of sufficiently high rank), there is some epimorphism $\chi : D \rightarrow H_n$ so that the following diagram commutes:

$$\begin{array}{ccc} & D & \\ \swarrow \chi & \downarrow & \\ H_n & \twoheadrightarrow & B \end{array}$$

where the horizontal map is obtained by applying δ to any of the C factors, and the vertical map is the composition $\beta \circ \tilde{\pi}_0$ followed by the quotient map $A \rightarrow B$. Let $\tilde{\chi} : (\widetilde{G_\Gamma \times C_2^I}) \rightarrow H_n$ be the map obtained by precomposing χ with the quotient map $(\widetilde{G_\Gamma \times C_2^I}) \rightarrow D$.

We define a map $\gamma : (\widetilde{G_\Gamma \times C_2^I}) \rightarrow A \times_B H_n$ by setting $\gamma(g) = ((\beta \circ \tilde{\pi}_0)(g), \tilde{\chi}(g))$. By the choice of χ , this defines a map to $A \times_B H_n$. Let $M = \ker(\beta \circ \tilde{\pi}_0)$ and let $M' =$

$\ker(\tilde{\chi})$. Since $B \cong \widetilde{(G_\Gamma \times C_2^I)}/MS_{pq}$, we have $MS_{pq} \supseteq M'$, and since $S_{pq} \subseteq M'$, and since $\widetilde{(G_\Gamma \times C_2^I)}/M' \cong H_n$ is a 2-group, it follows that $MM' = MS_{pq}$. And, by construction, $(\beta \circ \tilde{\pi}_0)$ and $\tilde{\chi}$ induce the same isomorphism $\widetilde{G_\Gamma \times C_2^I}/MS_{pq} \rightarrow B$. Thus, if $(a, c) \in A \times_B H_n$ and we choose g, h such that $(\beta \circ \tilde{\pi}_0)(g) = a$ and $\tilde{\chi}(h) = c$, then we have $gMM' = hMM'$, so there are $m \in M$ and $m' \in M'$ such that $mm' = g^{-1}h$, so if we set $k = gm = h(m')^{-1}$, then we have $\gamma(k) = ((\beta \circ \tilde{\pi}_0)(g), \tilde{\chi}(h)) = (a, c)$. This shows γ is an epimorphism, as desired. $\square$

Finally, we will record a fact about the way normal subgroups intersect in the maximal pro-2 quotient of $\widetilde{G_\Gamma \times C_2^I}$, which will be useful in later applications. First, we need the following fact, which seems to be folklore:

Fact 2.29. Suppose G is a profinite group with closed normal ℓ -Sylow subgroup S_ℓ . Then for all closed normal $A, B \trianglelefteq G$,

$$S_\ell A \cap S_\ell B = S_\ell(A \cap B).$$

Proof. Replacing G with the quotient $G/(A \cap B)$, it is enough to show that if $A, B \trianglelefteq G$ are closed normal subgroups with $A \cap B = 1$, then $S_\ell A \cap S_\ell B = S_\ell$. Since A and B are both normal, we must have $[A, B] \leq A \cap B = 1$ so $AB \cong A \times B$. Since S_ℓ is the unique pro- ℓ Sylow subgroup of G , every pro- ℓ subgroup of G (and thus of A, B , and AB) must lie in S_ℓ . This implies

$$S_\ell \cap AB = (S_\ell \cap A)(S_\ell \cap B),$$

since, in the direct product $A \times B$, the pro- ℓ Sylow is the product of the ℓ -Sylows of A and of B .

Fix $x \in S_\ell A \cap S_\ell B$. We can write $x = s_1 a = s_2 b$ for $s_1, s_2 \in S_\ell$, $a \in A$, $b \in B$. Then we have

$$ab^{-1} = s_1^{-1} s_2 \in S_\ell \cap AB = (S_\ell \cap A)(S_\ell \cap B).$$

Write $ab^{-1} = uv$ with $u \in (S_\ell \cap A)$ and $v \in (S_\ell \cap B)$. Since A and B commute and intersect in the trivial group, we must have $a = u \in S_\ell$ which entails that $x = s_1 a \in S_\ell$. This shows $S_\ell A \cap S_\ell B = S_\ell$ as desired. $\square$

Corollary 2.30. Suppose Γ is a graph and I is a set, and let S_{pq} denote the product of the p - and q -Sylow subgroups of $\widetilde{G_\Gamma \times C_2^I}$. Then for any closed normal subgroups $M, N \trianglelefteq \widetilde{G_\Gamma \times C_2^I}$, we have

$$S_{pq}M \cap S_{pq}N = S_{pq}(M \cap N).$$

Proof. Let $G = \widetilde{G_\Gamma \times C_2^I}$. We know that, in G , if we let S_p and S_q denote the p - and q -Sylow subgroups of G , respectively, we have that S_p and S_q are normal and $S_{pq} = S_p S_q$. Then we may apply Fact 2.29 twice to obtain

$$S_{pq}(M \cap N) = S_p(S_q(M \cap N)) = S_p(S_q M \cap S_q N) = S_p S_q M \cap S_p S_q N,$$

which gives the desired equality. $\square$

3. AN IDENTIFICATION THEOREM

In this section, we are interested in describing arbitrary models S of the theory $\text{Th}(S(\widetilde{G_\Gamma \times C_2^\mu}))$, for a given infinite graph Γ and a cardinal μ . In fact, we give a description of a broader class of structures which we call the *graph closed* complete systems, showing that they are closely related to groups produced by the CDM graph coding. Our ultimate aim is to prove that every model of this theory is isomorphic to one of the form $S(G_{\Gamma'} \times C_2^{\mu'})$ for some $\Gamma' \equiv \Gamma$ and some cardinal μ' . Given an arbitrary complete system S , we introduce two distinguished subsystems $S_- \subseteq S_+ \subseteq S$ and build up to the main identification theorem by proving identification lemmas for these subsystems. Each identification lemma has two parts. First, we show that the subsystem is more or less isomorphic to an inverse system of a group produced by CDM graph coding. Then, we show that given $S \preceq S' \models \text{Th}(S(\widetilde{G_\Gamma \times C_2^\mu}))$, a representation for one of the distinguished subsystems of S lifts to a compatible representation for the corresponding subsystem of S' .

3.1. Vertex width and graph closure. First, we fix notation for the theories that we are studying:

Definition 3.1. Suppose Γ is an infinite graph and μ is a cardinal (possibly finite, including 0).

- (1) We define $T_{\Gamma, \mu}$ to be the L_{IS} -theory of $S(\widetilde{G_\Gamma \times C_2^\mu})$.
- (2) Given $S \models T_{\text{IS}}$, we define S_- to be the subsystem of S generated by the following:

$$\{\alpha \in S : [\alpha] \cong D_p\} \cup \{\alpha \in S : [\alpha] \cong W\}.$$

- (3) Given $S \models T_{\text{IS}}$, we define S_+ to be the subsystem of S generated by

$$\{\alpha \in S : [\alpha] \cong C_2\} \cup \{\alpha \in S : [\alpha] \cong D_p\} \cup \{\alpha \in S : [\alpha] \cong W\},$$

or, equivalently, the subsystem of S generated by S_- and $\{\alpha \in S : [\alpha] \cong C_2\}$.

Definition 3.2. Suppose Γ is an infinite graph, μ is a cardinal, and $\alpha \in S \models T_{\Gamma, \mu}$ is an element with $[\alpha] \cong C_2$. We say α has *vertex width* n if n is the least integer ≥ 1 such that there are (pairwise $\sim$ -inequivalent) $\beta_0, \dots, \beta_{n-1} \in S$ with $[\beta_i] \cong D_p$ and $\alpha \geq \gamma$, where $\gamma = \bigwedge_{i < n} \beta_i$. We say that the β_i (or the classes $[\beta_i]$) *witness* that α has vertex width n . If there are no such β_i , we say that α has vertex width ∞ .

Note that, in the above definition, the β_i correspond to a size n set of vertices V_0 , and $\bigwedge_{i < n} \beta_i$ corresponds to the $D_p^{V_0}$ quotient of $\widetilde{G_\Gamma}$. Saying that α with $[\alpha] \cong C_2$ has vertex width n , then, means that α corresponds to a C_2 quotient of $D_p^{V_0}$ for a set V_0 of vertices with $|V_0| = n$ which is not a quotient of any $D_p^{V'}$ for any V' of size strictly smaller than n .

Here is the suggested way to think about vertex width: the group $\widetilde{G_\Gamma \times C_2^I}$ has as a quotient the group $C_2^V \times C_2^I$ which is an elementary abelian 2-group and thus an $\mathbb{F}_2$ -vector space. We can write it in additive notation as $\mathbb{F}_2^V \times \mathbb{F}_2^I$. The open normal subgroups of index 2 are in one-to-one correspondence with continuous linear functionals $(\mathbb{F}_2^V \times \mathbb{F}_2^I) \rightarrow \mathbb{F}_2$, thus are elements of the continuous dual of $\mathbb{F}_2^V \times \mathbb{F}_2^I$, which is a direct sum of the continuous duals of $\mathbb{F}_2^V$ and $\mathbb{F}_2^I$. In the

context of the structure in which we are working, however, these two factors behave differently. The continuous dual of $\mathbb{F}_2^V$ comes with a distinguished basis, namely the coordinate projections for each vertex $v \in V$. We will see in Lemma 3.4 below that the elements of this distinguished basis are definable in $S(\widetilde{G_\Gamma \times C_2^I})$. The kernels of these maps correspond exactly to $\sim$ -classes of elements α with $[\alpha] \cong C_2$ and which have vertex width 1. Elements of vertex width n , then, correspond precisely to elements of the continuous dual of $\mathbb{F}_2^V$ which can be expressed as a linear combination of n elements from this distinguished basis, but no fewer. Elements in the continuous dual of $\mathbb{F}_2^V \times \mathbb{F}_2^I$ which are not contained in the subspace $(\mathbb{F}_2^V)^*$, then, are the elements of infinite vertex width. This dictionary is spelled out in the following lemma:

Lemma 3.3. *Let $\Gamma = (V, R)$ be a graph and I a set, disjoint from V . Let $Q = \{[\alpha] : \alpha \in S(G_\Gamma \times C_2^I), [\alpha] \cong C_2\}$. Then there is a bijection $d : Q \rightarrow ((\mathbb{F}_2^V \times \mathbb{F}_2^I)^* - \{0\})$ satisfying the following:*

- (1) *For $[\alpha] \in Q$, α has vertex width n if and only if $d([\alpha])$ is a linear combination of n elements of $(\mathbb{F}_2^V)^*$ of the form π_v , where π_v denotes the coordinate projection to the v coordinate (and n is minimal with this property).*
- (2) *If $(\gamma_i)_{i < k}$ is a sequence of elements of $S(\widetilde{G_\Gamma \times C_2^I})$ with $[\gamma_i] \cong C_2$ for all i , then we have*

$$\gamma_i \not\geq \bigwedge_{j < i} \gamma_j$$

for all $i < k$ if and only if $\{d([\gamma_i]) : i < k\}$ are linearly independent in $(\mathbb{F}_2^V \times \mathbb{F}_2^I)^$.*

Proof. Let $\pi : \widetilde{G_\Gamma \times C_2^I} \rightarrow G_\Gamma \times C_2^I$ denote the universal Frattini cover, dual to the inclusion $S_+ \subseteq S = S(G_\Gamma \times C_2^I)$. We know that, since $\Phi(C_2) = 1$, every quotient of $\widetilde{G_\Gamma \times C_2^I}$ is a quotient of $G_\Gamma \times C_2^I$, that is, if an open normal subgroup of $\widetilde{G_\Gamma \times C_2^I}$ has index 2, then it contains $\ker(\pi)$. The quotient of $G_\Gamma \times C_2^I$ by the normal subgroup $C_q^R \times C_p^V$ is naturally isomorphic to $C_2^V \times C_2^I$, in a way that respects the coordinate projections.

Since C_2 has order 2, any open normal subgroup of $\widetilde{G_\Gamma \times C_2^I}$ of index 2 must contain $C_q^R \times C_p^V$. Thus, to establish the existence of the bijection d , it suffices to define a bijection from $\{[\alpha] : \alpha \in S(C_2^V \times C_2^I), [\alpha] \cong C_2\}$ to the continuous dual $(\mathbb{F}_2^V \times \mathbb{F}_2^I)^*$. We will switch to additive notation and write $C_2^V \times C_2^I$ as $\mathbb{F}_2^V \times \mathbb{F}_2^I$. Then if $\alpha \in S(\mathbb{F}_2^V \times \mathbb{F}_2^I)$ and $[\alpha] \cong C_2$, then $[\alpha] = [U]$ for some open subgroup $U \leq \mathbb{F}_2^V \times \mathbb{F}_2^I$ of index 2. There is a unique non-zero continuous linear functional $f_U : \mathbb{F}_2^V \times \mathbb{F}_2^I \rightarrow \mathbb{F}_2$ with kernel U . Conversely, if $f \in (\mathbb{F}_2^V \times \mathbb{F}_2^I)^*$ is non-zero, then $\ker(f)$ is an index 2 subgroup of $(\mathbb{F}_2^V \times \mathbb{F}_2^I)$ so $[\ker(f)] \in Q$. It follows that the map $[U] \mapsto f_U$ is the desired bijection.

Now we prove (1). Recall from Lemma 2.5(1) that each D_p quotient of $G_\Gamma \times C_2^I$ is obtained by quotienting by the kernel of a coordinate projection to some coordinate $v \in V$. It follows that $\alpha \in S(\mathbb{F}_2^V \times \mathbb{F}_2^I) \subseteq S(G_\Gamma \times C_2^I)$ has vertex width 1 if and only if $[\alpha] = [\ker(\pi_v)]$ for the coordinate projection $\pi_v : \mathbb{F}_2^V \times \mathbb{F}_2^I \rightarrow \mathbb{F}_2$ for some $v \in V$. More generally, if $\alpha \in S(G_\Gamma \times C_2^I)$ has vertex width n and $\{\gamma_i : i < n\}$ is a minimal set of witnesses with $[\gamma_i] \cong D_p$ for all $i < n$, then for each $i < n$, we can choose some $\delta_i \geq \gamma_i$ with $[\delta_i] \cong C_2$. Then since $[\alpha]$ is a 2-group and $\alpha \geq \bigwedge_{i < n} \gamma_i$, we must

have $\alpha \geq \bigwedge_{i < n} \delta_i$. From this it follows that an arbitrary α has vertex width n if and only if n is minimal such that there is a set $\{\delta_i : i < n\}$ where $[\delta_i] \cong C_2$ and δ_i has vertex width 1 for each i and where $\alpha \geq \bigwedge_{i < n} \delta_i$. Then the conclusion follows from (2).

So now we show (2). Fix a sequence $(\gamma_i)_{i < k}$ in $S(\widetilde{G_\Gamma \times C_2^I})$ with $[\gamma_i] \cong C_2$ and let $f_i = d([\gamma_i])$ for all $i < k$. We know from the above that, necessarily, $\gamma_i \in S(\mathbb{F}_2^V \times \mathbb{F}_2^I)$ (viewed as a subsystem of $S(\widetilde{G_\Gamma \times C_2^I})$) so $[\gamma_i] = [U_i]$ for some index 2 open subgroup $U_i \trianglelefteq (\mathbb{F}_2^V \times \mathbb{F}_2^I)$ and f_i is the unique continuous linear functional with kernel U_i .

Suppose first that for some $i < k$, $f_i = \sum_{j < i} c_j f_j$. Then set $X = \{j < i : c_j = 1\}$, so $f_i = \sum_{j \in X} f_j$. This clearly gives $U_i = \ker(f_i) \supseteq \bigcap_{j \in X} U_j$ so we get

$$\gamma_i \geq \bigwedge_{j \in X} \gamma_j \geq \bigwedge_{j < i} \gamma_j.$$

For the other direction, suppose now that $\gamma_i \geq \bigwedge_{j < i} \gamma_j$. Then choose $Y \subseteq \{0, \dots, i-1\}$ minimal such that $\gamma_i \geq \bigwedge_{j \in Y} \gamma_j$. We claim that $f_i = \sum_{j \in Y} f_j$. By the minimality of Y , for each $\ell \in Y$, we have $U_i \not\supseteq \bigcap_{j \in Y - \{\ell\}} U_j$, so we can pick some $v_\ell \in \bigcap_{j \in Y - \{\ell\}} U_j$ with $f_i(v_\ell) = 1$. It follows, also, that $f_\ell(v_\ell) = 1$ since otherwise $v_\ell \in \bigcap_{j \in Y} U_j \subseteq U_i$. The images of the vectors $(v_\ell)_{\ell \in Y}$ form a basis for $(\mathbb{F}_2^V \times \mathbb{F}_2^I) / \bigcap_{\ell \in Y} U_\ell$, so we obtain $f_i = \sum_{j \in Y} f_j$, as desired. $\square$

Lemma 3.4. *For all n , there is some L_{IS} -formula $\varphi_n(x)$ that defines, for any Γ, μ and $S \models T_{\Gamma, \mu}$, the α with $[\alpha] \cong C_2$ of vertex width n .*

Proof. For each n , there is a formula $\psi_n(x)$ that asserts that x has vertex width at most n , defined as follows:

$$x \in X_2 \wedge x \notin X_1 \wedge (\exists v_0, \dots, v_{n-1} \in X_{2p}) \left(\bigwedge_{i < j < n} v_i \not\sim v_j \wedge \bigwedge_{i < n} [v_i] \cong D_p \wedge \left(x \geq \bigwedge_{i < n} v_i \right) \right),$$

where $\bigwedge_{i < n} v_i$ in the rightmost conjunct denotes the meet of the n elements $v_0, \dots, v_{n-1}$. Then the desired formula $\varphi_n(x)$ can be defined by $\psi_1(x)$ when $n = 1$ and by $\psi_n(x) \wedge \neg \psi_{n-1}(x)$ for $n > 1$. $\square$

Lemma 3.5. *Fix a graph $\Gamma = (V, R)$ and a cardinal μ . Suppose $\alpha \in S \models T_{\Gamma, \mu}$, $[\alpha]_\sim \cong C_2$, and α has vertex width n . Suppose further that $\beta_i, \beta'_i \in S$ satisfy $[\beta_i] \cong [\beta'_i] \cong D_p$ for all $i < n$. Then if $\gamma = \bigwedge_{i < n} \beta_i$, $\gamma' = \bigwedge_{i < n} \beta'_i$, and $\alpha \geq \gamma$ and $\alpha \geq \gamma'$, then $\{[\beta_i]_\sim : i < n\} = \{[\beta'_i]_\sim : i < n\}$. In other words, if α has vertex width n , then the set of witnesses is well-defined.*

Proof. As the statement is elementary (for a fixed n), it suffices to prove this when $S = S(\widetilde{G_\Gamma \times C_2^I})$ for some I with $|I| = \mu$. This follows from Lemma 3.3(1) since having vertex width n is equivalent to corresponding to a linear combination of n elements which themselves correspond to elements of vertex width 1, under the bijection d defined there. As the elements of vertex width 1 are linearly independent, they are completely determined. We can also give a direct argument: by Corollary 2.6, each β_i and β'_i is an element of the subsystem $S(G_\Gamma) \subseteq S(\widetilde{G_\Gamma \times C_2^\mu})$. We know that, after possibly replacing each β_i with something in the same $\sim$ -class, we may assume $\beta_i = \ker(\pi_{v_i})$ and, likewise, $\beta'_i = \ker(\pi_{v'_i})$ for sets of vertices $U = \{v_i :$

$i < n\}$ and $U' = \{v'_i : i < n\}$ from Γ . Then we have $\bigwedge_{i < n} \beta_i = \bigcap_{v \in U} \ker(\pi_v)$ and $\bigwedge_{i < n} \beta'_i = \bigcap_{v \in U'} \ker(\pi_v)$. It follows, then, that

$$\gamma \vee \gamma' = \left(\bigcap_{v \in U} \ker(\pi_v) \right) \left(\bigcap_{v \in U'} \ker(\pi_v) \right) = \bigcap_{v \in (U \cap U')} \ker(\pi_v).$$

As $\alpha \geq \gamma$ and $\alpha \geq \gamma'$, we must, therefore, have $\alpha \geq \bigcap_{v \in (U \cap U')} \ker(\pi_v)$. By the minimality of n , we must have $U = U \cap U' = U'$. This shows $\{[\beta_i] : i < n\} = \{[\beta'_i] : i < n\}$. $\square$

Recall that if $\Gamma = (V, R)$ is a graph, then G_Γ can be described as a semi-direct product $C_q^R \rtimes D_p^V$ with the action described in Section 2. Then since $D_p = C_p \rtimes C_2$ and the action of $D_p \times D_p$ factors through the quotient $C_2 \times C_2$ in the definition of W , G_Γ may be further described as

$$(C_q^R \times C_p^V) \rtimes C_2^V,$$

where C_2^V acts on C_q^R in the induced action and C_2^V acts on C_p^V coordinate-wise. This gives a short exact sequence

$$1 \rightarrow P \rightarrow G_\Gamma \rightarrow C_2^V \rightarrow 1$$

where $P = C_q^R \times C_p^V$ is the product of the p - and q -Sylow subgroups of G_Γ . We will let $\xi_\Gamma : G_\Gamma \rightarrow C_2^V$ denote the quotient map (which, in coordinates, is just the projection map $((C_q^R \times C_p^V) \rtimes C_2^V) \rightarrow C_2^V$).

Lemma 3.6. *Suppose Γ is an infinite graph and $\alpha \in S(G_\Gamma)$. Then α has vertex width n if and only if there exists some induced subgraph $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ with $|V_0| = n$ and $[\alpha] = [N]$ for some open normal subgroup $N \trianglelefteq G_{\Gamma_0}$ with $N \supseteq P$, where P denotes the product of the p - and q -Sylow subgroups of G_{Γ_0} , and such that $\xi_{\Gamma_0}(N) \trianglelefteq C_2^{V_0}$ is the normal subgroup consisting of those $(g_v)_{v \in V_0} \in C_2^{V_0}$ such that the number of v with $g_v = -1$ is even.*

Proof. First, suppose α has vertex width n , and let $\beta_0, \dots, \beta_{n-1}$ list the witnesses, so we have $[\beta_i] \cong D_p$ and $[\beta_i] \neq [\beta_j]$ for all $i \neq j$. By Lemma 2.5, we have $[\beta_i] = [\ker(\pi_{v_i})]$ for some $v_i \in V$, for all $i < n$. Let $V_0 = \{v_i : i < n\}$ and let $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ be the induced subgraph with vertices V_0 . Then since $\alpha \geq \bigwedge_{i < n} \beta_i$, we may identify $[\alpha] = [N]$ for some normal subgroup $N \trianglelefteq G_{\Gamma_0}$. Note that we must have $N \supseteq P$, the product of the p - and q -Sylows, since $||[\alpha]| = 2$. So, setting $\bar{N} = \xi_{\Gamma_0}(N)$, we see ξ_{Γ_0} induces an isomorphism $G_{\Gamma_0}/N \rightarrow C_2^{V_0}/\bar{N}$. Moreover, we see that $(-1)_v \notin \bar{N}$ for all $v \in V_0$ since, if $(-1)_v \in \bar{N}$, then ξ_{Γ_0} would factor through the projection $C_2^{V_0} \rightarrow C_2^{V_0 - \{v\}}$, making $[\alpha]$ a quotient of $G_{\Gamma_0 - \{v\}}$, contradicting minimality of V_0 . Then, since $C_2^{V_0}/\bar{N} \cong C_2$, we have that, for all $v \neq v'$, the cosets $(-1)_v \bar{N}$ and $(-1)_{v'} \bar{N}$ are equal and their product is $\bar{N}$. It follows that $\bar{N}$ consists precisely of those $(g_v)_{v \in V_0} \in C_2^{V_0}$ such that the number of v with $g_v = -1$ is even.

For the other direction, assume we are given an induced subgraph $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ with n vertices such that $[\alpha] = [N] \cong C_2$ and $\xi_{\Gamma_0}(N) := \bar{N} \trianglelefteq C_2^{V_0}$ is the normal subgroup consisting of the $(g_v)_{v \in V_0}$ with the number of v with $g_v = -1$ even. Then, for any proper induced subgraph $\Gamma' \subsetneq \Gamma_0$, there is some $v \in V_0$ which is not in Γ' so $(-1)_v \in \xi_{\Gamma_0}(N \ker(\pi_{\Gamma_0 - \{v\}}))$ so $\xi_{\Gamma_0}(N \ker(\pi_{\Gamma_0 - \{v\}})) = C_2^{V_0}$. This shows the

minimality of Γ_0 , so $\{\ker(\pi_v) : v \in V_0\}$ is the set of witnesses establishing that α has vertex width n . $\square$

Definition 3.7. Fix a graph Γ and set I . Suppose $A \subseteq S(\widetilde{G_\Gamma \times C_2^I})$ is a small subset. We define the *graph closure* of A , denoted $\text{gcl}(A)$, to be the smallest subsystem containing A satisfying the following conditions:

- (1) If $\alpha \in \text{gcl}(A)$ satisfies $[\alpha]_\sim \cong C_2$ and α has finite vertex width, then the set of witnesses is contained in $\text{gcl}(A)$ as well.
- (2) If $[\beta_0]_\sim$ and $[\beta_1]_\sim$ are distinct $\sim$ -classes with $\beta_i \in \text{gcl}(A)$ and $[\beta_i]_\sim \cong D_p$ for $i = 0, 1$, then if $\gamma \leq \beta_0$, $\gamma \leq \beta_1$ and $[\gamma]_\sim \cong W$, then $\gamma \in \text{gcl}(A)$.

If $A = \text{gcl}(A)$, we say that A is *graph closed*.

Lemma 3.8. Fix a graph Γ and suppose $\delta \in S(G_\Gamma)$ with $1 \neq \delta$. Then we have

$$\delta \in \text{gcl}(\{\alpha : \alpha \geq \delta \text{ and } [\alpha] \cong C_2\}).$$

Proof. Fix $1 \neq \delta \in S(G_\Gamma)$ (where 1 denotes the trivial quotient class) and let $D = \text{gcl}(\{\alpha : \alpha \geq \delta \text{ and } [\alpha] \cong C_2\})$. Since $S(G_\Gamma)$ is the union of $S(G_{\Gamma_0})$ as Γ_0 ranges over finite induced subgraphs of Γ , we can choose some minimal finite $\Gamma_0 = (V_0, R_0)$ with $\delta \in S(G_{\Gamma_0})$. Then we have $[\delta] = G_{\Gamma_0}/N$ for some proper normal subgroup $N \trianglelefteq G_{\Gamma_0}$. By Lemma 2.9, if $P \trianglelefteq G_{\Gamma_0}$ denotes the product of the p - and q -Sylow subgroups of G_{Γ_0} , then G_{Γ_0}/NP is also a nontrivial quotient of G_{Γ_0} . Note that if for some $v \in V_0$, $(\beta)_v \in NP$ (where, recall, $(\beta)_v$ denotes the involution of G_{Γ_0} which is the involution in the v coordinate and 1 in all other vertex coordinates), then $\ker(\pi_{\Gamma_0 - \{v\}}) \subseteq NP$ so the natural projection

$$G_{\Gamma_0}/NP \rightarrow G_{\Gamma_0}/NP \ker(\pi_{\Gamma_0 - \{v\}})$$

is an isomorphism, which entails $\delta \in S(G_{\Gamma_0 - \{v\}})$, contradicting the minimality of Γ_0 .

Write $G_{\Gamma_0}/NP \cong C_2^{V_0}/\overline{N}$ via the isomorphism induced by the quotient map $\xi_{\Gamma_0} : G_{\Gamma_0} \rightarrow C_2^{V_0}$ defined before Lemma 3.6, where we are writing $\overline{N}$ for $\xi_{\Gamma_0}(NP)$. Then for each $v \in V_0$, we know $(-1)_v \notin \overline{N}$, so we may choose some normal $N_v \trianglelefteq C_2^{V_0}$ with $\overline{N} \subseteq N_v$, $[C_2^{V_0} : N_v] = 2$, and $(-1)_v \notin N_v$. Then we set $X_v = \{w \in V_0 : (-1)_w \notin N_v\}$. Then the natural projection map $C_2^{V_0} \rightarrow C_2^{X_v}$ induces an isomorphism $C_2^{V_0}/N_v \rightarrow C_2^{X_v}/H$, where H is the normal subgroup consisting of those $(g_w)_{w \in X_v}$ such that the number of w with $g_w = -1$ is even. Then $C_2^{V_0}/N_v$ corresponds to some α with $[\alpha] \cong C_2$ and $\alpha \geq \delta$. Moreover, by construction, α has vertex width $|X_v|$ with witnesses $\{\ker(\pi_w) : w \in X_v\}$, which contains $\ker(\pi_v)$. This shows $\ker(\pi_v) \in D$ for all $v \in V_0$. By definition of graph closure, any $\gamma \in S(G_\Gamma)$ with $[\gamma] \cong W$ corresponding to an edge between $v, v' \in V_0$, then, must also be in D . We therefore obtain $\delta \in S(G_{\Gamma_0}) \subseteq D$, as desired. $\square$

Suppose $\gamma_0, \dots, \gamma_{k-1} \in S \models T_{\Gamma, \mu}$ satisfy $[\gamma_i] \cong C_2$ for all $i < k$. Say $\gamma_0, \dots, \gamma_{k-1}$ are *independent* if $\gamma_i \not\geq \bigwedge_{j < i} \gamma_j$ for all $i < k$ (by Lemma 3.3(2), this is equivalent to $\{d([\gamma_i])\}_i$ being linearly independent, and is therefore independent of the ordering).

Lemma 3.9. Suppose $\gamma_0, \dots, \gamma_{k-1}, \alpha \in S \models T_{\Gamma, \mu}$ with $[\alpha] \cong [\gamma_i] \cong C_2$ for all $i < k$. If $\gamma_0, \dots, \gamma_{k-1}$ are independent and $\alpha \geq \bigwedge_{i < k} \gamma_i$, then we have

$$\gamma_i \geq \alpha \wedge \bigwedge_{j \neq i} \gamma_j$$

for all i .

Proof. In $\widetilde{S(G_\Gamma \times C_2^\mu)}$, this follows from Lemma 3.3(2), by the usual Steinitz exchange for vector spaces. For S , it follows by elementarity. $\square$

Lemma 3.10. *Fix a graph Γ and cardinal μ and suppose $A \subseteq S \models T_{\Gamma, \mu}$ is a small subset.*

- (1) $\text{gcl}(\text{gcl}(A)) = \text{gcl}(A)$.
- (2) $\Gamma(\text{gcl}(A))$ is an induced subgraph of Γ .
- (3) $\text{gcl}(A) \subseteq \text{acl}(A)$.

Proof. (1) It is immediate from the definitions that $\text{gcl}(A)$ is itself graph-closed.

(2) follows directly from the second graph closure condition: if $\beta_0, \beta_1 \in \text{gcl}(A)$ with $[\beta_i] \cong D_p$ and $[\beta_0] \neq [\beta_1]$, then any $\gamma \in S$ with $[\gamma] \cong W$ and $\gamma \leq \beta_0 \wedge \beta_1$ lies in $\text{gcl}(A)$. This is precisely the condition that $\Gamma(\text{gcl}(A))$ is an induced subgraph of Γ .

(3) We build $\text{gcl}(A)$ in ω stages. Set $A_0 = \langle A \rangle$ and, for each $n \in \omega$, define A_{n+1} to be the subsystem generated by A_n together with:

- all $\beta \in S$ with $[\beta] \cong D_p$ that witness the finite vertex width of some $\alpha \in A_n$ with $[\alpha] \cong C_2$ of finite vertex width; and
- all $\gamma \in S$ with $[\gamma] \cong W$ and $\gamma \leq \beta_0 \wedge \beta_1$ for some $\beta_0, \beta_1 \in A_n$ with $[\beta_i] \cong D_p$ and $[\beta_0] \sim \neq [\beta_1] \sim$.

Since any C_2 -element of $\bigcup_n A_n$ appears at some finite stage, $\text{gcl}(A) = \bigcup_{n < \omega} A_n$. We show $A_n \subseteq \text{acl}(A)$ for all $n \in \omega$ by induction; the base case is clear.

Assume $A_n \subseteq \text{acl}(A)$. For witnesses, suppose $\alpha \in A_n$ has $[\alpha] \cong C_2$ and vertex width k , and let β be a witness for α . By Lemma 3.5, the set of witness $\sim$ -classes $\{[\beta_i] : i < k\}$ is uniquely determined by α . It is easy to see there is a formula with parameter α defining the union of these k witness $\sim$ -classes, which is a finite set of $k \cdot |D_p|$ elements. In particular, β lies in this finite α -definable set, so $\beta \in \text{acl}(\alpha) \subseteq \text{acl}(A_n) \subseteq \text{acl}(A)$.

For W -elements, suppose $\beta_0, \beta_1 \in A_n$ have $[\beta_i] \cong D_p$ and $[\beta_0] \neq [\beta_1]$, and $\gamma \leq \beta_0 \wedge \beta_1$ with $[\gamma] \cong W$. By the definition of W , there is at most one $\sim$ -class $[\gamma'] \cong W$ with any representative γ' satisfying $\gamma' \leq \beta_0 \wedge \beta_1$. Hence γ lies in a $\{\beta_0, \beta_1\}$ -definable set of size at most $|W|$, so $\gamma \in \text{acl}(\beta_0, \beta_1) \subseteq \text{acl}(A_n) \subseteq \text{acl}(A)$. $\square$

3.2. First identification. Our first step will be to identify S_- for an arbitrary graph closed $S \subseteq S' \models T_{\Gamma, \mu}$:

Lemma 3.11. *Suppose Γ is an infinite graph and μ is a cardinal.*

- (1) *Suppose $S_* \models T_{\Gamma, \mu}$. Then there is an isomorphism $\Psi : (S_*)_- \rightarrow S(G_{\Gamma_*})$ where $\Gamma_* = (V_*, R_*) \models \text{Th}(\Gamma)$ is the graph interpreted in S_* , so*

$$\begin{aligned} V_* &= \{[\alpha] : \alpha \in S_*, [\alpha] \cong D_p\} \\ R_* &= \{([\alpha], [\alpha']) \in (V_*)^2 : [\alpha] \neq [\alpha'], (\exists \beta \in S_*)([\beta] \cong W \text{ and } \beta \leq \alpha \wedge \alpha')\}. \end{aligned}$$

The isomorphism can be chosen so that if $v = [\alpha] \in V_$, then $\Psi([\alpha]) = [\ker(\pi_v)]$ and if $r = ([\alpha], [\alpha']) \in R_*$ and $\beta \leq \alpha \wedge \alpha'$ satisfies $[\beta] \cong W$, then $\Psi([\beta]) = [\ker(\pi_r)]$.*

- (2) If $S = \text{gcl}(S) \subseteq S_* \models T_{\Gamma, \mu}$ and $\Psi : (S_*)_- \rightarrow S(G_{\Gamma_*})$ is an isomorphism as in (1), then $\Psi|_{S_-}$ yields an isomorphism $S_- \rightarrow S(G_{\Gamma(S)})$, the diagram

$$\begin{array}{ccc} S_- & \xrightarrow{\Psi|_{S_-}} & S(G_{\Gamma(S)}) \\ \downarrow & & \downarrow S(\pi_{\Gamma(S)}) \\ (S_*)_- & \xrightarrow{\Psi} & S(G_{\Gamma_*}). \end{array}$$

commutes, where the map $S_- \rightarrow (S_*)_-$ is the inclusion and $S(\pi_{\Gamma(S)})$ is dual to the projection $\pi_{\Gamma(S)} : G_{\Gamma_*} \rightarrow G_{\Gamma(S)}$.

- (3) If $S = \text{gcl}(S) \subseteq S_* \models T_{\Gamma, \mu}$ and $\Psi : S_- \rightarrow S(G_{\Gamma(S)})$ is an isomorphism as in (1), then Ψ extends to an isomorphism $\Psi_* : (S_*)_- \rightarrow S(G_{\Gamma_*})$ such that the diagram

$$\begin{array}{ccc} S_- & \xrightarrow{\Psi} & S(G_{\Gamma(S)}) \\ \downarrow & & \downarrow S(\pi_{\Gamma(S)}) \\ (S_*)_- & \xrightarrow{\Psi_*} & S(G_{\Gamma_*}). \end{array}$$

commutes, where the map $S_- \rightarrow (S_*)_-$ is the inclusion and $S(\pi_{\Gamma(S)})$ is dual to the projection $\pi_{\Gamma(S)} : G_{\Gamma_*} \rightarrow G_{\Gamma(S)}$.

Proof. (1) First, suppose $\Gamma_0 = (V_0, R_0)$ is a finite labeled graph with $V_0 = \{v_0, \dots, v_{n-1}\}$, then $S(G_{\Gamma} \times C_2^I)$ satisfies the first-order sentence φ_{Γ_0} which asserts that if $[\alpha_0], \dots, [\alpha_{n-1}] \cong D_p$ and, for $i < j$, there is some $\beta \leq \alpha_i \wedge \alpha_j$ with $[\beta] \cong W$ if and only if $(v_i, v_j) \in R_0$, then the subsystem generated by

$$\langle [\alpha_i] : i < n \rangle \cup \langle [\beta_{i,j}] : [\beta_{i,j}] \cong W, (v_i, v_j) \in R_0 \rangle$$

is isomorphic to $S(G_{\Gamma_0})$ via an isomorphism that maps $[\alpha_i] \mapsto [\ker(\pi_{v_i})]$ and, for $r = (v_i, v_j) \in R_0$, maps $[\beta_{i,j}] \mapsto [\ker(\pi_r)]$. Such a sentence exists since $S(G_{\Gamma_0})$ is a finite structure. Since $S_* \models T_{\Gamma, \mu}$, φ_{Γ_0} is true in S_* for each finite graph Γ_0 .

Now we construct the isomorphism Ψ , as in the statement. For each $v = [\alpha] \in V_*$, choose an isomorphism $\sigma_v : [\alpha] \rightarrow [\ker(\pi_v)]$, where π_v refers to the projection $\pi_v : G_{\Gamma_*} \rightarrow D_p$ to the v coordinate. For each $r = (v, v') = ([\alpha], [\alpha']) \in R_*$ and $\beta \leq \alpha \wedge \alpha'$ with $[\beta] \cong W$, choose some isomorphism $\sigma_r : [\beta] \rightarrow [\ker(\pi_r)]$ (where $\pi_r : G_{\Gamma_*} \rightarrow W$ is the projection to the r coordinate) such that the following diagram commutes:

$$\begin{array}{ccc} [\beta] & \xrightarrow{\sigma_r} & [\ker(\pi_r)] \\ \downarrow C & & \downarrow C \\ [\alpha] \times [\alpha'] & \xrightarrow{(\sigma_v, \sigma_{v'})} & [\ker(\pi_v)] \times [\ker(\pi_{v'})]. \end{array}$$

We will show that these choices determine the desired isomorphism $\Psi : (S_*)_- \rightarrow S(G_{\Gamma_*})$.

Let X be a finite subset of $\{\alpha \in S_* : [\alpha] \cong D_p\} \cup \{\alpha \in S_* : [\alpha] \cong W\}$ such that if $\alpha \not\leq \alpha' \in X$ and there is some $\beta \in S_*$ with $[\beta] \cong W$ and $\beta \leq \alpha \wedge \alpha'$, then $\beta \in X$. Let $\langle X \rangle$ be the subsystem of S_* generated by X and let $\gamma \in \langle X \rangle$ be a $\leq$ -minimal element (which exist because X is finite). Let $\{v_0, \dots, v_{n-1}\} = \{[\alpha_0], \dots, [\alpha_{n-1}]\}$ list the vertices of Γ_X . Then because $S_* \models \varphi_{\Gamma_X}$, defined above, there is an isomorphism $\chi_X : \langle X \rangle \rightarrow S(G_{\Gamma_X})$ with $\chi_X([\alpha_i]) = [\ker(\pi_{v_i})]$. For each

$i < n$, define $\tau_{v_i} : [\ker(\pi_{v_i})] \rightarrow [\ker(\pi_{v_i})]$ to be the automorphism $\sigma_{v_i} \circ (\chi_X|_{[\alpha]})^{-1}$. Likewise for each edge $r = (v_i, v_j)$ of Γ_X , define $\tau_r : [\ker(\pi_r)] \rightarrow [\ker(\pi_r)]$ to be $\sigma_r \circ (\chi_X|_{[\beta]})^{-1}$ for the unique class $[\beta]$ with $[\beta] \cong W$ and $\beta \leq \alpha_i \wedge \alpha_j$. This choice of τ_r is compatible with τ_{v_i} and τ_{v_j} , as in Corollary 2.16. This corollary gives us, then, a unique $\tau_X \in \text{Aut}(S(G_{\Gamma_X}))$ such that τ_X restricts to give τ_{v_i} or τ_r on the D_p and W classes corresponding to v_i and r , respectively. Define $\Psi_X = \tau_X \circ \chi_X$. Then we have $\Psi_X|_{[\alpha_i]} = \sigma_{v_i}$ and if $r = (v_i, v_j)$ and $\beta \leq \alpha_i \wedge \alpha_j$, then $\Psi_X|_{[\beta]} = \sigma_r$.

If we choose a larger such X' with $X \subseteq X'$, we get a commutative diagram:

$$\begin{array}{ccc} \langle X \rangle & \xrightarrow{\Psi_X} & S(G_{\Gamma_X}) \\ \downarrow & & \downarrow S(\pi_{\Gamma_X}) \\ \langle X' \rangle & \xrightarrow{\Psi_{X'}} & S(G_{\Gamma_{X'}}), \end{array}$$

since, by construction, Ψ_X and $\Psi_{X'}$ agree on the generators of $\langle X \rangle$ (here the map $\langle X \rangle \rightarrow \langle X' \rangle$ is the inclusion). It follows that we obtain an isomorphism on the union $\Psi : (S_*)_- \rightarrow S(G_{\Gamma_*})$, as desired.

(2) follows from the proof of (1). Restricting just to the finite X contained in

$$\{\alpha \in S : [\alpha] \cong D_p\} \cup \{\alpha \in S : [\alpha] \cong W\},$$

the resulting map on the union gives an isomorphism $\Psi_{S_-} : (S)_- \rightarrow S(G_{\Gamma(S)})$. Since, by Lemma 3.10(2), $\Gamma(S)$ is an induced subgraph of Γ_* , the image of Ψ_{S_-} is just the subsystem of $S(G_{\Gamma_*})$ dual to the quotient via $\pi_{\Gamma(S)}$, yielding the commutative diagram in (2).

(3) By (1) and (2), there is an isomorphism $\Xi : (S_*)_- \rightarrow S(G_{\Gamma_*})$ which restricts to S_- to yield an isomorphism $\Xi|_{S_-} : S_- \rightarrow S(G_{\Gamma(S)})$. Write $\Gamma(S) = (V_S, R_S)$. By construction, for each vertex $v \in V(S)$, $\Psi \circ \Xi^{-1}$ restricts to give an automorphism σ_v of $[\ker(\pi_v)]$ and, likewise, to give an automorphism σ_r of $[\ker(\pi_r)]$ for each edge $r \in R_S$. Choose an automorphism $\sigma_v : [\ker(\pi_v)] \rightarrow [\ker(\pi_v)]$ arbitrarily for each $v \in V_* \setminus V_S$. Then, for each $r \in R_* \setminus R_S$, choose an automorphism $\sigma_r : [\ker(\pi_r)] \rightarrow [\ker(\pi_r)]$ satisfying the compatibility conditions of Corollary 2.16. That corollary then gives us $\Sigma \in \text{Aut}(S(G_{\Gamma_*}))$ which restricts to σ_v and σ_r on each quotient $[\ker(\pi_v)]$ and $[\ker(\pi_r)]$ for all $v \in V_S$ and $r \in R_S$. Unwinding definitions, we get $(\Sigma \circ \Xi)|_{S_-} = \Psi$, so, setting $\Psi_* = \Sigma \circ \Xi$, we obtain the desired isomorphism. $\square$

3.3. Second identification.

Lemma 3.12. *Suppose Γ is an infinite graph, μ is a cardinal, and $S_0 = \text{gcl}(S_0) \subseteq S_1 = \text{gcl}(S_1) \subseteq S_* \models T_{\Gamma, \mu}$.*

- (1) *A sequence $(\alpha_i)_{i < \lambda}$ of elements of $(S_0)_+$ with $[\alpha_i] \cong C_2$ is a maximal such sequence satisfying $\alpha_i \notin \langle (S_0)_-, \alpha_{<i} \rangle$ for all $i < \lambda$ if and only if it is a maximal such sequence satisfying $\alpha_i \notin \langle (S_1)_-, \alpha_{<i} \rangle$.*
- (2) *We have the equality of subsystems $(S_1)_- \vee (S_0)_+ = (S_0)_-$.*

Proof. (1) Let Γ_i be the graph interpreted in S_i . By Lemma 3.11, we may identify $(S_i)_-$ with $S(G_{\Gamma_i})$ for $i = 0, 1$.

First, assume $(\alpha_i)_{i < \lambda}$ is maximal such that $\alpha_i \notin \langle (S_0)_-, \alpha_{<i} \rangle$ for all $i < \lambda$. If $\alpha_i \notin \langle (S_1)_-, \alpha_{<i} \rangle$ for all i , then $(\alpha_i)_{i < \lambda}$ is a maximal such sequence, since this is an even stronger constraint.

Fix $i < \lambda$ and suppose towards contradiction that $\alpha_i \in \langle (S_1)_-, \alpha_{<i} \rangle$. Since $[\alpha_i] \cong C_2$, we can find some $\beta_0, \dots, \beta_{k-1} \in (S_1)_-$ with $[\beta_j] \cong C_2$ and with each β_j of vertex width 1 for all j , as well as $\gamma_0, \dots, \gamma_{\ell-1}$, a subsequence of $\alpha_{<i}$ with

$$\alpha_i \geq \bigwedge_{j < k} \beta_j \wedge \bigwedge_{j < \ell} \gamma_j,$$

and we may suppose k, ℓ are minimal with this property (so, by assumption, $k \geq 1$). We may also suppose that α_i , as well as the β_j and γ_j are the identity coset of the normal subgroup in their respective equivalence classes.

Claim: There is some $\delta \in S_1$ with $[\delta] \cong C_2$ and

$$\delta \geq \left(\bigwedge_{j < k} \beta_j \right) \vee \left(\alpha_i \wedge \bigwedge_{j < \ell} \gamma_j \right).$$

Proof of Claim: We have

$$\left[\bigwedge_{j < k} \beta_j \wedge \bigwedge_{j < \ell} \gamma_j \right] \cong C_2^{k+\ell}.$$

If $\epsilon \geq \bigwedge_{j < k} \beta_j \wedge \bigwedge_{j < \ell} \gamma_j$, then ϵ corresponds to a normal subgroup of $C_2^{k+\ell}$ so we will prove the claim by studying the normal subgroups of this group. For the purposes of this claim only, we will switch to additive notation and identify this group with $\mathbb{F}_2^{k+\ell}$. Under this identification, we can identify each β_j with $\ker(\pi_j)$ where π_j is the projection onto the j th coordinate and, similarly, each γ_j can be identified with $\ker(\pi_{k+j})$.

It follows, then, that α_i can be identified with the kernel of some linear form $f : \mathbb{F}_2^{k+\ell} \rightarrow \mathbb{F}_2$. Our assumption that k and ℓ are minimal entails that $f(e_j) = 1$ where e_j denotes the j th standard basis vector, for $j < k + \ell$. It follows that α_i corresponds to the normal subgroup $\{(x_0, \dots, x_{k+\ell-1}) \in \mathbb{F}_2^{k+\ell} : \sum_j x_j = 0\}$. Then we have that $\alpha_i \wedge \bigwedge_{j < \ell} \gamma_j$ corresponds to the normal subgroup

$$\{(x_0, \dots, x_{k+\ell-1}) \in \mathbb{F}_2^{k+\ell} : \sum_j x_j = 0 \text{ and } x_k = \dots = x_{k+\ell-1} = 0\}.$$

Let δ be the element of S corresponding to the normal subgroup

$$\{(x_0, \dots, x_{k+\ell-1}) \in \mathbb{F}_2^{k+\ell} : \sum_{j < k} x_j = 0\}.$$

So we have $\delta \geq \alpha_i \wedge \bigwedge_{j < \ell} \gamma_j$. The kernel corresponding to δ contains the kernel corresponding to $\bigwedge_{j < k} \beta_j$, namely

$$\{(x_0, \dots, x_{k+\ell-1}) \in \mathbb{F}_2^{k+\ell} : x_0 = \dots = x_{k-1} = 0\}.$$

This shows $\delta \geq \bigwedge_{j < k} \beta_j$, which proves the claim. $\square$

Consider the δ of the claim. Then because $\delta \geq \alpha_i \wedge \bigwedge_{j < \ell} \gamma_j$, we have $\delta \in S_0$. Then since $\delta \geq \bigwedge_{j < k} \beta_j$ and each β_j has vertex width 1, we know that δ has finite vertex width. Since $S_0 = \text{gcl}(S_0)$, we deduce that $\beta_j \in (S_0)_-$ for all $j < k$. This entails $\alpha_i \in \langle (S_0)_-, \alpha_{<i} \rangle$, a contradiction.

The other direction is similar. Assume now that $(\alpha_i)_{i < \lambda}$ is a maximal sequence of elements of $(S_0)_+$ with $[\alpha_i] \cong C_2$ such that $\alpha_i \notin \langle (S_1)_-, \alpha_{<i} \rangle$ for all $i < \lambda$. If $(\alpha_i)_{i < \lambda}$ is not maximal such that $\alpha_i \notin \langle (S_0)_-, \alpha_{<i} \rangle$ for all $i < \lambda$, then there must

be some $\alpha_\lambda \in (S_0)_+$ with $[\alpha_\lambda] \cong C_2$ and $\alpha_\lambda \notin \langle (S_0)_-, \alpha_{<\lambda} \rangle$. By maximality, it must be the case that $\alpha_\lambda \in \langle (S_1)_-, \alpha_{<\lambda} \rangle$. So there are some $\beta_0, \dots, \beta_{k-1} \in (S_1)_-$ with $[\beta_j] \cong C_2$ and with each β_j of vertex width 1 for all j , as well as $\gamma_0, \dots, \gamma_{l-1}$, a subsequence of $\alpha_{<\lambda}$ with

$$\alpha_\lambda \geq \bigwedge_{j < k} \beta_j \wedge \bigwedge_{j < l} \gamma_j,$$

and we may suppose k, l are minimal with this property. Then the claim above produces a $\delta \in S_0$ with $[\delta] \cong C_2$ and satisfying both $\delta \geq \alpha_\lambda \wedge \bigwedge_{j < l} \gamma_j$ and $\delta \geq \bigwedge_{j < k} \beta_j$. So $S_0 = \text{gcl}(S_0)$ entails $\{\beta_j : j < k\} \subseteq S_0$ so $\alpha_\lambda \in S_0$, a contradiction.

(2) By Lemma 3.11, we may identify $(S_1)_-$ with $S(G_{\Gamma_1})$ for some induced subgraph $\Gamma_1 \subseteq \Gamma(S_*) \models \text{Th}(\Gamma)$. The containment $(S_0)_- \subseteq ((S_0)_+ \vee (S_1)_-)$ is clear, so we will prove the reverse containment. First, consider an $\alpha \in ((S_0)_+ \vee (S_1)_-)$ with $[\alpha] \cong C_2$. The fact that $\alpha \in (S_1)_-$ entails there are finitely many $\beta_0, \dots, \beta_{k-1} \in (S_1)_-$ with $[\beta_i] \cong D_p$ for $i < k$ and $\alpha \geq \bigwedge_{i < k} \beta_i$. So α has finite vertex width and thus, from $S_0 = \text{gcl}(S_0)$, we have each $\beta_i \in S_0$ (and thus in $(S_0)_-$) for all $i < k$. This shows $\alpha \in (S_0)_-$, since $(S_0)_-$ is upwards closed.

Next, by Lemma 3.8, if $\delta \in ((S_0)_+ \vee (S_1)_-)$ is arbitrary, we have $\delta \in \text{gcl}(\{\alpha : \alpha \geq \delta, [\alpha] \cong C_2\}) \subseteq \text{gcl}((S_0)_-) = (S_0)_-$, since $S_0 = \text{gcl}(S_0)$. $\square$

Lemma 3.13. *Suppose Γ is an infinite graph and μ is a cardinal.*

(1) *Suppose $S = \text{gcl}(S) \subseteq S_* \models T_{\Gamma, \mu}$. Then*

$$S_+ \cong S(G_{\Gamma(S)} \times C_2^I).$$

(2) *Suppose $S_0 = \text{gcl}(S_0) \subseteq S_1 = \text{gcl}(S_1) \subseteq S_* \models T_{\Gamma, \mu}$. Suppose further we are given an isomorphism $\Psi_0 : (S_0)_+ \rightarrow S(G_{\Gamma_0} \times C_2^{I_0})$, where Γ_0 is the graph interpreted in S_0 and I_0 is some set. Then there exist a set $I_1 \supseteq I_0$ and an isomorphism $\Psi_1 : (S_1)_+ \rightarrow S(G_{\Gamma_1} \times C_2^{I_1})$, where $\Gamma_1 \supseteq \Gamma_0$ is the graph interpreted in S_1 , such that the diagram*

$$\begin{array}{ccc} (S_0)_+ & \xrightarrow{\Psi_0} & S(G_{\Gamma_0} \times C_2^{I_0}) \\ \downarrow & & \downarrow S(\pi) \\ (S_1)_+ & \xrightarrow{\Psi_1} & S(G_{\Gamma_1} \times C_2^{I_1}). \end{array}$$

commutes, where the map $(S_0)_+ \rightarrow (S_1)_+$ is the inclusion and the map π is given by the projection $\pi_{\Gamma_0} : G_{\Gamma_1} \rightarrow G_{\Gamma_0}$ and the projection $\pi_{I_0} : C_2^{I_1} \rightarrow C_2^{I_0}$.

Proof. (1) By Lemma 3.11(1), we know $(S)_- \cong S(G_{\Gamma(S)})$. By Zorn's Lemma, we can pick a maximal sequence $(\alpha_i)_{i < \lambda}$ (for an ordinal λ) in S_* such that $[\alpha_i] \cong C_2$ and $\alpha_i \notin \langle S_-, \alpha_{<i} \rangle$ for all $i < \lambda$ (so, necessarily, each $\alpha_i \in S_+$). For each $i < \lambda$, let $S_i = \langle S_-, \alpha_{<i} \rangle$. We will prove by induction on $i < \lambda$ that there is an isomorphism

$$\varphi_i : G_{\Gamma(S)} \times C_2^i \rightarrow G(S_i).$$

We will also assume that for $j < i$, these isomorphisms cohere in the sense that the diagram commutes

$$\begin{array}{ccc} G_{\Gamma(S)} \times C_2^i & \xrightarrow{\varphi_i} & G(S_i) \\ \downarrow \pi'_{i,j} & & \downarrow \pi_{i,j} \\ G_{\Gamma(S)} \times C_2^j & \xrightarrow{\varphi_j} & G(S_j). \end{array}$$

where $\pi_{i,j} : G(S_i) \rightarrow G(S_j)$ is the epimorphism dual to the inclusion $S_j \subseteq S_i$ and $\pi'_{i,j} : G_{\Gamma(S)} \times C_2^i \rightarrow G_{\Gamma(S)} \times C_2^j$ is the projection obtained by deleting all of the $i \setminus j$ coordinates.

For $i = 0$, the existence of an isomorphism $\varphi_0 : G_{\Gamma(S)} \rightarrow G(S_-)$ follows from Lemma 3.11(1) and duality, as noted above. Note, for a given $i < \lambda$, since $\alpha_i \notin S_i$ and $[\alpha_i] \cong C_2$ has only 1 as a proper quotient, we have

$$[\alpha_i] \vee S_i = 1$$

(meaning the coarsest common quotient is trivial) and hence

$$G(S_{i+1}) = G(S_i \wedge [\alpha_i]) = G(S_i) \times [\alpha_i] \cong G_{\Gamma(S)} \times C_2^{i+1},$$

where the isomorphism between the latter two terms follows by induction. Finally, for limit i , we know that $G(S_i)$ is the inverse limit of S_j and $G_{\Gamma(S)} \times C_2^i$ is the inverse limit of $G_{\Gamma(S)} \times C_2^j$ for $j < i$ and these are isomorphic by induction and the coherence of the isomorphisms φ_j for $j < i$. Maximality of the sequence $(\alpha_i)_{i < \lambda}$ entails that $S_{<\lambda} = (S_*)_+$ and, taking $I = \lambda$, we have $G(S_+) \cong G_{\Gamma(S)} \times C_2^I$, as desired.

(2) Let Γ_0 and Γ_1 be the graphs obtained by the graph interpretation in S_0 and S_1 respectively, so Γ_0 is an induced subgraph of Γ_1 which is an induced subgraph of $\Gamma(S_*) \models \text{Th}(\Gamma)$. Replacing I_0 with $\lambda_0 = |I_0|$, we may assume $I_0 = \lambda_0$. We assume we are given an isomorphism $\Psi_0 : (S_0)_+ \rightarrow S(G_{\Gamma_0} \times C_2^{\lambda_0})$ and we will produce the isomorphism Ψ_1 as in the statement.

Define $\alpha_i \in (S_0)_+$ by $\alpha_i = \Psi_0^{-1}(\ker(\pi_i))$ where $\pi_i : G_{\Gamma_0} \times C_2^{\lambda_0} \rightarrow C_2$ is the projection to the i th coordinate, for each $i < \lambda_0$. Then we have $[\alpha_i] \cong C_2$, $\alpha_i \notin \langle (S_0)_-, \alpha_{<i} \rangle$ for all $i < \lambda_0$, and $\langle (S_0)_-, \alpha_{<\lambda_0} \rangle = (S_0)_+$.

By Lemma 3.12, $(\alpha_i)_{i < \lambda_0}$ is a maximal sequence of elements of $(S_0)_+$ with $[\alpha_i] \cong C_2$ and $\alpha_i \notin \langle (S_1)_-, \alpha_{<i} \rangle$ for all $i < \lambda_0$. Extend this sequence to $(\alpha_i)_{i < \lambda_1}$, a maximal sequence elements of $(S_1)_+$ such that $[\alpha_i] \cong C_2$ and $\alpha_i \notin \langle (S_1)_-, \alpha_{<i} \rangle$ for all $i < \lambda_1$. Then, as in (1), we have $(S_1)_+ = \langle (S_1)_-, \alpha_{<\lambda_1} \rangle$ and thus, by (1), $(S_1)_+ \cong S(G_{\Gamma_1} \times C_2^{\lambda_1})$.

Note that, by Lemma 3.12(2), we have $(S_1)_- \vee (S_0)_+ = (S_0)_-$. It follows, then, that

$$G((S_1)_- \wedge (S_0)_+) \cong G((S_1)_-) \times_{G((S_0)_-)} G((S_0)_+).$$

Note that the given isomorphism $\psi_0 : G_{\Gamma_0} \times C_2^{I_0} \rightarrow G((S_0)_+)$ dual to Ψ_0 induces an isomorphism $\overline{\psi_0} : G_{\Gamma_0} \rightarrow G((S_0)_-)$, dual to the restriction $\Psi_0|_{(S_0)_-}$. By Lemma 3.11(3) and duality, there is some isomorphism $\chi : G_{\Gamma_1} \rightarrow G((S_1)_-)$ such that the

diagram commutes:

$$\begin{array}{ccc} G_{\Gamma_1} & \xrightarrow{\chi} & G((S_1)_-) \\ \downarrow \pi_{\Gamma_0} & & \downarrow \\ G_{\Gamma_0} & \xrightarrow{\overline{\psi_0}} & G((S_0)_-). \end{array}$$

It follows, then, that χ and ψ_0 define a map (χ, ψ_0) from $G_{\Gamma_1} \times_{G_{\Gamma_0}} (G_{\Gamma_0} \times C_2^{\lambda_0}) \cong G_{\Gamma_1} \times C_2^{\lambda_0}$ to $G((S_1)_-) \times_{G((S_0)_-)} G((S_0)_+)$. Repeating the induction from (1), then, shows that we may find some $\psi_1 : G_{\Gamma_1} \times C_2^{\lambda_1} \rightarrow G((S_1)_+)$ such that the diagram

$$\begin{array}{ccc} G_{\Gamma_1} \times C_2^{\lambda_1} & \xrightarrow{\psi_1} & G((S_1)_+) \\ \downarrow \pi_{\lambda_0} & & \downarrow \\ G_{\Gamma_1} \times C_2^{\lambda_0} & \xrightarrow{(\chi, \psi_0)} & G((S_1)_- \wedge (S_0)_+). \end{array}$$

commutes, where π_{λ_0} denotes the projection obtained by removing the C_2 -coordinates indexed by $\lambda_1 \setminus \lambda_0$. This gives the desired isomorphism. $\square$

3.4. Third identification.

Lemma 3.14. *Let Γ be an infinite graph and μ a cardinal. Suppose $S_0 = \text{gcl}(S_0) \subseteq S_1 = \text{gcl}(S_1) \subseteq S_* \models T_{\Gamma, \mu}$. Then $S_0 \vee (S_1)_+ = (S_0)_+$.*

Proof. The inclusion $(S_0)_+ \subseteq S_0 \vee (S_1)_+$ is clear, so we prove the reverse inclusion.

By Lemma 3.13, we may identify $(S_1)_+$ with $S(G_{\Gamma_1} \times C_2^{I_1})$ and $(S_0)_+$ with $S(G_{\Gamma_0} \times C_2^{I_0})$ in such a way that inclusion $(S_0)_+ \rightarrow (S_1)_+$ is dual to the projection $\pi : G_{\Gamma_1} \times C_2^{I_1} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$. If $\alpha \in S_0 \vee (S_1)_+$, then, we may identify the class $[\alpha]$ with a quotient $A := (G_{\Gamma_1} \times C_2^{I_1})/N$ for some open normal subgroup $N \trianglelefteq G_{\Gamma_1} \times C_2^{I_1}$. As in the proof of Lemma 2.10, we define

$$V_* = \{v \in V_1 : \pi_v(N) \lesssim D_p \text{ or } v \text{ is incident to } r \text{ and } \pi_r(N) \cap C_q = 1\}.$$

We first claim that $V_* \subseteq V_0$. To see this note that if $\pi_v(N) \lesssim D_p$, then π_v induces an epimorphism from $(G_{\Gamma_1} \times C_2^{I_1})/N \rightarrow D_p/\pi_v(N)$. In the case that $\pi_v(N) = 1$, this quotient corresponds to a class $[\beta] \cong D_p$ in S_0 , which entails $v \in V_0$. The other possibility is that $\pi_v(N) = C_p$, in which case the quotient $D_p/\pi_v(N)$ corresponds to some class $[\beta] \cong C_2$ with $\beta \in S_0$ of vertex width 1 so $v \in V_0$ since $S_0 = \text{gcl}(S_0)$. Finally, if $\pi_r(N) \cap C_q = 1$, then, by Lemma 2.8(4), the quotient $W/\pi_r(N)$ necessarily has a C_2 quotient of vertex width 2, so both vertices of r would be contained in V_0 since $S_0 = \text{gcl}(S_0)$.

For every vertex $v \in V_1 \setminus V_0$ and $r \in R_1 \setminus R_0$, we have $\pi_v(N) = D_p$ and $\pi_r(N) \supseteq C_q$ so by Lemma 2.8, $(C_p)_v \subseteq N$ and $(C_q)_r \subseteq N$. Thus, the quotient $(G_{\Gamma_1} \times C_2^{I_1})/N$ can be identified with the quotient

$$(G_{\Gamma_0} \times C_2^{V_1 \setminus V_0} \times C_2^{I_1})/N'.$$

Let $\pi_1 : G_{\Gamma_0} \times C_2^{V_1 \setminus V_0} \times C_2^{I_1} \rightarrow C_2^{V_1 \setminus V_0} \times C_2^{I_1 \setminus I_0}$ be the projection. So $\ker(\pi_1) = G_{\Gamma_0} \times C_2^{I_0}$. The quotient

$$(G_{\Gamma_0} \times C_2^{V_1 \setminus V_0} \times C_2^{I_1})/N' \ker(\pi_1)$$

corresponds to a quotient $(C_2^{V_1 \setminus V_0} \times C_2^{I_1 \setminus I_0})/N''$. If this were nontrivial, it would have a C_2 quotient not contained in $S(G_{\Gamma_0} \times C_2^{I_0}) = (S_0)_+$ and, therefore, not in S_0 which is impossible. This shows that this quotient is trivial, which entails that the factor $C_2^{V_1 \setminus V_0} \times C_2^{I_1 \setminus I_0}$ is contained in N' . Therefore, A is a quotient of $G_{\Gamma_0} \times C_2^{I_0}$ as desired. This shows $\alpha \in (S_0)_+$. $\square$

Theorem 3.15. *Let Γ be an infinite graph and μ a cardinal.*

- (1) *Suppose $S_0 = \text{gcl}(S_0) \subseteq S_1 \models T_{\Gamma, \mu}$. Then the epimorphism $G(S_0) \rightarrow G((S_0)_+)$ dual to the inclusion $(S_0)_+ \subseteq S_0$ is a Frattini cover (though not necessarily the universal Frattini cover).*
- (2) *If $S_* \models T_{\Gamma, \mu}$, then*

$$S_* \cong S(\widetilde{G_{\Gamma_*} \times C_2^I})$$

for some $\Gamma_ \equiv \Gamma$ and some set I and the epimorphism $G(S_*) \rightarrow G((S_*)_+)$ dual to the inclusion $(S_*)_+ \subseteq S_*$ is the universal Frattini cover.*

- (3) *Suppose $S_0 \preceq S_1 \models T_{\Gamma, \mu}$ and $\Gamma_0 \preceq \Gamma_1$ are the graphs interpreted in S_0 and S_1 respectively. Suppose further we are given an isomorphism $\Psi_0 : S_0 \rightarrow S(G_{\Gamma_0} \times C_2^{I_0})$ for some set I_0 . Then there are a set I_1 and an isomorphism $\Psi_1 : S_1 \rightarrow S(G_{\Gamma_1} \times C_2^{I_1})$ such that the diagram*

$$\begin{array}{ccc} S_0 & \xrightarrow{\Psi_0} & \widetilde{S(G_{\Gamma_0} \times C_2^{I_0})} \\ \downarrow & & \downarrow S(\tilde{\pi}) \\ S_1 & \xrightarrow{\Psi_1} & \widetilde{S(G_{\Gamma_1} \times C_2^{I_1})} \end{array}$$

commutes, where the map $S_0 \rightarrow S_1$ is the inclusion and the map $\tilde{\pi}$ is a lift of the projection $\pi : G_{\Gamma_1} \times C_2^{I_1} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$.

Proof. (1) We first make an observation about the theory $T_{\Gamma, \mu}$. Suppose B is a finite quotient of $\widetilde{G_{\Gamma} \times C_2^{\mu}}$. Then by Fact 1.5, $A := B/\Phi(B)$ is a (finite) quotient of $G_{\Gamma} \times C_2^{\mu}$. By Lemma 2.10, there is some natural number $N = N(|A|)$ such that A is a quotient of $G_{\Gamma_0} \times C_2^l$, itself a quotient of $G_{\Gamma} \times C_2^{\mu}$ where $\Gamma_0 = (V_0, R_0) \subseteq \Gamma$ is an induced subgraph with $|V_0| \leq N$ and also $l \leq N$. It follows, then, that $S(\widetilde{G_{\Gamma} \times C_2^{\mu}})$ satisfies an L_{IS} -sentence which naturally asserts the following:

$$(*)_B \left\{ \begin{array}{l} \text{For every } \beta \in X_{|B|} \text{ with } [\beta] \cong B, \text{ there is some } \alpha \geq \beta \text{ with} \\ [\alpha] \cong A \text{ and there are, for some } k, k' \leq N, \gamma_0, \dots, \gamma_{k-1} \in X_{2p} \\ \text{with each } [\gamma_i] \cong D_p \text{ and } \delta_0, \dots, \delta_{k'-1} \in X_2 \text{ with each } [\delta_i] \cong \\ C_2 \text{ such that if, for some } k'' \leq \binom{N}{2}, \epsilon_0, \dots, \epsilon_{k''-1} \in X_{4p^2q} \\ \text{enumerate representatives of all classes } [\epsilon] \cong W \text{ such that there} \\ \text{are some } i < j < k \text{ with } \epsilon \leq \gamma_i \wedge \gamma_j, \text{ then} \\ \\ \alpha \geq \left(\bigwedge_{i < k} \gamma_i \wedge \bigwedge_{i' < k'} \delta_{i'} \wedge \bigwedge_{i'' < k''} \epsilon_{i''} \right). \end{array} \right.$$

To see this, just note that the γ_i correspond to the kernels $\ker(\pi_v)$ for each $v \in V_0$, the $\epsilon_{i''}$ correspond to the kernels $\ker(\pi_r)$ for each $r \in R_0$, and the $\delta_{i'}$ correspond to

the C_2 quotient obtained by projecting $G_{\Gamma_0} \times C_2^l$ to each C_2 coordinate (so, $\widetilde{G_{\Gamma_0} \times C_2^l}$, here, $k' = \ell$). The equivalence class of their meet is the smallest quotient of $G_{\Gamma} \times C_2^{\mu}$ that has them all as a quotient, which must be $\widetilde{G_{\Gamma_0} \times C_2^l}$. Therefore, the sentence $(*)_B$ simply restates that every B quotient of $\widetilde{G_{\Gamma} \times C_2^{\mu}}$ has the property that its further quotient $B/\Phi(B)$ is itself a quotient of $G_{\Gamma_0} \times C_2^l$ for an induced subgraph $\Gamma_0 \subseteq \Gamma$ with at most N vertices and with $l \leq N$.

Now to conclude, suppose $\beta \in S_0$ and $[\beta] \cong B$. Since $S_0 \subseteq S_1$ and $S_1 \equiv S(G_{\Gamma} \times C_2^{\mu})$, $S(G_{\Gamma} \times C_2^{\mu})$ must also contain an element whose equivalence class is isomorphic to B and S_1 satisfies the sentence $(*)_B$. For the associated bound N , we know there is some induced subgraph $\Gamma_* \subseteq \Gamma_1$ with at most N vertices and some $l \leq N$ such that $G((S_1)_+)$ has $G_{\Gamma_*} \times C_2^l$ as a quotient and $[\beta]/\Phi([\beta])$ is a quotient of this group. Let $\gamma \in S_0$ be an element with $[\gamma] = [\beta]/\Phi([\beta])$. Then, since $[\beta]/\Phi([\beta])$ is a quotient of $G_{\Gamma_1} \times C_2^{I_1}$, we have $\gamma \in (S_1)_+$. By Lemma 3.14, then, we get $\gamma \in (S_0)_+$. We have thus shown that every finite quotient of $G(S_0)$ is a Frattini cover of a finite quotient of $G((S_0)_+)$. Thus, we know, by Fact 1.6, that the map $G(S_0) \rightarrow G((S_0)_+)$ is a Frattini cover, completing the proof.

(2) By (1), the epimorphism $\widetilde{G(S_*)} \rightarrow G((S_*)_+)$ induced by the inclusion $(S_*)_+ \rightarrow S_*$ is a Frattini cover. Since $S(G_{\Gamma} \times C_2^{\mu})$ is projective and $S_* \models T_{\Gamma, \mu}$, we also have S_* is projective by Fact 1.7. This shows $G(S_*) \rightarrow G((S_*)_+)$ is the universal Frattini cover.

(3) Let $\psi_0 : \widetilde{G_{\Gamma_0} \times C_2^{I_0}} \rightarrow G(S_0)$ be the isomorphism dual to Ψ_0 . We have an epimorphism $\widetilde{G(S_0)} \rightarrow G((S_0)_+)$ dual to the inclusion and an epimorphism $G_{\Gamma_0} \times C_2^{I_0} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$, by the definition of the universal Frattini cover, so there is a unique χ_0 such that the following diagram commutes.

$$\begin{array}{ccc} \widetilde{G_{\Gamma_0} \times C_2^{I_0}} & \xrightarrow{\psi_0} & G(S_0) \\ \downarrow & & \downarrow \\ G_{\Gamma_0} \times C_2^{I_0} & \xrightarrow{\chi_0} & G((S_0)_+). \end{array}$$

By Lemma 3.13(2), there are $\Gamma_1 \supseteq \Gamma_0$ and $I_1 \supseteq I_0$ and an isomorphism $\chi_1 : G_{\Gamma_1} \times C_2^{I_1} \rightarrow G((S_1)_+)$ such that there is a commutative diagram

$$\begin{array}{ccccc} G(S_1) & \longrightarrow & G((S_1)_+) & \xrightarrow{\chi_1^{-1}} & G_{\Gamma_1} \times C_2^{I_1} \\ \downarrow & & \downarrow & & \downarrow \pi \\ G(S_0) & \longrightarrow & G((S_0)_+) & \xrightarrow{\chi_0^{-1}} & G_{\Gamma_0} \times C_2^{I_0}. \end{array}$$

where π is the projection. By (1), we have isomorphisms $\theta_i : \widetilde{G_{\Gamma_i} \times C_2^{I_i}} \rightarrow G(S_i)$ for $i = 0, 1$. Let $\rho : G(S_1) \rightarrow G(S_0)$ denote the epimorphism dual to the inclusion

$S_0 \subseteq S_1$ and define $\tilde{\pi} = \theta_0^{-1} \circ \rho \circ \theta_1$. Then the diagram

$$\begin{array}{ccccccc}
 \widetilde{G_{\Gamma_1} \times C_2^{I_1}} & \xrightarrow{\theta_1} & G(S_1) & \longrightarrow & G((S_1)_+) & \xrightarrow{\psi_1} & G_{\Gamma_1} \times C_2^{I_1} \\
 \downarrow \tilde{\pi} & & \downarrow \rho & & \downarrow & & \downarrow \pi \\
 \widetilde{G_{\Gamma_0} \times C_2^{I_0}} & \xrightarrow{\theta_0} & G(S_0) & \longrightarrow & G((S_0)_+) & \xrightarrow{\psi_0} & G_{\Gamma_0} \times C_2^{I_0}
 \end{array}$$

commutes, which entails that $\tilde{\pi}$ is a lift of π . Then we are done by duality. $\square$

4. MODEL-THEORETIC ANALYSIS

4.1. Characterizing the theories. Let $p_\infty(x)$ be the partial type in the X_2 sort axiomatized by $\{[x] \cong C_2\} \cup \{\neg\varphi_n(x) : n < \omega\}$, where $\varphi_n(x)$ is the formula from Lemma 3.4 defining the elements of vertex width n . A C_2 -element is a realization of p_∞ if and only if it is an element of infinite vertex width. Given a model $S \models T_{\Gamma, \mu}$, define $\dim_\infty(S)$ to be the maximal cardinality κ of a sequence $(\alpha_i)_{i < \kappa}$ in $p_\infty(S)$ (or, equivalently, in $p_\infty(S_+)$) with $\alpha_i \notin \langle S_-, \alpha_{<i} \rangle$. By Lemma 3.3, any two such sequences have the same cardinality, as the length of such a sequence is equal to the dimension of the continuous dual of $\ker(G(S_+) \rightarrow G(S_-))$, viewed as an $\mathbb{F}_2$ -vector space. The following lemma, together with compactness, entails that p_∞ is a consistent partial type, whenever Γ is an infinite graph:

Lemma 4.1. *Suppose Γ is an infinite graph. Then, for all natural numbers n , there is some $\alpha \in S(\widetilde{G_\Gamma})$ such that $[\alpha]_\sim \cong C_2$ and the vertex width of α is n .*

Proof. The $n = 1$ case is clear so we may assume $n \geq 2$. Let V_n be an arbitrary set of n vertices from Γ . We know that $D_p^{V_n}$ is a quotient of $\widetilde{G_\Gamma}$ and, quotienting further by the normal subgroup $C_p^{V_n}$, we obtain $C_2^{V_n}$ as a quotient. Let N denote the normal subgroup of $C_2^{V_n}$ consisting of those $\bar{g} = (g_v)_{v \in V_n}$ such that the number of v with $g_v = -1$ is even. It is easy to check that this is a normal subgroup, $C_2^{V_n}/N \cong C_2$, and the natural map $C_2^{V_n} \rightarrow C_2^{V_n}/N$ does not factor through a projection $C_2^{V_n} \rightarrow C_2^{V'}$ for any proper $V' \subseteq V_n$. $\square$

Remark 4.2. By the above Lemma, when Γ is an infinite graph, the theory $T_{\Gamma, \mu}$ is not $\aleph_0$ -categorical for any μ . This is because there are elements of any model of $T_{\Gamma, \mu}$ of arbitrary finite vertex width and by Lemma 3.4 these satisfy different formulas in the X_2 -sort. This implies by the Ryll-Nardzewski theorem that $T_{\Gamma, \mu}$ is not $\aleph_0$ -categorical.

Lemma 4.3. *Suppose Γ is an infinite graph and μ is a cardinal. A model $S \models T_{\Gamma, \mu}$ is determined up to isomorphism by the isomorphism type of the graph $\Gamma(S)$ and $\dim_\infty(S)$.*

Proof. Using the notation for the subsystems defined in Definition 3.1, we know that if $S \models T_{\Gamma, \mu}$, then $S_- \cong S(G_{\Gamma(S)})$ and $S_+ \cong S(G_{\Gamma(S)} \times C_2^I)$ for some set I with $|I| = \dim_\infty(S)$. Hence if $S, S' \models T_{\Gamma, \mu}$ satisfy that $\Gamma(S) \cong \Gamma(S')$ and $\dim_\infty(S) = \dim_\infty(S')$ then for some Γ_* and set I_* we have $S_+ \cong S(G_{\Gamma_*} \times C_2^{I_*}) \cong S'_+$ and thus both S and S' are isomorphic to $S(\widetilde{G_{\Gamma_*} \times C_2^{I_*}})$ and therefore isomorphic to one another. $\square$

Now we come to the main point of this subsection, which is to prove that the theory $T_{\Gamma, \mu}$ is independent of μ :

Proposition 4.4. *If Γ and Γ' are infinite elementarily equivalent graphs and μ and μ' are cardinals (possibly finite or even 0), then $T_{\Gamma, \mu} = T_{\Gamma', \mu'}$.*

Proof. By a standard forcing and absoluteness argument (see, e.g., [HK23]), we may assume that for some regular uncountable cardinal κ , all complete theories in a countable language have a saturated model of cardinality κ . We will show that saturated models of size κ of $T_{\Gamma, \mu}$ and $T_{\Gamma', \mu'}$ are isomorphic.

First, we note that if $S \models T_{\Gamma, \mu}$ is a κ -saturated model, then $\dim_{\infty}(S) \geq \kappa$. By Lemma 4.1, p_{∞} is a consistent partial type so, by saturation, $\dim_{\infty}(S) \geq 1$. Moreover, if $\lambda < \kappa$ and we are given a sequence $J = (\alpha_i)_{i < \lambda}$ of elements of $p_{\infty}(S)$ with $\alpha_i \notin \langle S_{-}, \alpha_{<i} \rangle$ for all $i < \lambda$, then consider the partial type over J which asserts that for each $n < \omega$ and each finite $w \subseteq \lambda$, there are not n C_2 -elements of vertex width 1 such that x is greater than or equal to the common meet of these n C_2 -elements and $(\alpha_i)_{i \in w}$ in the partial order $\geq$ on S . Because there are infinitely many C_2 -elements in S , this is a consistent partial type over a set of parameters of size $< \kappa$, which must be realized in S . Additionally, if α_* is a realization, then $\alpha_* \notin \langle S_{-}, \alpha_{<\lambda} \rangle$. It follows that $\dim_{\infty}(S) \geq \kappa$.

To conclude, suppose $S \models T_{\Gamma, \mu}$ and $S' \models T_{\Gamma', \mu'}$ are both saturated models of size κ . Then, the interpreted graphs $\Gamma(S)$ and $\Gamma(S')$ are elementarily equivalent and saturated of size κ so $\Gamma(S) \cong \Gamma(S')$. Moreover, by the above, we have $\dim_{\infty}(S) = \kappa = \dim_{\infty}(S')$. Thus, by Lemma 4.3, $S \cong S'$. In particular, S and S' have the same theory, so $T_{\Gamma, \mu} = T_{\Gamma', \mu'}$. $\square$

From here on out, we will drop the subscript μ and refer to the theory of $S(\widetilde{G}_{\Gamma})$ as T_{Γ} . Let $\bar{\kappa}$ denote a sufficiently large cardinal. Given an infinite graph Γ , we will choose some Γ_* which is $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous, and we will denote by $\mathbb{S}_{\Gamma} \models T_{\Gamma}$ the model

$$\mathbb{S}_{\Gamma} = S(\widetilde{G_{\Gamma_*} \times C_2^{\bar{\kappa}}}).$$

This will turn out (see Lemma 4.7 below) to be a $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous model of T_{Γ} , and thus can serve as a monster model of T_{Γ} . We will use the word *small* to mean of cardinality $< \bar{\kappa}$. If $\Gamma_0 \subseteq \Gamma_*$ is an induced subgraph, then the subsystem generated by the D_p -elements corresponding to the vertices of Γ_0 and the W -elements corresponding to the edges of Γ_0 may be canonically identified with $S(G_{\Gamma_0})$. Moreover, by Corollary 2.24, the subsystem generated by the α such that $[\alpha]$ is a Frattini pq -cover of some $[\beta]$ for $\beta \in S(G_{\Gamma_0})$ can be canonically identified with $S_{(pq)}(\widetilde{G_{\Gamma_0}})$. Thus, we will refer to these subsystems of $\mathbb{S}_{\Gamma}$ by $S(G_{\Gamma_0})$ and $S_{(pq)}(\widetilde{G_{\Gamma_0}})$. Note, there are, in general, many subsystems $S \supseteq S(G_{\Gamma_0})$ of $\mathbb{S}_{\Gamma}$ such that the induced epimorphism $G(S) \rightarrow G(S(G_{\Gamma_0}))$ is a universal Frattini cover so we cannot write $S(\widetilde{G_{\Gamma_0}})$ to unambiguously refer to a subsystem of $\mathbb{S}_{\Gamma}$.

4.2. Types and algebraic closure. For the rest of the section, we will work with respect to a fixed infinite graph Γ with $\mathbb{S}_{\Gamma} = S(\widetilde{G_{\Gamma_*} \times C_2^{\bar{\kappa}}})$.

Lemma 4.5. *Suppose $S \subseteq \mathbb{S}_{\Gamma}$ is a small subsystem.*

- (1) *S is graph closed if and only if S_+ is graph closed if and only if S_- is graph closed.*

- (2) If $\Gamma_0 \subseteq \Gamma_*$ is an induced subgraph and $(\gamma_i)_{i < \lambda}$ is an independent sequence of C_2 -elements of $\mathbb{S}_\Gamma$ of infinite vertex width, then $\langle S(G_{\Gamma_0}), (\gamma_i)_{i < \lambda} \rangle$ is a graph closed subsystem of $\mathbb{S}_\Gamma$.
- (3) If S is graph closed and $\Gamma(S) \subseteq \Gamma' \subseteq \Gamma_*$ where Γ' is an induced subgraph, then $\langle S, S(G_{\Gamma'}) \rangle$ is graph closed.

Proof. (1) This is immediate from the definitions, since any C_2 -element of finite vertex width in S necessarily is an element of S_+ , and additionally any D_p - or W -elements of S also live in S_+ . Thus S is graph closed if and only if S_+ is graph closed.

(2) Let $S' = \langle S(G_{\Gamma_0}), (\gamma_i)_{i < \lambda} \rangle$. Write $\Gamma_0 = (V_0, R_0)$. Because each epimorphism $G(S') \rightarrow C_2$ must factor through the quotient of $G(S')$ by the product of the p - and q -Sylow subgroups, we have that if $\alpha \in S'$ is a C_2 -element, $\alpha \in \langle S(C_2^{V_0}), (\gamma_i)_{i < \lambda} \rangle = \langle (\alpha_v)_{v \in V_0}, (\gamma_i)_{i < \lambda} \rangle$, where α_v is a C_2 -element of vertex width 1 corresponding to the vertex v for each $v \in V_0$.

Suppose now that $\alpha \in S'$ is a C_2 -element of finite vertex width n and let $(\beta_i)_{i < n}$ be a sequence of C_2 -elements of vertex width 1 corresponding to the witnesses. Then using Lemma 3.3, we have, for some finite sets $X \subseteq V_0$, $Y \subseteq \lambda$

$$d([\alpha]) = \sum_{x \in X} d([\alpha_x]) + \sum_{y \in Y} d([\gamma_y]) = \sum_{i < n} d([\beta_i]),$$

which entails

$$\sum_{y \in Y} d([\gamma_y]) = \sum_{i < n} d([\beta_i]) - \sum_{x \in X} d([\alpha_x]),$$

but if non-zero, the sum on the left-hand side corresponds to a C_2 -element of infinite vertex width, while the one on the right corresponds to a C_2 -element of finite vertex width, so both sums must be zero. This shows $d([\alpha]) = \sum_{x \in X} d([\alpha_x])$, or, in other words, $\alpha \geq \bigwedge_{x \in X} \alpha_x$, so $\alpha \in S(G_{\Gamma_0}) \subseteq S'$. Since Γ_0 is an induced subgraph of Γ_* , we get, then, that S' is graph closed.

(3) It suffices to consider the case that the vertex set of Γ' contains exactly one vertex v not contained in $\Gamma(S)$, since then we may obtain the general case by iteration (possibly transfinitely). Let α_v be a C_2 -element of $\mathbb{S}_\Gamma$ of vertex width 1 corresponding to v . Then since $\alpha_v \notin S$, we have $G(\langle S, \alpha_v \rangle) \cong G(S) \times C_2$. Thus the Frattini subgroup $\Phi(G(\langle S, \alpha_v \rangle))$ is just $\Phi(G(S))$. Since $\Phi(C_2) = 1$, any epimorphism $G(\langle S, \alpha_v \rangle) \rightarrow C_2$ must factor through the quotient $G(\langle S, \alpha_v \rangle) / \Phi(G(S))$ which, by duality, entails that if $\alpha \in \langle S, \alpha_v \rangle$ is a C_2 -element, then $\alpha \in \langle S_+, \alpha_v \rangle$. By Lemma 3.13, we have $S_+ = \langle S(G_{\Gamma(S)}), (\gamma_i)_{i < \lambda} \rangle$ where $(\gamma_i)_{i < \lambda}$ is an independent sequence of elements of infinite vertex width. Then by the proof of (2), we see that if $\alpha \in \langle S_+, \alpha_v \rangle$ is a C_2 -element of finite vertex width, then the witnesses must correspond to vertices in Γ' . Since any C_2 -element of $\langle S, S(G_{\Gamma'}) \rangle$ must be contained in $\langle S_+, \alpha_v \rangle$ and Γ' was chosen to be an induced subgraph of Γ_* , we see $\langle S, S(G_{\Gamma'}) \rangle$ is graph closed. $\square$

Lemma 4.6. (1) Suppose S_0 and S_1 are small subsystems of $\mathbb{S}_\Gamma$ with $\text{gcl}(S_i) = S_i$ for $i = 0, 1$. If $\Xi : S_0 \rightarrow S_1$ is an isomorphism such that $\Gamma(\Xi) : \Gamma(S_0) \rightarrow \Gamma(S_1)$ is partial elementary with respect to $\text{Th}(\Gamma)$, then Ξ is partial elementary, with respect to T_Γ .

- (2) Suppose $S \models T_\Gamma$ is a small model of T_Γ . If $\Psi : S \rightarrow \mathbb{S}_\Gamma$ is an embedding of L_{IS} -structures such that $\Gamma(\Psi)$ is elementary, with respect to $\text{Th}(\Gamma)$, then Ψ is an elementary embedding.

Proof. (1) Write Γ_i for $\Gamma(S_i)$ for $i = 0, 1$. Note that we have $(S_i)_- = S(\Gamma_i)$ for $i = 0, 1$. Because $\Gamma(\Xi) : \Gamma_0 \rightarrow \Gamma_1$ is partial elementary and Γ_* is strongly $\bar{\kappa}$ -homogeneous, there is some $\sigma \in \text{Aut}(\Gamma)$ extending $\Gamma(\Xi)$. We know that Ξ restricts to $S(\Gamma_0) \subseteq S(\Gamma_0 \times C_2^{I_0})$ to give an isomorphism between $S(\Gamma_0)$ and $S(\Gamma_1)$. Thus, by Lemma 2.14, there is some $\Theta : S(\Gamma_*) \rightarrow S(\Gamma_*)$ such that $\Gamma(\Theta) = \sigma$ and Θ extends $\Xi|_{S(\Gamma_0)}$. By Lemma 3.12(2), $(\mathbb{S}_\Gamma)_- \vee S_i = (S_i)_-$ for $i = 0, 1$ entails that

$$\langle (\mathbb{S}_\Gamma)_-, S_i \rangle = (\mathbb{S}_\Gamma)_- \wedge S_i \cong S(\Gamma_* \times_{G_{\Gamma_i}} G(S_i))$$

for $i = 0, 1$. Therefore, there is a unique map

$$\Psi_0 : \langle S(\Gamma_*), S_0 \rangle \rightarrow \langle S(\Gamma_*), S_1 \rangle$$

extending Ξ and Θ . Choose a maximal sequence $(\alpha_i)_{i < \gamma}$ of elements in S_0 with $[\alpha_i] \cong C_2$ and $\alpha_i \notin \langle (S_0)_-, \alpha_{<i} \rangle$. Then, again using the fact that $(\mathbb{S}_\Gamma)_- \vee S_0 = (S_0)_-$, we have $\alpha_i \notin \langle (\mathbb{S}_\Gamma)_-, \alpha_{<i} \rangle$ for all $i < \gamma$, so we may extend this sequence to $(\alpha_i)_{i < \bar{\kappa}}$ which is a maximal sequence of elements in $(\mathbb{S}_\Gamma)_+$ such that $[\alpha_i] \cong C_2$ and $\alpha_i \notin \langle (\mathbb{S}_\Gamma)_-, \alpha_{<i} \rangle$ for all $i < \bar{\kappa}$. Define $\alpha'_i = \Xi(\alpha_i)$ for $i < \gamma$. Then $\alpha'_i \notin \langle (S_1)_-, \alpha'_{<i} \rangle$ for all $i < \gamma$, and we may likewise extend this sequence to a maximal sequence $(\alpha'_i)_{i < \bar{\kappa}}$ in $(\mathbb{S}_\Gamma)_+$ such that $\alpha'_i \notin \langle (\mathbb{S}_\Gamma)_-, \alpha'_{<i} \rangle$. By maximality, we have

$$\langle (\mathbb{S}_\Gamma)_-, \alpha_{<\bar{\kappa}} \rangle = \langle (\mathbb{S}_\Gamma)_-, \alpha'_{<\bar{\kappa}} \rangle = (\mathbb{S}_\Gamma)_+.$$

Moreover, using Fact 1.4(3) and the fact that $\Phi(C_2) = 1$, we have

$$\alpha_i \notin \tilde{S}$$

for any subsystem $\tilde{S} \supseteq \langle (\mathbb{S}_\Gamma)_-, \alpha_{<i} \rangle$ such that the induced map $G(\tilde{S}) \rightarrow G(\langle (\mathbb{S}_\Gamma)_-, \alpha_{<i} \rangle)$ is Frattini, for all $i < \bar{\kappa}$. In particular, since $G(S_0) \rightarrow G((S_0)_+)$ is necessarily Frattini, we have $\alpha_i \notin \langle (\mathbb{S}_\Gamma)_-, S_0, \alpha_{<i} \rangle$ for all $\gamma \leq i < \bar{\kappa}$. By an identical argument, we have $\alpha'_i \notin \langle (\mathbb{S}_\Gamma)_-, S_1, \alpha'_{<i} \rangle$ for all $\gamma \leq i < \bar{\kappa}$. Thus, the map Ψ_0 extends to a map $\Psi_1 : \langle (\mathbb{S}_\Gamma)_+, S_0 \rangle \rightarrow \langle (\mathbb{S}_\Gamma)_+, S_1 \rangle$, defined by mapping $\alpha_i \mapsto \alpha'_i$ for all $i < \bar{\kappa}$, since $\langle (\mathbb{S}_\Gamma)_+, S_0 \rangle = \langle (\mathbb{S}_\Gamma)_-, S_0, (\alpha_i)_{\gamma \leq i < \bar{\kappa}} \rangle$ and $\langle (\mathbb{S}_\Gamma)_+, S_1 \rangle = \langle (\mathbb{S}_\Gamma)_-, S_1, (\alpha'_i)_{\gamma \leq i < \bar{\kappa}} \rangle$. Then Ψ_1 lifts to an isomorphism Ψ_2 between universal Frattini covers of the domain and codomain, which is an automorphism $\Psi_2 : \mathbb{S}_\Gamma \rightarrow \mathbb{S}_\Gamma$.

(2) Given a small $S \models T_\Gamma$, by Theorem 3.15, we can, without loss of generality, assume $S = S(\Gamma_0 \times C_2^{I_0})$ for some small $\Gamma_0 \models \text{Th}(\Gamma)$ and small set $I_0 \subseteq \bar{\kappa}$. Since Γ_* is $\bar{\kappa}$ -saturated, we can find some $\Gamma' \cong \Gamma_*$ such that $\Gamma_0 \preceq \Gamma'$. The projection $\pi : G_{\Gamma'} \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ lifts to an epimorphism $\tilde{\pi} : G_{\Gamma'} \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$. Then $S(\tilde{\pi}) : S(G_{\Gamma_0} \times C_2^{I_0}) \rightarrow S(G_{\Gamma'} \times C_2^{\bar{\kappa}})$ is an L_{IS} -embedding, so there is some $S' \supseteq S$ and an isomorphism $\Theta : S' \rightarrow S(G_{\Gamma'} \times C_2^{\bar{\kappa}})$ such that $\Theta|_S = \Psi$. Moreover, since $\Gamma' \cong \Gamma_*$, it is clear that $S(G_{\Gamma'} \times C_2^{\bar{\kappa}}) \cong \mathbb{S}_\Gamma$, so there is an isomorphism $\Theta' : S' \rightarrow \mathbb{S}_\Gamma$. Thus, we have $S \equiv \Theta'(S)$. And, by (1), $\Psi(S) \equiv S$. This shows $S \equiv \Psi(S)$, so Ψ is elementary. $\square$

Lemma 4.7. *The structure $\mathbb{S}_\Gamma$ is a monster model of T_Γ —that is, $\mathbb{S}_\Gamma$ is $\bar{\kappa}$ -saturated and strongly $\bar{\kappa}$ -homogeneous.*

Proof. By standard model theory, to show that $\mathbb{S}_\Gamma$ is $\bar{\kappa}$ -saturated, we must show that it is $\bar{\kappa}$ -universal and $\bar{\kappa}$ -homogeneous (which is, of course, implied by being $\bar{\kappa}$ -strongly homogeneous). So, towards showing universality, fix some $S \models T_\Gamma$ with

$|S| \leq \bar{\kappa}$. By Theorem 3.15, we may assume $S = \widetilde{S(G_{\Gamma_0} \times C_2^{I_0})}$ for some graph $\Gamma_0 = (V_0, R_0) \models \text{Th}(\Gamma)$ and set I_0 with $|\Gamma_0| \leq \bar{\kappa}$ and $|I_0| \leq \bar{\kappa}$. Since Γ was chosen to be $\bar{\kappa}$ -saturated, there is an elementary embedding $\iota_0 : \Gamma_0 \rightarrow \Gamma$ and, additionally, there is some injection $\iota_1 : I_0 \rightarrow \bar{\kappa}$. There is naturally an embedding $\Psi : S(G_{\Gamma_0} \times C_2^{I_0}) \rightarrow (\mathbb{S}_\Gamma)_-$ with $\Psi([\ker(\pi_v)]) = [\ker(\pi_{\iota_0(v)})]$ for all $v \in V_0$ and $\Psi([\ker(\pi_i)]) = [\ker(\pi_{\iota_1(i)})]$ for all $i \in I_0$. If $\psi : G_\Gamma \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ is dual to Ψ , then we know there is a lift $\psi' : \widetilde{G_\Gamma \times C_2^{\bar{\kappa}}} \rightarrow \widetilde{G_{\Gamma_0} \times C_2^{I_0}}$ so the dual epimorphism $\Psi' : S \rightarrow \mathbb{S}_\Gamma$ gives an L_{IS} -embedding which induces a $\text{Th}(\Gamma)$ -elementary embedding on interpreted graphs. By Lemma 4.6(2), we get Ψ' is elementary. This shows $\mathbb{S}_\Gamma$ is $\bar{\kappa}$ -universal.

If $A, B \subseteq \mathbb{S}_\Gamma$ and $\Theta : A \rightarrow B$ is a partial elementary isomorphism, then Θ extends to an isomorphism between $\text{acl}(A)$ and $\text{acl}(B)$ so strong $\bar{\kappa}$ -homogeneity follows from Lemma 4.6(1), using that the graph closure is contained in the algebraic closure, by Lemma 3.10(3). $\square$

Lemma 4.8. *Suppose $S = \text{gcl}(S)$ is a small subsystem of $\mathbb{S}_\Gamma$. Then there is a unique subsystem S' with $S \subseteq S' \subseteq \mathbb{S}_\Gamma$ and such that the epimorphism $G(S') \rightarrow G(S)$ induced by the inclusion $S \subseteq S'$ is a universal pq -Frattini cover of $G(S)$.*

Proof. By Lemma 3.13, we may represent S_+ as $S(G_{\Gamma_0} \times C_2^{I_0})$ and $(\mathbb{S}_\Gamma)_+$ as $S(G_\Gamma \times C_2^{\bar{\kappa}})$ so that $\Gamma_0 \subseteq \Gamma$ is an induced subgraph, $I_0 \subseteq \bar{\kappa}$ is a subset, and the epimorphism $G_\Gamma \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma_0} \times C_2^{I_0}$ is the projection to the Γ_0 and I_0 coordinates. By Corollary 2.24 and duality, there is a unique subsystem S'' with $S_+ \subseteq S'' \subseteq \mathbb{S}_\Gamma$ such that the epimorphism $G(S'') \rightarrow G(S_+)$ dual to the inclusion $S_+ \subseteq S''$ is a pq -universal Frattini cover.

Moreover, by Theorem 3.15(1), the map $G(S) \rightarrow G(S_+)$ induced by the inclusion $S_+ \subseteq S$ is a Frattini cover (although not necessarily the universal one). If N is the kernel of this epimorphism, then, since $N \subseteq \Phi(G(S))$, we know N is a direct product of two closed normal subgroups N_2 and N_{pq} where N_2 is a 2-group and N_{pq} has only finite quotients whose order is divisible by p or q [RZ10, Proposition 2.3.8]. Let S_2 and S_{pq} be the subsystems of S corresponding to the quotients $G(S)/N_2$ and $G(S)/N_{pq}$. Then we have $S = S_2 \wedge S_{pq}$ and $S_+ = S_2 \vee S_{pq}$. By uniqueness of the subsystem corresponding to the universal pq -Frattini cover of S_+ , we must have $S_{pq} \subseteq S''$. Note that $S'' \vee S_2 = S_+$.

Let $S' = S \wedge S''$. We clearly have $S \subseteq S'$ and we claim that the epimorphism $G(S') \rightarrow G(S)$ is a universal pq -Frattini cover. The p - and q -Sylows of $G(S')$ are isomorphic to the p - and q -Sylows of $G(S'')$ since the kernel of the epimorphism $G(S') \rightarrow G(S'')$ induced by the inclusion $S'' \subseteq S'$ is a 2-group. Since S'' is a universal pq -Frattini cover, the p - and q -Sylows of $G(S'')$ and thus of $G(S')$ are free, which entails that $G(S')$ is pq -projective. Moreover, from $S' = S_2 \wedge S''$ and $S_+ = S_2 \vee S''$, we have

$$G(S') \cong G(S'') \times_{G(S_+)} G(S_2)$$

and, recall, we also have

$$G(S'') \cong G(S_{pq}) \times_{G(S_+)} G(S_2).$$

Thus if $\rho : G(S'') \rightarrow G(S_{pq})$ denotes the epimorphism dual to the inclusion $S_{pq} \subseteq S''$, we have that the inclusion $S'' \subseteq S'$ induces $(\rho, \text{id}_{G(S_2)}) : G(S'') \times_{G(S_+)} G(S_2) \rightarrow$

$G(S_{pq}) \times_{G(S_+)} G(S_2)$, which is Frattini, since ρ is. Thus $G(S') \rightarrow G(S)$ is a universal pq -Frattini cover.

Uniqueness follows because we did not make any choices in the decomposition of S' . More precisely, suppose $S \subseteq S_* \subseteq \mathbb{S}_\Gamma$ and the epimorphism $G(S_*) \rightarrow G(S)$ dual to the inclusion $S \subseteq S_*$ is a universal pq -Frattini cover, then $S_* = S'$. Since this epimorphism $G(S_*) \rightarrow G(S)$ is Frattini and the epimorphism $G(S') \rightarrow G(S_+)$ dual to the inclusion $S_+ \subseteq S$ is also Frattini, the composition $G(S_*) \rightarrow G(S_+)$ is also Frattini. Then the kernel of this map splits as a direct product of closed normal subgroups $N_{2,*}$ and $N_{pq,*}$ as above, corresponding to subsystems $S_{*,2}$ and $S_{*,pq}$ with $S_{*,2} \vee S_{*,pq} = S_+$ and $S_{*,2} \wedge S_{*,pq} = S_*$. Then, as above, $G(S_{*,pq})$ is pq -projective and the induced map $G(S_{*,pq}) \rightarrow G(S_+)$ is pq -Frattini so this map is the universal pq -Frattini cover of S_+ which is unique, so $S_{*,pq} = S''$ and, likewise, the fact that $G(S_*) \rightarrow G(S)$ is a pq -Frattini cover entails $S_{*,2} = S_2$, so we obtain $S_* = S_{*,pq} \wedge S_{*,2} = S'' \wedge S_2 = S'$. $\square$

Definition 4.9. Given a set $A \subseteq \mathbb{S}_\Gamma$, define $\text{gcl}_{pq}(A)$ to be the unique subsystem S with $\text{gcl}(A) \subseteq S \subseteq \mathbb{S}_\Gamma$ such that the epimorphism $G(S) \rightarrow G(\text{gcl}(A))$, dual to the inclusion $\text{gcl}(A) \subseteq S$, is a universal pq -Frattini cover (the existence and uniqueness of which is guaranteed by Lemma 4.8).

Lemma 4.10. For any small set $A \subseteq \mathbb{S}_\Gamma$, $\text{gcl}_{pq}(A) \subseteq \text{acl}(A)$.

Proof. By Lemma 3.10(3), it suffices to prove that $\text{gcl}_{pq}(S) \subseteq \text{acl}(S)$ in the case that $S = \text{gcl}(S)$. Identify S_+ with $S(G_{\Gamma(S)} \times C_2^I)$ for some set I . By Fact 2.19, the quotient $[\alpha]$ is a pq -Frattini cover of a quotient of $G(S)$, so we know that if N is the product of the p - and q -Sylow subgroups of $\Phi([\alpha])$, there is some $\beta \in S$ with $\beta \geq \alpha$ and $[\beta] \cong [\alpha]/\Phi([\alpha])$. Then by Theorem 3.15(1), there is some $\gamma \in S_+$ with $[\gamma] \cong [\beta]/\Phi([\beta])$ and $\gamma \geq \beta$. Then there is a finite induced subgraph $\Gamma_0 \subseteq \Gamma(S)$ and a finite subset $I_0 \subseteq I$ such that $\gamma \in S(G_{\Gamma_0} \times C_2^{I_0}) \subseteq S_+$. Set $S_0 = \text{gcl}(S(G_{\Gamma_0} \times C_2^{I_0}), \beta)$ and observe that $\Gamma(S_0) = \Gamma_0$ since $[\beta]/\Phi([\beta]) = [\gamma] \subseteq S(G_{\Gamma_0} \times C_2^{I_0})$. Since $G_{\Gamma_0} \times C_2^{I_0}$ is a finite group, we know that ${}_{pq}(G_{\Gamma_0} \times C_2^{I_0})$ is a bounded profinite group and thus has only finitely many quotients isomorphic to $[\alpha]$. By Lemma 4.8, there is a unique subsystem $S_0 \subseteq S'_0 \subseteq \mathbb{S}_\Gamma$ such that the epimorphism $G(S'_0) \rightarrow G(S_0)$ is the universal pq -Frattini cover of $G(S_0)$. By Fact 2.19, if $\alpha' \in \mathbb{S}_\Gamma$ is any element with $[\alpha'] \cong [\alpha]$ and $\alpha' \leq \beta$ such that the canonical map $[\alpha'] \rightarrow [\beta]$ is a pq -Frattini cover, then $[\alpha']$ must be a quotient of $G(S'_0)$ or, in other words, $\alpha' \in S'_0$. This means there are only finitely many such α' and since $\beta \in S$, this establishes $\alpha \in \text{acl}(S)$. $\square$

We write $a \downarrow_C^a b$ to mean $\text{acl}(aC) \cap \text{acl}(bC) = \text{acl}(C)$. We say a sequence $I = (a_i)_{i < \omega}$ is $\downarrow^a$ -Morley over C if I is C -indiscernible and $a_i \downarrow_C^a a_{<i}$ for all i .

Proposition 4.11. Suppose $S \subseteq \mathbb{S}_\Gamma$ is a subsystem. Then $S = \text{acl}(S)$ if and only if $S = \text{gcl}_{pq}(S)$ and $\Gamma(S) = \text{acl}_\Gamma(\Gamma(S))$, where acl_Γ denotes algebraic closure in $\text{Th}(\Gamma)$.

Proof. By Lemma 4.10, $S = \text{acl}(S)$ entails $S = \text{gcl}_{pq}(S)$. Moreover, it is clear that $\text{acl}(S) = S$ entails $\Gamma(S) = \text{acl}_\Gamma(\Gamma(S))$. So we assume $\text{gcl}_{pq}(S) = S$ and $\Gamma(S) = \text{acl}_\Gamma(\Gamma(S))$ and we prove $\text{acl}(S) = S$. By Lemma 3.13, we may, without loss of generality, assume $S_+ = S(G_{\Gamma(S)} \times C_2^I)$ for some subset $I \subseteq \bar{\kappa}$ and the projection $\pi : G_\Gamma \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma(S)} \times C_2^I$ is dual to the inclusion $S_+ \subseteq (\mathbb{S}_\Gamma)_+$.

Pick any $\alpha \in \mathbb{S}_\Gamma \setminus S$. Let $\beta \in \mathbb{S}_\Gamma$ satisfy $\beta \geq \alpha$ and $[\beta] \cong [\alpha]/\Phi([\alpha])$. Then, necessarily, $\beta \in (\mathbb{S}_\Gamma)_+ = S(G_\Gamma \times C_2^{\bar{\kappa}})$. Choose some small induced subgraph Γ_0 with $\Gamma(S) \subseteq \Gamma_0 \subseteq \Gamma$ and small set I_0 with $I \subseteq I_0 \subseteq \bar{\kappa}$ such that $\beta \in S(G_{\Gamma_0} \times C_2^{I_0})$. In Γ , choose an $\downarrow^a$ -Morley sequence $(\Gamma_i)_{i < \omega}$ over $\Gamma(S)$ starting with Γ_0 . Also choose some sequence $(I_i)_{i < \omega}$ of subsets of $\bar{\kappa}$ such that $|I_i| = |I_0|$ and $I_i \cap I_j = I$ for all $i < j$.

We claim $S(G_{\Gamma_0} \times C_2^{I_0}) \cap S(G_{\Gamma_1} \times C_2^{I_1}) \subseteq S_+$. For an induced subgraph $\Gamma_* \subseteq \Gamma$ and subset $I_* \subseteq \bar{\kappa}$, let $\pi_{\Gamma_*, I_*} : G_\Gamma \times C_2^{\bar{\kappa}} \rightarrow G_{\Gamma_*} \times C_2^{I_*}$ denote the projection. Then, for each $i < \omega$, we have

$$S(G_{\Gamma_i} \times C_2^{I_i}) = S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{\Gamma_i, I_i})),$$

and we can calculate, for $i < j$,

$$\begin{aligned} S_i \cap S_j &= S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{\Gamma_i, I_i})) \vee S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{\Gamma_j, I_j})) \\ &= S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{\Gamma_i, I_i}) \ker(\pi_{\Gamma_j, I_j})) \\ &= S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{(\Gamma_i \cap \Gamma_j), (I_i \cap I_j)})) \\ &\subseteq S(G_\Gamma \times C_2^{\bar{\kappa}} / \ker(\pi_{\Gamma(S), I})) \\ &= S_+. \end{aligned}$$

Moreover, by Lemma 3.14, $S_i \cap S = S_+$, and so

$$\langle S_i, S \rangle = S_i \wedge S = S \left((G_{\Gamma_i} \times C_2^{I_i}) \times_{G((S)_+)} G(S) \right).$$

Therefore, if we pick an isomorphism $\Psi_i : S(G_{\Gamma_0} \times C_2^{I_0}) \rightarrow S(G_{\Gamma_i} \times C_2^{I_i})$ which is the identity on $S(G_{\Gamma(S)} \times C_2^I) = S_+$, it extends to an isomorphism $\Psi'_i : \langle S, S_0 \rangle \rightarrow \langle S, S_i \rangle$ which is the identity on S . By Lemma 4.5(3), $\langle S, S_i \rangle$ is graph closed and, therefore, by Lemma 4.6(1), each Ψ'_i extends to some $\Theta_i \in \text{Aut}(\mathbb{S}_\Gamma/S)$. Let $\beta_i = \Theta_i(\beta)$ and $\alpha_i = \Theta_i(\alpha)$ for $i < \omega$. Since $S_i \cap S_j = S$ for $i \neq j$, we have either $\beta_i = \beta \in S$ for all i , or $\beta_i \in S_i \setminus S$ for all i and are therefore pairwise distinct.

In the second case, in which the β_i are pairwise distinct, we can easily conclude: since $[\alpha]$ is a finite group, it can only have finitely many quotients and therefore $\{\alpha_i : i < \omega\}$ is an infinite set. It follows that $\alpha \notin \text{acl}(S)$.

In the first case, in which $\beta_i = \beta \in S$ for all i , we have that $[\alpha]$ is a Frattini cover of a quotient of $G(S)$, so $[\alpha]$ is a quotient of the universal Frattini cover of S . Let S' be any subsystem of $\mathbb{S}_\Gamma$ with $S \subseteq S'$, $\alpha \in S'$, and the epimorphism $G(S') \rightarrow G(S)$ is a universal Frattini cover of S . For any n , by Lemma 2.28 and duality, we can find $S' = S'_0, \dots, S'_{n-1}$ which are pairwise isomorphic over S with $S'_i \cap S'_j = S$ for $i \neq j$ (in the lemma $A = G(S)$). Now $(S'_i)_- = S_-$ for all i so S'_i is graph closed by Lemma 4.5(1). Then, for each i , we can define some $\Xi_i \in \text{Aut}(\mathbb{S}_\Gamma/S)$ with $\Xi_i(S') = S'_i$ for each $i < n$. Setting $\alpha_i = \Xi_i(\alpha)$, we find n distinct conjugates of α over S . Since n is arbitrary, we conclude $\alpha \notin \text{acl}(S)$. Thus $S = \text{acl}(S)$. $\square$

Remark 4.12. The description of algebraic closure is a little unwieldy but the general idea for how to produce the algebraic closure of a set X goes as follows: first take the substructure generated by X and consider all of the C_2 -elements. Some of these will have finite vertex width, so you throw in the D_p witnesses, and then you throw in the W quotients corresponding to edges so you get an induced subgraph $\Gamma(\text{gcl}(X))$. That graph may not be algebraically closed in the theory of Γ , so you replace it with a larger graph $\text{acl}_\Gamma(\Gamma(\text{gcl}(X)))$. At this stage, you have that

$\langle S(G_{\text{acl}_\Gamma(\text{gcl}(X))}, X_2(\langle X \rangle)) \rangle$ corresponds to $S(G_{\text{acl}_\Gamma(\text{gcl}(X))} \times C_2^\gamma)$, where the C_2^γ factor is generated by the C_2 -elements of infinite vertex width, which have not been swallowed up by the group associated to the graph. Then to get the full algebraic closure, one only needs to throw in the finite quotients of the universal Frattini cover of this group.

4.3. The maximal pro-2 quotient. In $\mathbb{S}_\Gamma$, let D denote the subsystem of $\mathbb{S}_\Gamma$ corresponding to the quotient of $G_{\Gamma_*} \times C_2^{\overline{\kappa}}$ by the (normal) product of the p - and q -Sylow subgroups, which we will denote S_{pq} . The group $G(D)$ is a free pro-2 group of infinite rank. The subsystem D can be viewed as a many-sorted definable subset of $\mathbb{S}_\Gamma$. When S is a subsystem of $\mathbb{S}_\Gamma$, we write $D(S)$ for the elements of D which are in S , which may be alternatively described as the inverse system corresponding to $S_{pq}N$, where N is the closed normal subgroup corresponding to S .

It will turn out that the analysis of the model-theoretic properties of T_Γ will essentially reduce to the analysis of the model-theoretic properties of the induced structure on $D(\mathbb{S}_\Gamma)$, which may be explicitly described relative to Γ . Before we start the analysis, we will point out the useful fact that passing to D commutes with $\wedge$ and $\vee$ for subsystems of $\mathbb{S}_\Gamma$:

Lemma 4.13. *Suppose S_1 and S_2 are subsystems of $\mathbb{S}_\Gamma$. Then we have*

$$D(S_1 \vee S_2) = D(S_1) \vee D(S_2) \text{ and } D(S_1 \wedge S_2) = D(S_1) \wedge D(S_2).$$

Proof. Let $N_1, N_2 \trianglelefteq G(\mathbb{S}_\Gamma)$ be the closed normal subgroups of $G(\mathbb{S}_\Gamma)$ corresponding to the subsystems S_1 and S_2 , respectively. By duality, we have to show $(S_{pq}N_1)(S_{pq}N_2) = S_{pq}N_1N_2$, which is clear, and we have to show $S_{pq}N_1 \cap S_{pq}N_2 = S_{pq}(N_1 \cap N_2)$, which was established in Corollary 2.30. $\square$

Let L_H be L_{IS} together with a new unary predicate H . We will view D as an L_H -structure by giving it the L_{IS} -structure it inherits as a subsystem of $\mathbb{S}_\Gamma$ and interpreting $H(D)$ as the collection of elements $\alpha \in X_2(D)$ with vertex width 1 such that α corresponds to the identity element of the two element group $[\alpha]$ (we are just making a canonical choice of element from each $\sim$ class of elements of vertex width 1). By Lemma 3.4, this is a reduct of the induced structure on D .

We will show that, viewed as an L_H -structure, D is an H -structure in the sense of Berenstein and Vasiliev [BV16]:

Definition 4.14. Suppose T is a supersimple theory and $\mathcal{Q}(x)$ is a partial 1-type over $\emptyset$. A structure $(M, H(M))$ consisting of a model $M \models T$ together with the interpretation $H(M)$ of a new unary predicate H is called an H -structure if it satisfies the following conditions:

- (1) For all $a \in H(M)$, $a \models \mathcal{Q}$.
- (2) For all n , if $a_0, \dots, a_{n-1} \in H(M)$ is a sequence of distinct elements, then the a_i are non-forking independent over $\emptyset$ in T .
- (3) (Density/coheir property) If $A \subseteq M$ is finite and $q \in S_1(A)$ is type containing $\mathcal{Q}$ which does not fork over $\emptyset$, then q has a realization in $H(M)$.
- (4) (Co-density/extension property) If $A \subseteq M$ is finite and $q \in S_1(A)$, then there is some $a \in M$ with $a \models q$ and $a \perp_A^f H(M)$.

We write T_H for the L_H -theory of $(M, H(M))$ when $(M, H(M))$ is an H -structure. The theory T_H does not depend on the choice of $(M, H(M))$ (though it does depend on the choice of $\mathcal{Q}$) [BCV17, Corollary 2.6].

We will show that the subsystem D , viewed as an L_H -structure, satisfies the conditions of Definition 4.14. But first, we will need a description of forking independence in $\text{Th}_{L_{\text{IS}}}(D)$. Fortunately, this has been essentially characterized already in Chatzidakis's detailed analysis of the model theory of inverse systems of profinite groups with the embedding property:

Fact 4.15. [Cha98, Proposition 4.1] Suppose G is a super-projective profinite group and $T = \text{Th}_{L_{\text{IS}}}(S(G))$. Then in any model of T , we have, for subsystems $A \subseteq B$,

$$\alpha \downarrow_A^f B \iff \alpha \vee B \in \text{acl}(A) \iff \alpha \vee B \in \text{acl}(\alpha \vee A).$$

Lemma 4.16. *Suppose G is an infinite rank free pro-2 group and $A \subseteq S(G)$ is a finite subset. Then $\text{acl}(A) = \langle A \rangle$.*

Proof. We will assume $A = \langle A \rangle$, i.e. A is a finite subsystem of $S(G)$, and then we will show that $\text{acl}(A) = A$. Choose any $\beta \in S(G) \setminus A$ and let $B = \langle A, \beta \rangle$, which is also a finite subsystem of $S(G)$. Fix any $n \geq 2$, and define $H_n = G(B) \times_{G(A)} G(B) \times_{G(A)} \dots \times_{G(A)} G(B)$ to be the n -fold fiber product of $G(B)$ with itself, over $G(A)$, taken with respect to the epimorphism $G(B) \rightarrow G(A)$ dual to the inclusion $A \subseteq B$. Let $\pi : H_n \rightarrow G(A)$ be the obvious map to A . Note that H_n is also a finite pro-2 group and thus an image of G . Then, by the superprojectivity of G , we obtain an epimorphism $\varphi : G \rightarrow H_n$ making the diagram commute,

$$\begin{array}{ccc} & G & \\ \swarrow \varphi & \downarrow & \\ H_n & \xrightarrow{\pi} & G(A) \end{array}$$

where the vertical arrow is the epimorphism $G \rightarrow G(A)$ dual to the inclusion $A \subseteq S(G)$. Let N be the kernel of φ . Then the quotient map $G \rightarrow G(A)$ factors into $G \rightarrow G/N \rightarrow G(A)$. If we let C be the subsystem of $S(G)$ corresponding to the quotient G/N , then we see $A \subseteq C$ and $G(C) \cong H_n$. Fix an isomorphism $G(C) \cong H_n$ and, for each $i = 1, \dots, n$, let B_i be the subsystem of C containing A dual to the quotient $G(C) \rightarrow G(B)$ obtained by projecting $G(C)$ to the i th coordinate. Then we have $B_i \cong_A B$ for all i , and we may write $B_i = \langle A, \beta_i \rangle$ for some $\beta_i \in B_i$. By [Cha98, Theorem 2.4], we obtain $\beta_i \equiv_A \beta$ for all i and, since $B_i \vee B_j = A$ for all $i \neq j$, we have $\beta_i \neq \beta_j$. Since n was taken to be arbitrary, we get $\beta \notin \text{acl}(A)$. $\square$

Corollary 4.17. *If G is a free pro-2 group of infinite rank, then for subsystems $A \subseteq B \subseteq S(G)$,*

$$\alpha \downarrow_A^f B \iff \alpha \vee B \in A \iff (\alpha \vee B) \geq (\alpha \vee A).$$

Proof. Immediate by Fact 4.15 and Lemma 4.16, using that free pro-2 groups are superprojective. $\square$

From here on out, we will write $\downarrow^D$ to denote non-forking independence in D , viewed as an L_{IS} -structure.

Fact 4.18. [Cha19, Theorem 4.6] If G is a superprojective profinite group (so, in particular, if G is a free or free pro- p profinite group), then $\text{Th}(S(G))$ weakly eliminates imaginaries.

We use $\downarrow^D$ to denote non-forking independence in $\text{Th}_{L_{\text{IS}}}(D)$ rather than $\text{Th}_{L_H}(D)$, so $\downarrow^D$ has the description given in Corollary 4.17. The point of referring to this relation as non-forking independence is just to remind the reader that it has the familiar properties, such as symmetry and transitivity, which we will use freely.

Lemma 4.19. *The subsystem D , viewed as an L_H -structure is a H -structure, with respect to the (complete) type $\mathcal{Q}(x) = \{x \in X_2 \wedge x \notin X_1 \wedge P(x, x, x)\}$, asserting that x is a C_2 -element and the identity element of its $\sim$ -class.*

Proof. We have to check that $(D, H(D))$ satisfies conditions (1)-(4) of Definition 4.14. Condition (1) is satisfied by the choice of $H(D)$. It is also clear from Lemma 3.3 that (2) is satisfied, since, when viewed as elements of the continuous dual $(\mathbb{F}_2^V)^*$, any n $\sim$ -inequivalent C_2 -elements of vertex width 1 are linearly independent. Thus if $\{\alpha_i : i < n\}$ is a set of distinct elements from $H(D)$, then we have $\alpha_i \notin \langle \alpha_{<i} \rangle$ for all $i < n$ and thus, by Corollary 4.17, $\alpha_i \downarrow^D \alpha_{<i}$, showing (2). For (3), note that if A is a finite subsystem of D and $q \in S_1(A)$ is a non-forking extension of $\mathcal{Q}$, then q can be axiomatized by

$$q(x) = \{x \in X_2 \wedge x \notin X_1 \wedge P(x, x, x) \wedge x \neq \beta : \beta \in A\},$$

and this will be realized by any element of $H(D)$ outside of $H(D) \cap A$.

The condition (4) is more involved. Let A be a finite subset of D and $q \in S_1(A)$. We may assume $A = \langle A \rangle$. Let $B = A \vee \langle H(D) \rangle$.

Claim: There is a subsystem D' with $A \subseteq D' \subseteq D$ such that $G(D')$ is a free pro-2 group of infinite rank and $D' \vee \langle H(D) \rangle = B$.

Proof of Claim: Let F be the subsystem of D corresponding to $G(D)/\Phi(G(D))$. Then $\langle H(D) \rangle \subseteq F$ and $G(F)$ is an elementary abelian 2-group (isomorphic to $C_2^V \times C_2^{\bar{K}}$). Let $C = A \vee F$. So $B \subseteq C$ and $G(B)$ and $G(C)$ are both finite elementary abelian 2-groups. Moreover, $G(A) \rightarrow G(C)$ is a Frattini cover.

Let $(\alpha_i)_{i \in I_0}$ be a maximal sequence of independent C_2 -elements in B from $H(D)$. Then $B = \langle (\alpha_i)_{i \in I_0} \rangle$ and we may extend this sequence to a maximal sequence of independent C_2 -elements $(\alpha_i)_{i \in I_0 \cup I_1}$ in $H(D)$ (which is, necessarily, an enumeration of all of $H(D)$) and a maximal sequence of independent C_2 -elements $(\alpha_i)_{i \in I_0 \cup I_2}$ in C . Then by our assumption that $A \downarrow_B^D H(D)$, we have that $(\alpha_i)_{i \in I_0 \cup I_1 \cup I_2}$ is an independent sequence of C_2 -elements in $\langle C, H(D) \rangle$. Finally, extend the sequence to $(\alpha_i)_{i \in I_0 \cup I_1 \cup I_2 \cup I_3}$, a maximal independent sequence of C_2 -elements in F . Thus F is generated by this sequence and $G(F) \cong C_2^{I_0 \cup I_1 \cup I_2 \cup I_3}$ via the isomorphism dual to the map $\ker(\pi_i) \mapsto \alpha_i$ for all i . Note that we have

$$(\alpha_i)_{i \in I_0 \cup I_2} \downarrow_{\emptyset}^D (\alpha_i)_{i \in I_1 \cup I_3},$$

and hence

$$C \downarrow_{\emptyset}^D (\alpha_i)_{i \in I_1 \cup I_3}.$$

Moreover, by the choice of $C = A \vee F$, we have

$$A \underset{C}{\downarrow}^D F$$

which entails $A \underset{C}{\downarrow}^D \langle (\alpha_i)_{i \in I_1 \cup I_3} \rangle$, by transitivity. This shows $G(\langle A, (\alpha_i)_{i \in I_1 \cup I_3} \rangle) \cong G(A) \times C_2^{I_1 \cup I_3}$. Thus we may identify $G(\langle A, (\alpha_i)_{i \in I_1 \cup I_3} \rangle)$ with $G(A) \times C_2^{I_1 \cup I_3}$, $G(F)$ with $C_2^{I_0 \cup I_1 \cup I_2 \cup I_3}$, $G(C)$ with $C_2^{I_0 \cup I_2}$ and the induced map $G(\langle A, (\alpha_i)_{i \in I_1 \cup I_3} \rangle) \rightarrow G(F)$ with the product of the induced map $G(A) \rightarrow G(C)$ and the identity map $C_2^{I_1} \times C_2^{I_3} \rightarrow C_2^{I_1} \times C_2^{I_3}$.

Let $x_i \in G(F)$ be the element which is -1 in the i coordinate and 1 in every other coordinate. Then the sequence $\langle x_i : i \in I_0 \cup I_2 \rangle$ is a minimal set of generators for $G(C)$. Since the map $G(A) \rightarrow G(C)$ is Frattini, the sequence $\langle x_i : i \in I_0 \cup I_2 \rangle$ lifts to a set $\langle x'_i : i \in I_0 \cup I_2 \rangle$ of elements of $G(A)$ which form a minimal set of generators. Then since the map $G(D) \rightarrow G(\langle A, F \rangle)$ is Frattini, the set $\{x_i : i \in I_1 \cup I_3\} \cup \{x'_i : i \in I_0 \cup I_2\}$ lifts to $\{y_i : i \in I_0 \cup I_1 \cup I_2 \cup I_3\}$, a set of minimal generators in $G(D)$. Then any subset of the y_i generate a free pro-2 group. Define a map $G(D) \rightarrow \langle y_i : i \in I_0 \cup I_2 \cup I_3 \rangle$ by sending each $y_i \mapsto y_i$ for $i \in I_0 \cup I_2 \cup I_3$ and mapping $y_i \mapsto 1$ for all $i \in I_1$. Let D' be the subsystem of D dual to this quotient.

Since the y_i lift the x_i , we have that $D' \vee F = \langle \alpha_i : i \in I_0 \cup I_2 \cup I_3 \rangle$, and thus $D' \vee \langle H(D) \rangle = \langle \alpha_i : i \in I_0 \cup I_2 \rangle = B$. This proves the claim. $\square$

Now the given q must be realized since $\text{Th}(D')$ is totally categorical so every model of its theory is $\aleph_0$ -saturated. This proves (4). $\square$

Definition 4.20. Suppose $(M, H(M))$ is an H -structure. Then a subset $A \subseteq M$ is called H -independent if $A \underset{H(A)}{\downarrow}^D H(M)$.

We need some general facts about H -structures:

Fact 4.21. Suppose $(M, H(M))$ is an H -structure where M is an L -structure and $(M, H(M))$ is an expansion of M to the language $L_H = L \cup \{H\}$.

- (1) [BCV17, Proposition 4.5] If C is an H -independent subset of M , and a is a finite tuple from M , then there is a unique smallest finite subset $H_0 \subseteq H(M)$ such that $a \underset{CH_0}{\downarrow}^f H(M)$. This set will be referred to as the H -basis of a over C , denoted $\text{HB}(a/C)$. When A is a (possibly infinite) set, we write $\text{HB}(A/C)$ for the union of the H -bases of all finite tuples from A .

- (2) [BCV17, Proposition 2.5] If a and a' are both H -independent tuples, then

$$\text{tp}_{L_H}(a) = \text{tp}_{L_H}(a') \iff \text{tp}_L(a, H(a)) = \text{tp}_L(a', H(a')).$$

- (3) [BCV17, Corollary 4.15] If $A \subseteq M$, then the algebraic closure of A in the H -structure $(M, H(M))$, denoted $\text{acl}_H(A)$, satisfies

$$\text{acl}_H(A) = \text{acl}(A, \text{HB}(A))$$

where acl refers to algebraic closure in the underlying theory T , without the H predicate.

- (4) [BCV17, Theorem 5.3, Proposition 5.9] If the underlying theory T is superstable, then T_H is also superstable and non-forking independence in T_H , denoted $\underset{C}{\downarrow}^H$, is characterized by

$$a \underset{C}{\downarrow}^H B \iff \text{HB}(a/B) = \text{HB}(a/C) \text{ and } a \underset{CH}{\downarrow} B,$$

when $C = \text{acl}_H(C) \subseteq B = \text{acl}_H(B)$, where $\perp$ denotes non-forking independence in T .

We now introduce a third language L_R which extends L_H with the addition of a new binary relation symbol R . Then D will be viewed as an L_R -structure by interpreting R so that R^D consists of the pairs $(h, h') \in H(D)^2$ such that h and h' are C_2 -elements coming from vertices in Γ which are connected by an edge. This makes sense since $H(D)$ is, by definition, the set of C_2 -elements of $\mathbb{S}_\Gamma$ which have vertex width 1 and thus each lie above a unique D_p -element associated to a vertex of the graph. Note that the graph $(H(D), R^D)$ may be naturally identified with Γ . We will eventually show that the L_R -structure on D gives the full induced structure on D coming from $\mathbb{S}_\Gamma$ and that certain model-theoretic properties of Γ are transferred to the L_R -structure D . This will be a key step towards concluding that model-theoretic properties are transferred from Γ to $\mathbb{S}_\Gamma$.

Lemma 4.22. *If a and a' are H -independent tuples from D , then the following are equivalent:*

- (1) $a \equiv^{L_R} a'$.
- (2) $a \equiv^{L_H} a'$ and $H(a) \equiv^\Gamma H(a')$.
- (3) $(a, H(a)) \equiv^{L_{IS}} (a', H(a'))$ and $H(a) \equiv^\Gamma H(a')$.

Proof. (1) $\implies$ (2) $\implies$ (3) is immediate, so we prove (3) $\implies$ (1). The proof will follow [?, Proposition 2.5]. Write $a = (a_0, a_1, h)$ where a_0 is independent over H , $h = H(a)$, and $a_1 \in \langle a_0, h \rangle$. Similarly, write $a' = (a'_0, a'_1, h')$ in a corresponding manner. It suffices to show that, given any $b \in D$, there are h_1, h'_1 , and b' such that ah_1b and $a'h'_1b'$ are each H -independent, $a_0a_1hh_1b \equiv^{L_{IS}} a'_0a'_1h'h'_1b'$, $b \in H$ if and only if $b' \in H$, and $H(ah_1b) \equiv^\Gamma H(a'h'_1b')$.

Case 1: $b \in H$ and $b \in \langle a \rangle$.

In this case, since a is H -independent, we must have $b \in H(a)$ so there is, by assumption, a corresponding $b' \in H(a')$. We can, in this case, take h_1 and h'_1 to be empty. It is clear that all the conditions are satisfied.

Case 2: $b \in H$, $b \notin \langle a \rangle$.

Pick $b' \in H$ with $H(a)b \equiv^\Gamma H(a')b'$. Note that we cannot have $b' \in \langle a' \rangle$ since, as in Case 1, this would entail that $b' \in h'$ and thus $b \in h$, contrary to our assumption. Under our assumptions, we have $G(\langle a, b \rangle) \cong G(\langle a \rangle) \times C_2 \cong G(\langle a' \rangle) \times C_2 \cong G(\langle a', b' \rangle)$ so the map $a \mapsto a'$ extends to an L_{IS} -isomorphism $\langle a, b \rangle \rightarrow \langle a', b' \rangle$ sending b to b' and inducing a partial Γ -elementary map on the H -elements. In this case too we can take $h_1 = h'_1 = \emptyset$ and all the conditions are satisfied.

Case 3: $b \notin H$.

We can find a finite tuple h_1 such that $b \perp_{ah_1}^D H$. We may assume h_1 is disjoint from a , which entails that ah_1 is also H -independent. Then, by Case 2, we can find some h'_1 from H with $a_0a_1hh_1 \equiv^{L_{IS}} a'_0a'_1h'h'_1$ and $H(ah_1) \equiv^\Gamma H(a'h'_1)$. Now, by the co-density/extension property for H -structures, we can find some b' with $a_0a_1hh_1b \equiv^{L_{IS}} a'_0a'_1h'h'_1b'$ and $b' \perp_{a'h'_1}^D H$. Then $a'h'_1b'$ is H -independent, so we are done. $\square$

Corollary 4.23. *Suppose $S_0 = \text{acl}(S_0)$ is a small subsystem of $\mathbb{S}_\Gamma$.*

- (1) *Suppose D_1 and D_2 are subsystems of D containing $D(S_0)$ with $\text{acl}_H(D_i) = D_i$ for $i = 1, 2$. If $D_1 \equiv_{D(S_0)}^{L_R} D_2$ in D , then $D_1 \equiv_{S_0} D_2$ in $\mathbb{S}_\Gamma$.*

- (2) If $S_0 \subseteq S_1 = \text{gcl}(S_1) \subseteq \mathbb{S}_\Gamma$ and D' is a subsystem of D with $D' \equiv_{D(S_0)}^{L_R} D(S_1)$, then there is some $S' \subseteq \mathbb{S}_\Gamma$ such that $S_1 \equiv_{S_0} S'$ in $\mathbb{S}_\Gamma$ and $D(S') = D'$.

Proof. (1) First, note that, for $i = 1, 2$, $D(S_0) = D_i \vee S_0$, since the containments $D(S_0) \subseteq D_i$ and $D(S_0) \subseteq S_0$ are obvious and $D_i \vee S_0$ is a subsystem corresponding to a pro-2 quotient of $G(S_0)$ so is, by definition of D , contained in $D(S_0)$. Then, for $i = 1, 2$, we have

$$G(\langle D_i, S_0 \rangle) \cong G(D_i) \times_{G(D(S_0))} G(S_0).$$

It follows, by duality, that the identity map on S_0 and the isomorphism $D_1 \rightarrow D_2$ over $D(S_0)$ extend uniquely to an isomorphism $\langle D_1, S_0 \rangle \rightarrow \langle D_2, S_0 \rangle$ fixing S_0 . By Lemma 4.22, this map is elementary.

(2) Let $\sigma : D(S_1) \rightarrow D'$ be an isomorphism of subsystems which is partial elementary in D (viewed as an L_R -structure). By (1) and the strong homogeneity of $\mathbb{S}_\Gamma$, σ extends to some $\tilde{\sigma} \in \text{Aut}(\mathbb{S}_\Gamma/S_0)$. We put $S' = \tilde{\sigma}(S_1)$. $\square$

Lemma 4.24. Suppose $D_* = \text{acl}_H(D_*) \subseteq D$ is a small subsystem and D_0, D_1 are subsystems of D containing D_* with $D_0 \downarrow_{D_*H(D_0)}^D H$ and $D_1 \downarrow_{D_*H(D_1)}^D H D_0$. Then $H(\langle D_0, D_1 \rangle) \subseteq H(D_0)H(D_1)$.

Proof. Note that, since $D_* = \text{acl}_H(D_*)$, it is, in particular H -independent, i.e. $D_* \downarrow_{H(D_*)}^D H$, and thus, by base monotonicity, $D_* \downarrow_{H(D_0)}^D H$. Then, the assumption that $D_0 \downarrow_{D_*H(D_0)}^D H$, together with transitivity, entails $D_0 \downarrow_{H(D_0)}^D H$. By base monotonicity again, we get $D_0 \downarrow_{H(D_0)H(D_1)}^D H$.

Next, since $D_1 \downarrow_{D_*D_0H(D_1)}^D H$, we have $D_1 \downarrow_{D_0H(D_1)}^D H$ so, by transitivity, we get $D_0D_1 \downarrow_{H(D_0)H(D_1)}^D H$. This entails that

$$H \cap \langle D_0, D_1 \rangle = H(\langle D_0, D_1 \rangle) \subseteq \langle H(D_0), H(D_1) \rangle.$$

Since H consists of independent elements, $H \cap \langle H(D_0), H(D_1) \rangle = H(D_0)H(D_1)$, so we get $H(\langle D_0, D_1 \rangle) \subseteq H(D_0)H(D_1)$ as desired. $\square$

Lemma 4.25. Let $S \subseteq \mathbb{S}_\Gamma$ be a small subsystem.

- (1) Suppose $S = \text{gcl}(S)$. Then, identifying Γ and H , we have $\Gamma(S) = \text{HB}(D(S))$. More generally, if $C \subseteq D$ is a small H -independent subset, then

$$\text{HB}(D(S)/C) = \Gamma(S) - \Gamma(C).$$

- (2) The subsystem S is graph closed if and only if S satisfies the following:
- (a) If $\alpha \geq \beta$, $\alpha \in D(S)$ is a C_2 -element and β is a D_p -element, then $\beta \in S$.
 - (b) $\Gamma(S) \subseteq \Gamma$ is an induced subgraph
 - (c) $D(S)$ is acl_H -closed in D .
- (3) Suppose $\text{acl}_H(D(S)) = D(S)$. Then $D(S) = D(\text{gcl}(S))$ and thus $H(S) = H(\text{gcl}(S))$.

Proof. (1) By Lemma 3.12 and Lemma 3.14, we have $S \vee (\mathbb{S}_\Gamma)_- = S(G_{\Gamma(S)})$ and, thus, $D(S) \downarrow_{H(D(S))}^D H$, since unraveling definitions yields $H(D(S(G_{\Gamma(S)}))) = H(D(S))$. This shows $\text{HB}(D(S)) \subseteq H(D(S))$. Conversely, if $\text{HB}(D(S)) \subsetneq H(D(S))$, then we can pick some $\alpha \in H(D(S)) - \text{HB}(D(S))$. Since the elements of H are

independent, we must have $\alpha \notin \langle \text{HB}(D(S)) \rangle$ but $\alpha \in D(S) \cap H$, which contradicts $D(S) \downarrow_{\text{HB}(D(S))}^D H$. This shows the desired equality. The argument for the ‘more generally’ clause in the statement is identical.

(2) If $S = \text{gcl}(S)$, then conditions (a) and (b) are clear from the definition of graph closure and condition (c) stating $\text{acl}_H(D(S)) = D(S)$ follows by (1) and the description of algebraic closure in H -structures (Fact 4.21). For the other direction, assume $\text{acl}_H(D(S)) = D(S)$. We first need to show that if α is a C_2 -element of S of vertex width n and $\beta_1, \dots, \beta_n$ are the n distinct C_2 -elements of vertex 1 associated to the witnesses, then $\beta_1, \dots, \beta_n \in S$. We may assume α and each β_i are the identity element of its equivalence class, hence $\alpha \in D(S)$ and $\beta_i \in H$ for $i = 1, \dots, n$. Since $\alpha \in \langle \beta_1, \dots, \beta_n \rangle$ and $\alpha \downarrow_{H(D(S))}^D H$, we must have $\beta_1, \dots, \beta_n \in H(D(S))$. Then, by condition (a), it follows that if $\alpha \in D(S)$ is a C_2 -element of finite vertex width, then the D_p -elements that witness this are also in S . Finally, since $\Gamma(S)$ is an induced subgraph, we have that all conditions of being graph closed are satisfied so $S = \text{gcl}(S)$.

(3) The inclusion $H(S) \subseteq H(\text{gcl}(S))$ is immediate from $S \subseteq \text{gcl}(S)$, so we prove the reverse inclusion. Under the identification of $\Gamma(\mathbb{S}_\Gamma)$ and H , let $\Gamma' = (H(D(S)), R')$ denote the induced subgraph of $\Gamma(\mathbb{S}_\Gamma)$ with vertex set $H(D(S))$.

It is immediate from the definition of graph closure that $\langle S, S(G_{\Gamma'}) \rangle = S \wedge S(G_{\Gamma'})$ is graph closed. Moreover, $D(S(G_{\Gamma'})) = \langle H(D(S)) \rangle$. Thus, Lemma 4.13 gives

$$D(S \wedge S(G_{\Gamma'})) = D(S) \wedge \langle H(D(S)) \rangle = D(S),$$

since $H(D(S)) \subseteq D(S)$. $\square$

Corollary 4.26. *Suppose $\text{acl}(S_i) = S_i \subseteq \mathbb{S}_\Gamma$ are small subsystems for $i = 0, 1, 2$. Then $D(S_1) \downarrow_{D(S_0)}^H D(S_2)$ if and only if $D(S_1) \downarrow_{D(S_0)H}^D D(S_2)$.*

Proof. If $D(S_1) \downarrow_{D(S_0)}^H D(S_2)$, then $D(S_1) \downarrow_{D(S_0)H}^D D(S_2)$ by Fact 4.21(4). On the other hand, if $D(S_1) \downarrow_{D(S_0)H}^D D(S_2)$ then, again by Fact 4.21(4), in order to show $D(S_1) \downarrow_{D(S_0)}^H D(S_2)$, we just need to show that $\text{HB}(D(S_1)/D(S_0)) = \text{HB}(D(S_1)/D(S_2))$. By our hypothesis, $S_1 \cap S_2 = S_0$ and hence $\Gamma(S_1) \cap \Gamma(S_2) = \Gamma(S_0)$. Thus, by Lemma 4.25(1), we have

$$\text{HB}(D(S_1)/D(S_0)) = \Gamma(S_1) - \Gamma(S_0) = \Gamma(S_1) - \Gamma(S_2) = \text{HB}(D(S_1)/D(S_2)),$$

which completes the proof. $\square$

Lemma 4.27. *Suppose S_0, S_1 , and S_2 are small subsystems of $\mathbb{S}_\Gamma$ such that $S_0 = \text{gcl}(S_0) \subseteq S_1, S_2$ and $D(S_i) = \text{acl}_H(D(S_i))$ for $i = 0, 1, 2$. Suppose we are given an isomorphism $\Psi : S_1 \rightarrow S_2$ over S_0 which restricts to an L_R -partial elementary map $D(S_1) \rightarrow D(S_2)$ over $D(S_0)$. Then Ψ extends to an isomorphism $\Xi : \text{gcl}(S_1) \rightarrow \text{gcl}(S_2)$. In particular, Ψ is partial elementary.*

Proof. Under the identification of H and $\Gamma(\mathbb{S}_\Gamma)$, let Γ_i be the induced subgraph of $\Gamma(\mathbb{S}_\Gamma)$ with vertices $H(D(S_i))$ for $i = 0, 1, 2$. Then the map Ψ induces an isomorphism $f : \Gamma_1 \rightarrow \Gamma_2$ over Γ_0 (which is partial elementary with respect to $\text{Th}(\Gamma)$).

We first argue that if β is a D_p -element of $\mathbb{S}_\Gamma$ corresponding to a vertex of Γ_1 , then we may extend Ψ to some $\Psi' : \langle S_1, \beta \rangle \rightarrow \langle S_2, \beta' \rangle$ for some D_p -element $\beta' \in \mathbb{S}_\Gamma$ corresponding to the vertex $f(v)$. There is nothing to show if $\beta \in S_1$, so we may assume $\beta \notin S_1$, so $[\beta] \vee S_1 = [\alpha_v]$ where α_v is a C_2 -element in $H(D(S_1))$. Then let

$\alpha_{f(v)} = \Psi(\alpha_v) \in H(D(S_2))$ and let β' be a D_p -element corresponding to the vertex $f(v)$ (so with $\beta' \leq \alpha_{f(v)}$). Now we must have $\beta' \notin S_2$ so we have $[\beta'] \vee S_2 = [\alpha_{f(v)}]$. Since we have

$$G([\beta'] \vee S_2) \cong G([\beta] \vee S_1) \cong D_p \times_{C_2} G(S_1)$$

we may extend Ψ to an isomorphism $\Psi' : \langle S_1, \beta \rangle \rightarrow \langle S_2, \beta' \rangle$, an isomorphism over S_0 .

Applying this inductively to all vertices of Γ_1 , we may assume that we have extended Ψ to some $\Psi' : S'_1 \rightarrow S'_2$ over S_0 such that if $\beta \in \mathbb{S}_\Gamma$ is a D_p -element corresponding to a vertex of Γ_1 , then $\beta \in S'_1$ (which entails the corresponding property for S'_2). Now we need to extend to handle the W -elements corresponding to edges, so fix some $\gamma \in \mathbb{S}_\Gamma$, a W -element corresponding to an edge $r = (v, v')$ with v, v' vertices in Γ_1 . We want to extend Ψ' to some $\Psi'' : \langle S'_1, \gamma \rangle \rightarrow \langle S'_2, \gamma' \rangle$ where γ' is a W -element associated to the edge $r' = (f(v), f(v'))$. As above, we may assume $\gamma \notin S'_1$. Let $\beta_v, \beta_{v'}, \beta_{f(v)}$, and $\beta_{f(v')}$ be the D_p -elements associated to the vertices $v, v', f(v)$, and $f(v')$, respectively, which we may choose so that $\Psi'(\beta_v) = \beta_{f(v)}$ and $\Psi'(\beta_{v'}) = \beta_{f(v')}$. By the construction of S'_1 and S'_2 , we have

$$\begin{aligned} G(S'_1 \vee \gamma) &= [\beta_v \wedge \beta_{v'}] \cong D_p \times D_p \\ G(S'_2 \vee \gamma) &= [\beta_{f(v)} \wedge \beta_{f(v')}] \cong D_p \times D_p \end{aligned}$$

so we get

$$G(\langle S'_1, \gamma \rangle) \cong G(S'_1) \times_{(D_p \times D_p)} W \cong G(S'_2) \times_{(D_p \times D_p)} W \cong G(\langle S'_2, \gamma' \rangle).$$

By Lemma 2.12, the isomorphism $[\beta_{f(v)} \wedge \beta_{f(v')}] \rightarrow [\beta_v \wedge \beta_{v'}]$ induced by duality by Ψ' is induced by an isomorphism $[\gamma'] \rightarrow [\gamma]$. This, together with the isomorphism $\psi' : G(S'_2) \rightarrow G(S'_1)$ dual to Ψ' combine to yield an isomorphism $G(\langle S'_2, \gamma' \rangle) \rightarrow G(\langle S'_1, \gamma \rangle)$ which, dualizing again, yield the desired extension $\Psi'' : \langle S'_1, \gamma \rangle \rightarrow \langle S'_2, \gamma' \rangle$, which is still over S_0 .

Applying this inductively to all edges of Γ_1 , using Lemma 2.12, we obtain the desired extension $\Xi : \text{gcl}(S_1) \rightarrow \text{gcl}(S_2)$ over S_0 . The ‘in particular’ clause of the statement follows by Lemma 4.6, since isomorphisms of graph closed subsystems which induce elementary maps on graphs are partial elementary. $\square$

Lemma 4.28. *Suppose $S_i = \text{acl}(S_i)$ for $i = 0, 1, 2$ and $S_0 \subseteq S_1, S_2$. If $D(S_1) \vee D(S_2) = D(S_0)$, then $S_1 \vee S_2 = S_0$.*

Proof. We will prove the lemma via a series of claims. Fix $\alpha \in S_1 \cap S_2$. Our goal is to show that $\alpha \in S_0$.

Claim 1: Suppose $[\alpha] \cong G_{\Gamma_0}$ for some induced subgraph $\Gamma_0 \subseteq \Gamma(\mathbb{S}_\Gamma)$. Then $\alpha \in S_0$.

Proof of Claim 1: Let $\beta \geq \alpha$ be an element with $[\beta] \cong C_2^{V_0}$ corresponding to the quotient of $[\alpha]$ by the normal product of the p - and q -Sylow subgroups. Then $\beta \in D(S_1 \vee S_2) = D(S_1) \vee D(S_2) = D(S_0)$. Then $\alpha \in \text{gcl}(\beta)$ so $\alpha \in S_0$ since S_0 is, in particular, graph closed. $\square$

From this, it will easily follow when α corresponds to a quotient that also has an additional C_2^ℓ factor:

Claim 2: Suppose $[\alpha] \cong G_{\Gamma_0} \times C_2^\ell$ for some induced subgraph $\Gamma_0 \subseteq \Gamma(\mathbb{S}_\Gamma)$. Then $\alpha \in S_0$.

Proof of Claim 2: Let $\beta, \gamma \geq \alpha$ be chosen so that $[\beta] \cong G_{\Gamma_0}$, $[\gamma] \cong C_2^\ell$, and $\alpha = \beta \wedge \gamma$. Then $\beta \in S_1 \vee S_2$ so, by Claim 1, $\beta \in S_0$. Also, since $[\gamma]$ is a 2-group, we have $\gamma \in D(S_1 \vee S_2) = D(S_0) \subseteq S_0$ so we have $\gamma \in S_0$. Then $\alpha = \beta \wedge \gamma \in S_0$. $\square$

Now we handle the case of Frattini covers of the groups handled in Claim 2:
Claim 3: Suppose $\beta \geq \alpha$ and $[\beta] \cong [\alpha]/\Phi([\alpha]) \cong G_{\Gamma_0} \times C_2^\ell$ for some finite induced subgraph $\Gamma_0 \subseteq \Gamma$ and $\ell < \omega$, so the canonical map $[\alpha] \rightarrow [\beta]$ is a Frattini cover. Then $\alpha \in S_0$.

Proof of Claim: Because $\Phi([\alpha])$ is nilpotent, we can decompose $\Phi([\alpha]) = \Phi_2([\alpha])\Phi_{pq}([\alpha])$ where $\Phi_2([\alpha])$ is a 2-group and the order of $\Phi_{pq}([\alpha])$ only divisible by p and q . Then we find δ and ϵ satisfying the following:

- (1) $\alpha \leq \delta, \epsilon \leq \beta$.
- (2) $\alpha = \delta \wedge \epsilon, \beta = \delta \vee \epsilon$.
- (3) $[\delta] \cong [\alpha]/\Phi_2([\alpha])$ so the canonical map $[\delta] \rightarrow [\beta]$ is pq -Frattini.
- (4) $[\epsilon] \cong [\alpha]/\Phi_{pq}([\alpha])$ so the canonical map $[\epsilon] \rightarrow [\beta]$ is 2-Frattini.

By Claim 2, we know $\beta \in S_0$ so we have $\delta \in S_0$ by the description of algebraic closure in Proposition 4.11.

Choose $\epsilon' \geq \epsilon$ with $[\epsilon']$ corresponding to the quotient of $[\epsilon]$ by the normal product of the p - and q -Sylow subgroups. If we let K_1 be the kernel of the canonical map $[\epsilon] \rightarrow [\beta]$ and let K_2 denote the kernel of the canonical map $[\epsilon] \rightarrow [\epsilon']$, we have $K_1 \cap K_2 = 1$ since K_1 is a 2-group and K_2 is only divisible by primes p and q . Thus, $[\epsilon] \cong [\epsilon']/(K_1 \cap K_2)$ which entails $\epsilon = \beta \wedge \epsilon'$. Now since $[\epsilon']$ is a 2-group, we have $\epsilon' \in D(S_1 \vee S_2) = D(S_0)$. Thus $\epsilon = \beta \wedge \epsilon' \in S_0$. Then, finally, we obtain $\alpha = \delta \wedge \epsilon \in S_0$, proving the claim. $\square$

Now we conclude by considering the case of arbitrary α . Let $\beta \in S_1 \cap S_2$ be an element with $\alpha \leq \beta$ and $[\beta] \cong [\alpha]/\Phi([\alpha])$, so the canonical map $[\alpha] \rightarrow [\beta]$ is a Frattini cover and $\beta \in (\mathbb{S}_\Gamma)_+$. Thus, by Lemma 2.10, there exists $\beta' \leq \beta$ with $[\beta'] \cong G_{\Gamma_0} \times C_2^\ell$ for some finite induced subgraph $\Gamma_0 \subseteq \Gamma(\mathbb{S}_\Gamma)$ and $\ell < \omega$ and, by inspection of the proof, we see that $\beta' \in \text{acl}(\beta)$ so we have $\beta' \in S_1 \vee S_2$. By Fact 1.4(3), any Frattini cover of $[\beta]$ will be a quotient of a Frattini cover of $[\beta']$ so are reduced to considering the case $\beta = \beta'$, but this has been handled by Claim 3. $\square$

4.4. Preservation results. Define an independence relation $\downarrow^*$ on $\mathbb{S}_\Gamma$ as follows: we declare $A \downarrow_C^* B$ if and only if

- $\text{acl}(AC) \cap \text{acl}(BC) = \text{acl}(C)$, i.e. $A \downarrow_C^a B$.
- $D(\text{acl}(AC)) \downarrow_{HD(\text{acl}(C))}^D D(\text{acl}(BC))$.

By Corollary 4.26, the second bullet point could be equivalently written $D(\text{acl}(AC)) \downarrow_{D(\text{acl}(C))}^H D(\text{acl}(BC))$.

With this notation, we may state more simply the following key proposition:

Proposition 4.29 (Relative Stationarity). *Suppose we are given small subsystems $S_i = \text{acl}(S_i) \subseteq \mathbb{S}_\Gamma$ for $i = 0, 1, 2$ and $S_0 \subseteq S_1 \cap S_2$. Suppose, further, that we are given some S'_1 satisfying the following conditions:*

- (1) $S_1 \downarrow_{S_0}^* S_2$ and $S'_1 \downarrow_{S_0}^* S_2$.
- (2) *There is an L_{IS} -isomorphism $S_1 \rightarrow S'_1$ over S_0 which induces a partial elementary map $\Gamma(S_1) \rightarrow \Gamma(S'_1)$ over $\Gamma(S_2)$.*

Then $S_1 \equiv_{S_2} S'_1$.

Proof. Since $S_1 \vee S_2 = S_0$, $S'_1 \vee S_2 = S_0$ and S_1 is isomorphic to S'_1 over S_0 , we have that the given isomorphism $S_1 \rightarrow S'_1$ over S_0 extends to an isomorphism

$\Psi : \langle S_1, S_2 \rangle \rightarrow \langle S'_1, S_2 \rangle$ over S_2 . We need to check that this is partial elementary over S_0 . We will check the conditions of Lemma 4.27.

For simplicity, write D_i for $D(S_i)$ for $i = 0, 1, 2$ and D'_1 for $D(S'_1)$. Since $S_i = \text{acl}(S_i)$, we know that $\text{acl}_H(D(S_i)) = D(S_i)$ for all i . In particular, this entails that D_i is H -independent, so, by symmetry and base monotonicity, we have

$$D_i \underset{D_0 H(D_i)}{\downarrow}^D H.$$

for $i = 1, 2$ since $D_0 \subseteq D_1, D_2$. By our hypothesis and transitivity, then, we get

$$D_2 \underset{D_0 H(D_2)}{\downarrow}^D D_1 H.$$

This means we can apply Lemma 4.24 to obtain $H(\langle D_1, D_2 \rangle) = H(D_1)H(D_2)$. Similarly, we have $H(\langle D'_1, D_2 \rangle) = H(D'_1)H(D_2)$. In particular, we have $\text{acl}_H(\langle D_1, D_2 \rangle) = \langle D_1, D_2 \rangle$ and $\text{acl}_H(\langle D'_1, D_2 \rangle) = \langle D'_1, D_2 \rangle$. Moreover, by Lemma 4.13, we know $D(\langle S_1, S_2 \rangle) = \langle D_1, D_2 \rangle$ and $D(\langle S'_1, S_2 \rangle) = \langle D'_1, D_2 \rangle$.

By our independence assumptions and stationarity of $\downarrow^D$, we have $\langle D(S_1)D(S_2) \rangle \equiv_{D(S_0)} \langle D(S'_1)D(S_2) \rangle$ in the L_{IS} -structure D , and by Lemma 4.22, $\langle D(S_1), D(S_2) \rangle$ and $\langle D(S'_1), D(S_2) \rangle$ are H -independent. Then, by the equivalence (2) $\Leftrightarrow$ (1) in Lemma 4.22, the given L_{IS} -isomorphism $S_1 \rightarrow S'_1$ over S_0 induces an L_R -partial elementary map $D(S_1) \rightarrow D(S'_1)$ over $D(S_2)$. Lemma 4.27, then, entails that Ψ is partial elementary, completing the proof. $\square$

Remark 4.30. An equivalent formulation of relative stationarity, which is sometimes easier to use, is the following: given algebraically closed subsystems S_1, S'_1, S_2, S'_2 which contain an algebraically closed S_0 , if $S_1 \equiv_{S_0} S'_1$, $S_2 \equiv_{S_0} S'_2$, $S_1 \downarrow_{S_0}^* S_2$, $S'_1 \downarrow_{S_0}^* S'_2$ and $\Gamma(S_1)\Gamma(S_2) \equiv_{\Gamma(S_0)} \Gamma(S'_1)\Gamma(S'_2)$, then $S_1 S_2 \equiv_{S_0} S'_1 S'_2$. To see that this follows from relative stationarity, one can, by an automorphism, pick S''_1 such that $S''_1 S'_2 \equiv_{S_0} S_1 S_2$. Then, by relative stationarity, we get $S''_1 \equiv_{S'_2} S'_1$, so we have

$$S_1 S_2 \equiv_{S_0} S''_1 S'_2 \equiv_{S_0} S'_1 S'_2.$$

Definition 4.31. (1) We say a theory T has SOP_1 if there is a formula $\varphi(x, y)$ and a collection of tuples $(a_\eta)_{\eta \in 2^{<\omega}}$ satisfying the following conditions:

- (a) (Consistency for paths) For all $\eta \in 2^\omega$, $\{\varphi(x; a_{\eta|_i}) : i < \omega\}$ is a consistent set of formulas.
 - (b) (Inconsistency for certain incomparables) If $\eta \perp \nu \in 2^{<\omega}$, $\eta \supseteq (\eta \wedge \nu) \cap \langle 0 \rangle$, and $\nu = (\eta \wedge \nu) \cap \langle 1 \rangle$, then $\{\varphi(x, a_\eta), \varphi(x, a_\nu)\}$ is inconsistent.
- (2) We say a theory T has SOP_n for $n \geq 3$ if there is a formula $\varphi(x, y)$ and an indiscernible sequence $(a_i)_{i < \omega}$ satisfying the following conditions:
- (a) (Order property) $\models \varphi(a_i, a_j) \iff i < j$,
 - (b) (No loops of size n) The set of formulas

$$\{\varphi(x_i, x_{i+1}) : i < n-2\} \cup \{\varphi(x_{n-1}, x_0)\}$$

is inconsistent.

There is also a property SOP_2 , defined exactly the same as SOP_1 , except condition (2) is replaced by saying $\{\varphi(x, a_\eta), \varphi(x, a_\nu)\}$ is inconsistent for *all* incomparable pairs $\eta \perp \nu \in 2^{<\omega}$. It is known by [Mut26] that a theory has SOP_1 if and only if it has SOP_2 so we will not give a separate treatment for the property SOP_2 .

Definition 4.32. The following definitions make sense in an arbitrary complete theory T :

- (1) For $M \models T$, we write $a \downarrow_M^u b$ to mean $\text{tp}(a/Mb)$ is finitely satisfiable in M . If $(a_i)_{i < \omega}$ is an M -indiscernible sequence with $a_i \downarrow_M^u a_{<i}$, we say $(a_i)_{i < \omega}$ is a *coheir sequence* over M .
- (2) For $M \models T$, we say a formula $\varphi(x; a)$ *Kim-divides over M* if there is some coheir sequence $(a_i)_{i < \omega}$ over M with $a_0 = a$ such that $\{\varphi(x; a_i) : i < \omega\}$ is inconsistent.
- (3) We write $a \downarrow_M^K b$ to mean that $\text{tp}(a/Mb)$ contains no formula that Kim-divides over M .

Remark 4.33. The above definition of Kim-independence differs from the “official definition” since it is defined in terms of coheir sequences instead of Morley sequences in a global invariant type. These notions are easily seen to be equivalent in NSOP₁ theories, and it will be useful for us to work with coheir sequences since coheir sequences remain coheir sequences in any reduct (or, more generally, the image of a coheir sequence in an interpreted structure is again a coheir sequence).

Lemma 4.34. Suppose $S_* \models T_\Gamma$ and S_{**} and S_0 are small algebraically closed subsystems of $\mathbb{S}_\Gamma$ with $S_{**} \downarrow_{S_*}^K S_0$. Then $S_{**} \downarrow_{S_*}^* S_0$.

Proof. Fix a coheir sequence $(S_i)_{i < \omega}$ over S_* starting with S_0 . As $S_{**} \downarrow_{S_*}^K S_0$, there is some $S'_{**} \equiv_{S_0} S_{**}$ with $(S_i)_{i < \omega}$ indiscernible over S'_{**} . Then, in particular, $D(S_i)_{i < \omega}$ is a coheir sequence over $D(S_*)$ which is $D(S'_{**})$ -indiscernible in D viewed as an L_H -structure. Thus, by Kim’s Lemma in the simple theory $\text{Th}_{L_H}(D)$, we have $D(S'_{**}) \downarrow_{D(S_*)}^H D(S_0)$, so $D(S_{**}) \downarrow_{D(S_*)}^H D(S_0)$. It is also completely standard that Kim-independence entails $S_{**} \cap S_0 = S_*$, so we get $S_{**} \downarrow_{S_*}^* S_0$ (one can see this by applying the above argument to the reduct of $\mathbb{S}_\Gamma$ to the language of equality instead of the interpretation of the L_H -structure D). $\square$

Fact 4.35. [CR16, Theorem 5.1] The theory T is NSOP₁ if and only if, given any $M \models T$, if $b_0 c_0 \equiv_M b_1 c_1$, $c_1 \downarrow_M^u c_0$, and $c_i \downarrow_M^u b_i$ for $i = 0, 1$, then there is b such that $bc_0 \equiv_M bc_1 \equiv_M b_0 c_0$.

Theorem 4.36. Recall that $T_\Gamma = \text{Th}(\mathbb{S}_\Gamma)$.

- (1) The theory T_Γ is NSOP₁ if and only if $\text{Th}(\Gamma)$ is NSOP₁. Moreover, if $\text{Th}(\Gamma)$ is NSOP₁, $S_* \models T_\Gamma$, and S_0 and S_1 are small algebraically closed subsystems containing S_* , we have

$$a \downarrow_{S_*}^K b \iff a \downarrow_{S_*}^* b \text{ and } \Gamma(\text{acl}(a, S_*)) \downarrow_{\Gamma(S_*)}^K \Gamma(\text{acl}(b, S_*)),$$

where, on the left-hand side $\downarrow^K$ refers to Kim-independence in T_Γ and, on the right-hand side, $\downarrow^K$ refers to Kim-independence in $\text{Th}(\Gamma)$.

- (2) The theory T_Γ is simple if and only if $\text{Th}(\Gamma)$ is simple.
- (3) The theory T_Γ is stable if and only if $\text{Th}(\Gamma)$ is stable.

Proof. (1) Since the graph is interpretable in $\mathbb{S}_\Gamma$, we clearly have that if $\text{Th}(\Gamma)$ has SOP₁, then T_Γ must as well. Thus it suffices to go the other direction.

So now we assume $\text{Th}(\Gamma)$ is NSOP₁, and we will use Fact 4.35 to show that T_Γ is NSOP₁. Note that if $a \downarrow_{S_*}^u b$ then also $\text{acl}(a, S_*) \downarrow_{S_*}^u \text{acl}(b, S_*)$. Thus it suffices to

show that, given algebraically closed subsystems $S_{**}, S'_{**}, S_0, S_1$ all containing S_* , if $S_{**}S_0 \equiv_{S_*} S'_{**}S_1$, $S_0 \downarrow_{S_*}^u S_{**}$, $S_1 \downarrow_{S_*}^u S'_{**}$, and $S_1 \downarrow_{S_*}^u S_0$, then there is some S''_{**} with

$$S''_{**}S_0 \equiv_{S_*} S''_{**}S_1 \equiv_{S_*} S_{**}S_0.$$

In this situation, since the graph is interpreted in each subsystem, we have the following:

- $\Gamma(S_*) \models \text{Th}(\Gamma)$.
- $\Gamma(S_{**})\Gamma(S_0) \equiv_{\Gamma(S_*)} \Gamma(S'_{**})\Gamma(S_1)$.
- $\Gamma(S_0) \downarrow_{\Gamma(S_*)}^u \Gamma(S_{**})$.
- $\Gamma(S_1) \downarrow_{\Gamma(S_*)}^u \Gamma(S'_{**})$.
- $\Gamma(S_1) \downarrow_{\Gamma(S_*)}^u \Gamma(S_0)$.

Thus, by Fact 4.35 and our assumption that $\text{Th}(\Gamma)$ is NSOP₁, there is some Γ''_{**} (which we can assume is in $\Gamma(\mathbb{S}_\Gamma)$) such that

$$\Gamma''_{**}\Gamma(S_0) \equiv_{\Gamma(S_*)} \Gamma''_{**}\Gamma(S_1) \equiv_{\Gamma(S_*)} \Gamma(S_{**})\Gamma(S_0).$$

Under the identification of H and $\Gamma(\mathbb{S}_\Gamma)$, we have, then,

$$\Gamma''_{**}H(D(S_0)) \equiv_{H(D(S_*))}^{L_R} \Gamma''_{**}H(D(S_1)) \equiv_{H(D(S_*))}^{L_R} H(D(S_{**}))H(D(S_0))$$

in the L_R -structure D .

Claim: We can upgrade the previous equation to having the same type over $D(S_*)$, that is:

$$\Gamma''_{**}H(D(S_0)) \equiv_{D(S_*)}^{L_R} \Gamma''_{**}H(D(S_1)) \equiv_{D(S_*)}^{L_R} H(D(S_{**}))H(D(S_0)).$$

Proof of Claim: Because S_* was chosen to be a model, it is in particular algebraically closed so $D(S_*)$ is H -independent. So we have $D(S_*) \downarrow_{H(D(S_*))}^D H$. Then, in particular, we have, by base monotonicity, we have that the tuples $D(S_*)\Gamma''_{**}H(D(S_0))$, $D(S_*)\Gamma''_{**}H(D(S_1))$, and $D(S_*)H(D(S_{**}))H(D(S_0))$ are all H -independent. Thus, by Lemma 4.22, it suffices to show

$$\Gamma''_{**}H(D(S_0)) \equiv_{D(S_*)}^D \Gamma''_{**}H(D(S_1)) \equiv_{D(S_*)}^D H(D(S_{**}))H(D(S_0)).$$

Note that, by hypothesis, we have $H(D(S_{**})) \downarrow_{H(D(S_*))}^D H(D(S_0))$ so, by invariance, we also have $\Gamma''_{**} \downarrow_{H(D(S_*))}^D H(D(S_0))$. Since $D(S_*) \downarrow_{H(D(S_*))}^D H$ we also have, by base monotonicity, $\Gamma''_{**} \downarrow_{H(D(S_0))}^D D(S_0)$ and $H(D(S_{**})) \downarrow_{H(D(S_0))}^D D(S_0)$ so, by transitivity, we have $\Gamma''_{**} \downarrow_{H(D(S_*))}^D D(S_*)H(D(S_0))$ and $H(D(S_{**})) \downarrow_{H(D(S_*))}^D D(S_*)H(D(S_0))$. We know $\Gamma''_{**} \equiv_{H(D(S_*))} H(D(S_{**}))$ and $\text{acl}(H(D(S_{**}))) = \langle H(D(S_{**})) \rangle = \text{dcl}(H(D(S_{**})))$ so Γ''_{**} and $H(D(S_{**}))$ have the same type over the of $H(D(S_*))$ and, by weak elimination of imaginaries in D (Fact 4.18), types over algebraically closed sets are stationary. Thus by stationarity, we obtain

$$\Gamma''_{**}D(S_*)H(D(S_0)) \equiv_{H(D(S_*))}^D H(D(S_{**}))D(S_*)H(D(S_0))$$

and thus $\Gamma''_{**}H(D(S_0)) \equiv_{D(S_*)}^D H(D(S_{**}))H(D(S_0))$. An identical argument, replacing S_0 and S_1 and S_{**} with S'_{**} shows that $\Gamma''_{**}H(D(S_1)) \equiv_{D(S_*)}^D H(D(S'_{**}))H(D(S_1))$. Since, by assumption, $H(D(S_{**}))H(D(S_0)) \equiv_{H(D(S_*))} H(D(S'_{**}))H(D(S_1))$, this proves the claim. $\square$

Since $\Gamma_{**}'' \equiv_{D(S_*)}^{L_R} D(S_{**})$, we can, by an automorphism, pick D_{**}'' in D so that $H(D_{**}'') = \Gamma_{**}''$ and $D_{**}'' \equiv_{D(S_*)} D(S_{**})$. By extension, we may assume that D_{**}'' has been chosen so that $D_{**}'' \downarrow_{\Gamma_{**}'' D(S_*)}^D HD(S_0)D(S_1)$. Finally, by Corollary 4.23, we have $D_{**}'' \equiv_{S_*} D(S_{**})$ in $\mathbb{S}_\Gamma$, so we may choose, by an automorphism, some S_{**}'' in $\mathbb{S}_\Gamma$ with $D(S_{**}'') = D_{**}''$.

By transitivity, we have $D_{**}'' \downarrow_{D(S_*)}^D D(S_0)$ and $D_{**}'' \downarrow_{D(S_*)}^D D(S_1)$ so by Lemma 4.28, we have $S_{**}'' \downarrow_{S_*}^* S_0$ and $S_{**}'' \downarrow_{S_*}^* S_1$. Then, by Proposition 4.29, we have $S_{**}'' S_0 \equiv_{S_*} S_{**} S_0$ and $S_{**}'' S_1 \equiv_{S_*} S_{**} S_1$. Thus, by Fact 4.35, we conclude that T_Γ is NSOP₁.

Now that we know T_Γ is NSOP₁, we know

$$a \downarrow_{S_*}^K b \iff \text{acl}(a, S_*) \downarrow_{S_*}^K \text{acl}(b, S_*),$$

by [KR20, Corollary 5.17]. Thus, we have $a \downarrow_{S_*}^K b$ implies $a \downarrow_{S_*}^* b$ by Lemma 4.34. For the other direction, we set $S_{**} = \text{acl}(a, S_*)$ and $S_0 = \text{acl}(b, S_*)$, and assume $S_{**} \downarrow_{S_*}^* S_0$ and $\Gamma(S_{**}) \downarrow_{\Gamma(S_*)}^K \Gamma(S_0)$. Then we take a coheir sequence $(S_i)_{i < \omega}$ over S_* starting with S_0 . The above argument (applied to $(S_i)_{i < \omega}$ rather than to just S_0 and S_1) produces some S_{**}' with $S_{**}' S_i \equiv_{S_*} S_{**} S_0$ for all i , which entails that $S_{**}' \downarrow_{S_*}^K S_0$, completing the characterization of Kim-independence.

(2) By [KR20, Proposition 8.8], it suffices to show that if $\text{Th}(\Gamma)$ is simple, then $\downarrow^K$ satisfies base monotonicity over models. So we assume $\text{Th}(\Gamma)$ is simple and we fix $S_* \preceq S_0 \preceq S_1 \models T_\Gamma$ and $S_2 = \text{acl}(S_2) \supseteq S_1$. We suppose $S = \text{acl}(S, S_*)$ is a small subsystem of $\mathbb{S}_\Gamma$ with $S \downarrow_{S_*}^K S_2$. Setting $S' = \text{acl}(S, S_1)$, we must show $S' \downarrow_{S_1}^K S_2$. We will make use of the characterization of Kim-independence in (1).

By Lemma 4.13, $D(S') = \langle D(S), D(S_1) \rangle$. By base monotonicity of $\downarrow^H$, we have $D(S) \downarrow_{D(S_1)}^H D(S_2)$ and thus $D(S') \downarrow_{D(S_1)}^H D(S_2)$. In particular, we have $D(S') \vee D(S_2) = D(S_1)$ so $S' \vee S_2 = S_1$ by Lemma 4.28, which shows $S' \downarrow_{S_1}^* S_2$. Moreover, by Lemma 4.24, we have $H(\langle D(S), D(S_1) \rangle) = H(D(S))H(D(S_1))$ so, using the identification of H and $\Gamma(\mathbb{S}_\Gamma)$, we have $\Gamma(S') = \Gamma(S)\Gamma(S_1)$. Thus, by base monotonicity of Kim-independence in $\text{Th}(\Gamma)$, we have $\Gamma(S') \downarrow_{\Gamma(S_1)}^K \Gamma(S_2)$.

(3) In a stable or simple theory, non-forking independence agrees with Kim-independence. If $\text{Th}(\Gamma)$ is stable then it follows from (2) and Proposition 4.29 that non-forking independence in T_Γ is stationary over models, which implies that the theory T_Γ is stable. $\square$

Theorem 4.37. *Fix $n \geq 3$. Suppose Γ is a graph and $\text{Th}(\Gamma)$ is NSOP_n. Then $T_\Gamma = \text{Th}(S(\widetilde{G}_\Gamma))$ is NSOP_n.*

Proof. Suppose $S_* \preceq \mathbb{S}_\Gamma$ is a small submodel and $(S_i)_{i < \omega}$ is a coheir sequence of algebraically closed subsystems of $\mathbb{S}_\Gamma$ over S_* , with S_* contained in each S_i . Suppose $p(X, Y) = \text{tp}(S_0, S_1/S_*)$ and (S_i, S_j) realize $p(X, Y)$ if and only if $i < j$. Then we must show

$$p(X_0, X_1) \cup p(X_1, X_2) \cup \dots \cup p(X_{n-2}, X_{n-1}) \cup p(X_{n-1}, X_0)$$

is consistent.

Let $q(z, w) = \text{tp}_\Gamma(\Gamma(S_0), \Gamma(S_1)/\Gamma(S_*))$. By our assumption that $\text{Th}(\Gamma)$ is NSOP_n , we can find $\Gamma_0, \Gamma_1, \dots, \Gamma_{n-1}$ in $\Gamma(\mathbb{S}_\Gamma)$ such that $(\Gamma_i, \Gamma_{i+1}) \models q$ for all $i = 0, \dots, n-2$ and also $(\Gamma_{n-1}, \Gamma_0) \models q$.

By the choice of q (and the S_* -indiscernibility of $(S_i)_{i < \omega}$), we have $\Gamma_i \equiv_{\Gamma(S_*)} \Gamma(S_0)$ for all $i < n$. Using the identification of H and Γ and Lemma 4.22, we have

$$\Gamma_i \equiv_{H(D(S_*))}^{LR} H(D(S_0))$$

in D . Since S_* is algebraically closed, we have $D(S_*) \downarrow_{H(D(S_*))}^D H$ so, in particular, we have

$$D(S_*) \downarrow_{H(D(S_*))}^D \Gamma_i H(D(S_0)).$$

Since $\text{acl}(H(D(S_*))) = \langle H(D(S_*)) \rangle = \text{dcl}(H(D(S_*)))$ and types over algebraically closed sets are stationary (by Fact 4.18), we obtain

$$D(S_*)H(D(S_*))\Gamma_i \equiv^D D(S_*)H(D(S_*))H(D(S_0)).$$

Additionally, we know both $D(S_*)H(D(S_*))\Gamma_i$ and $D(S_*)H(D(S_*))H(D(S_0))$ are H -independent so we get $\Gamma_i \equiv_{D(S_*)}^{LR} H(D(S_0))$ by Lemma 4.22.

Pick $D_0 \equiv_{D(S_*)} D(S_0)$ with $H(D(S_0)) = \Gamma_0$. By extension, we can assume $D_0 \downarrow_{D(S_*)\Gamma_0}^D H$. Then, repeating this inductively, for each $i < n$, we can pick $D_i \equiv_{D(S_*)} D(S_0)$ with $H(D_i) = \Gamma_i$ and $D_i \downarrow_{D(S_*)\Gamma_i}^D H D_{<i}$.

By Corollary 4.23, we have $D_i \equiv_{S_*} D(S_0)$ in $\mathbb{S}_\Gamma$ for all $i < n$. Then, by an automorphism, we can pick some $S'_i \equiv_{S_*} S_0$ with $D(S'_i) = D_i$ for all $i < n$. By construction, we have $S'_i \downarrow_{S_*}^* S'_{i+1}$ for all $i < n-1$ and $S'_{n-1} \downarrow_{S_*}^* S'_0$. Moreover, for all $i < n-1$,

$$\Gamma(S'_i)\Gamma(S'_{i+1}) \equiv_{\Gamma(S_*)} \Gamma(S'_{n-1})\Gamma(S'_0) \equiv_{\Gamma(S_*)} \Gamma(S_0)\Gamma(S_1).$$

Thus, by relative stationarity, Proposition 4.29 (and Remark 4.30), we have $(S'_i, S'_{i+1}) \models p$ for all $i < n-1$ and $(S'_{n-1}, S'_0) \models p$. This shows T_Γ is NSOP_n . $\square$

5. PAC FIELDS

5.1. Preliminaries on PAC fields. If K is a field, we write K^s for the separable closure of K , K^{alg} for the algebraic closure of K , and we write $\mathcal{G}(K)$ for the absolute Galois group of K .

The following fact lets us realize the groups we have constructed in the previous sections as absolute Galois groups of PAC fields:

Fact 5.1. [Ax68] [LvdD81] The class of projective profinite groups is equal to the class of absolute Galois groups of PAC fields.

Write $\downarrow^{\text{SCF}}$ for non-forking independence in the (stable) theory of separably closed fields. The following theorem of Chatzidakis reduces the study of amalgamation in PAC fields to the study of amalgamation in the inverse system of the Galois group:

Fact 5.2. [Cha19, Theorem 2.1] Let F be a PAC field and let A, B, C_1 , and C_2 be algebraically closed subsets of F containing a common algebraically closed subset E , which contains a p -basis if the degree of imperfection of F is finite. Assume $A \cap B = E$, $C_1 \downarrow_E^{\text{SCF}} A$, $C_2 \downarrow_E^{\text{SCF}} B$, and there is an E^s -isomorphism $\varphi : C_1^s \rightarrow C_2^s$

such that $\varphi(C_1) = C_2$. Suppose further that there is $S_0 \subseteq S(\mathcal{G}(F))$ and elementary isomorphisms

$$\begin{aligned} S\Psi_1 : \langle S(\mathcal{G}(C_1)), S(\mathcal{G}(A)) \rangle &\rightarrow \langle S_0, S(\mathcal{G}(A)) \rangle \\ S\Psi_2 : \langle S(\mathcal{G}(C_2)), S(\mathcal{G}(B)) \rangle &\rightarrow \langle S_0, S(\mathcal{G}(B)) \rangle \end{aligned}$$

such that

- (1) $S\Psi_1$ is the identity on $S(\mathcal{G}(A))$, $S\Psi_2$ is the identity on $S(\mathcal{G}(B))$, and $S\Psi_i(S(\mathcal{G}(C_i))) = S_0$,
- (2) if $S\Phi : S(\mathcal{G}(C_1)) \rightarrow S(\mathcal{G}(C_2))$ is the morphism double dual to φ , then

$$S\Psi_2 \circ S\Phi = S\Psi_1|_{S(\mathcal{G}(C_1))}.$$

Then in some elementary extension F' of F , there is $C \models \text{tp}(C_1/A) \cup \text{tp}(C_2/B)$ with $C \downarrow_E^{\text{SCF}} AB$ and $S(\mathcal{G}(C)) = S_0$.

Finally, we have a preservation theorem for the SOP_n hierarchy (for $n \geq 3$):

Fact 5.3. [Cha19, Theorem 2.9] Suppose $n \geq 3$. If F is a PAC field and $\text{Th}(S(\mathcal{G}(F)))$ is NSOP_n , then $\text{Th}(F)$ is NSOP_n .

5.2. NSOP₁ and Kim-independence. Interpretability of $S(\mathcal{G}(F))$ in (F^{alg}, F) is proved in [Cha02, Proposition 5.5]. The “moreover” clause is clear from the proof.

Fact 5.4. Both F and $S(\mathcal{G}(F))$ are interpretable in (K, F) where K is any algebraically closed field containing F . Call the interpretation π . Moreover, if $L \subseteq F$ is a subfield such that F is a regular extension of L , then the restriction of π to (K, L) produces an interpretation of $S(\mathcal{G}(L))$, which embeds into $S(\mathcal{G}(F))$ via the natural restriction map $\mathcal{G}(F) \rightarrow \mathcal{G}(L)$.

Lemma 5.5. Let F be a large sufficiently saturated and homogeneous field (i.e. a monster model of its theory) and $M \prec F$ a small elementary substructure. Suppose $A = \text{acl}(A)$, $B = \text{acl}(B)$ are subsets of F with $M \subseteq A \cap B$.

- (1) If $A \equiv_M B$ in F , then $S(\mathcal{G}(A)) \equiv_{S(\mathcal{G}(M))} S(\mathcal{G}(B))$.
- (2) If $(A_i)_{i \in I}$ is an I -indexed indiscernible over M in $\text{tp}(A/M)$, then $(S(\mathcal{G}(A_i)))_{i \in I}$ is an I -indexed indiscernible over $S(\mathcal{G}(M))$.
- (3) If $A \downarrow_M^u B$ in F , then $S(\mathcal{G}(A)) \downarrow_{S(\mathcal{G}(M))}^u S(\mathcal{G}(B))$ in $S(\mathcal{G}(M))$.

Proof. (1) If $A \equiv_M B$ in F , then there is an automorphism $\sigma \in \text{Aut}(F/M)$ with $\sigma(A) = B$. The map σ has an extension $\tilde{\sigma}$ to F^{alg} which is, then, an automorphism of the pair (F^{alg}, F) taking A to B and fixing M pointwise. It follows $A \equiv_M B$ in the pair (F^{alg}, F) . Since $A = \text{acl}(A)$ and $B = \text{acl}(B)$, we know F is a regular extension of A and of B (see, e.g., [Cha99, Section 1.17]). By Fact 5.4, we have $S(\mathcal{G}(A)) \equiv_{S(\mathcal{G}(M))} S(\mathcal{G}(B))$.

(2) Suppose $(A_i)_{i \in I}$ is an I -indexed indiscernible over M . Given two k -tuples $\vec{i} = (i_0, \dots, i_{k-1})$ and $\vec{j} = (j_0, \dots, j_{k-1})$ from I with $\text{qftp}(\vec{i}) = \text{qftp}(\vec{j})$, we know $A_{i_0} \dots A_{i_{k-1}} \equiv_M A_{j_0} \dots A_{j_{k-1}}$ so $\text{acl}(A_{i_0} \dots A_{i_{k-1}}) \equiv_M \text{acl}(A_{j_0} \dots A_{j_{k-1}})$. Then by (1) $S(\mathcal{G}(\text{acl}(A_{i_0} \dots A_{i_{k-1}}))) \equiv_{S(\mathcal{G}(M))} S(\mathcal{G}(\text{acl}(A_{j_0} \dots A_{j_{k-1}})))$, which implies $(S(\mathcal{G}(A_i)))_{i \in I}$ is an I -indexed indiscernible over $S(\mathcal{G}(M))$.

(3) In *any* theory, if π is an interpretation of the structure X in the structure Y , and $A \downarrow_C^u B$ in Y , then $\pi(A) \downarrow_{\pi(C)}^u \pi(B)$. It follows that if $A \downarrow_M^u B$ in F , then $S(\mathcal{G}(A)) \downarrow_{S(\mathcal{G}(M))}^u S(\mathcal{G}(B))$ by Fact 5.4. $\square$

Our candidate for Kim-independence in a PAC field with NSOP_1 absolute Galois group will be the conjunction of non-forking independence in the ambient separably closed field and Kim-independence in the sense of the Galois group.

Definition 5.6. Suppose F is a field and $\text{Th}(S(\mathcal{G}(F)))$ is NSOP_1 . If a, b are tuples in some elementary extension of F , we say a and b are *weakly independent* over F if, letting $A = \text{acl}(aF)$ and $B = \text{acl}(bF)$,

- (1) $A \downarrow_F^{\text{SCF}} B$
- (2) $S(\mathcal{G}(A)) \downarrow_{S(\mathcal{G}(F))}^K S(\mathcal{G}(B))$.

Remark 5.7. This differs from the definition given by Chatzidakis [Cha19] in the context where $\text{Th}(S(\mathcal{G}(F)))$ is simple, as we use Kim-independence and Chatzidakis used non-forking independence in the definition of weak independence. If the base is a model, these definitions agree, and the definition we give here is well-behaved in the broader context of NSOP_1 Galois groups.

In order to show that the theory of a PAC field with NSOP_1 Galois group (in the inverse system language) is NSOP_1 , we will use the following characterization of NSOP_1 in terms of a weak variant of the independence theorem:

Proposition 5.8. *Suppose F is a PAC field and $\text{Th}(S(\mathcal{G}(F)))$ is NSOP_1 . Then $\text{Th}(F)$ is NSOP_1 .*

Proof. Let $L \models \text{Th}(F)$. Suppose $a_0 b_0 \equiv_L a_1 b_1$ with $b_i \downarrow_L^u a_i$ for $i = 0, 1$ and $b_1 \downarrow_L^u b_0$. Let $A_i = \text{acl}(a_i L)$ and $B_i = \text{acl}(b_i L)$ for $i = 0, 1$. Then we have $B_i \downarrow_L^u A_i$ for $i = 0, 1$ and $B_1 \downarrow_L^u B_0$. Then by Lemma 5.5(3), we know $S(\mathcal{G}(B_i)) \downarrow_{S(\mathcal{G}(L))}^u S(\mathcal{G}(A_i))$ for $i = 0, 1$ and $S(\mathcal{G}(B_1)) \downarrow_{S(\mathcal{G}(L))}^u S(\mathcal{G}(B_0))$. Then, because $\text{Th}(S(\mathcal{G}(L)))$ is NSOP_1 , there is some S_* such that $S_* S(\mathcal{G}(B_0)) \equiv_{S(\mathcal{G}(L))} S_* S(\mathcal{G}(B_1)) \equiv_{S(\mathcal{G}(L))} S(\mathcal{G}(A_0)) S(\mathcal{G}(B_0))$, by Fact 4.35. Then by Fact 5.2, there is some A_* with $A_* B_0 \equiv_L A_* B_1 \equiv_L A_0 B_0$. In particular, letting a_* be the subtuple of A_* corresponding to a_0 in A_0 , we have $a_* b_0 \equiv_L a_* b_1 \equiv_L a_0 b_0$. By the other direction of Fact 4.35, we have shown $\text{Th}(F)$ is NSOP_1 . $\square$

Theorem 5.9. *Suppose F is a PAC field and $\text{Th}(S(\mathcal{G}(F)))$ is NSOP_1 . Then given $L \models \text{Th}(F)$, $a \downarrow_L^K b$ if and only if a and b are weakly independent over L .*

Proof. Let $A = \text{acl}(aL)$ and $B = \text{acl}(bL)$. By Proposition 5.8, we already know $\text{Th}(F)$ is NSOP_1 and therefore $a \downarrow_L^K b$ if and only if $A \downarrow_L^K B$ [KR20, Corollary 5.17], so we show $A \downarrow_L^K B$.

First, assume $A \downarrow_L^K B$ and we will show A and B are weakly independent over L . Let $(B_i)_{i < \omega}$ be a Morley sequence in a global type finitely satisfiable in L extending $\text{tp}(B/L)$. As $A \downarrow_L^K B$, we may assume $(B_i)_{i < \omega}$ is A -indiscernible. Then by Kim's lemma in the stable theory SCF, we know $A \downarrow_L^{\text{SCF}} B$. Also $(S(\mathcal{G}(B_i)))_{i < \omega}$ is a Morley sequence in a global type finitely satisfiable in $S(\mathcal{G}(L))$ which is moreover $S(\mathcal{G}(A))$ -indiscernible. As $\text{Th}(S(\mathcal{G}(L)))$ is NSOP_1 , this implies $S(\mathcal{G}(A)) \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(B))$ by Kim's lemma for Kim-dividing. In other words, A and B are weakly independent over L .

Now assume A and B are weakly independent over L and we will show $A \downarrow_L^K B$. Let $(B_i)_{i < \omega}$ be an L -finitely satisfiable Morley sequence over L with $B_0 = B$.

Put $C_{n,0} = \text{acl}(B_0, \dots, B_{2^n-1})$ and $C_{n,1} = \text{acl}(B_{2^n}, \dots, B_{2^{n+1}-1})$. Note that $C_{n,1} \downarrow_L^u C_{n,0}$ and $C_{n,0} \equiv_L C_{n,1}$ for all $n < \omega$. By induction on n , we will choose A_n so that $A_0 = A$ and for all $n \geq 0$ the following conditions are satisfied:

- (1) $A_{n+1} \equiv_{C_{n,0}, C_{n,1}} A_n$.
- (2) A_n is weakly independent from $\text{acl}(C_{n,0}, C_{n,1}) = C_{n+1,0}$ over L .
- (3) $A_n C_{n,0} \equiv_L A_n C_{n,1}$.

Suppose we are at stage $n \geq 0$ of the induction. Pick A'_n so that $A_n C_{n+1,0} \equiv_L A'_n C_{n+1,1}$. Then, in particular, $S(\mathcal{G}(A_n)) \equiv_{S(\mathcal{G}(L))} S(\mathcal{G}(A'_n))$, $S(\mathcal{G}(A_n)) \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+1,0}))$, and $S(\mathcal{G}(A'_n)) \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+1,1}))$. Moreover, because $C_{n+1,1} \downarrow_L^u C_{n+1,0}$, we have $S(\mathcal{G}(C_{n+1,1})) \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+1,0}))$. As $\text{Th}(S(\mathcal{G}(F)))$ is NSOP_1 , we may apply the independence theorem for $\downarrow^K$ to obtain S so that $S \equiv_{S(\mathcal{G}(C_{n+1,0}))} S(\mathcal{G}(A_n))$, $S \equiv_{S(\mathcal{G}(C_{n+1,1}))} S(\mathcal{G}(A'_n))$ and $S \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+1,0})) S(\mathcal{G}(C_{n+1,1}))$. Note that $S(\mathcal{G}(C_{n+2,0})) \supseteq S(\mathcal{G}(C_{n+1,0})) S(\mathcal{G}(C_{n+1,1}))$ so, by extension, we may assume S has been chosen so that $S \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+2,0}))$. By Fact 5.2, there is A_{n+1} with $A_{n+1} \equiv_{C_{n+1,0}} A_n$, $A_{n+1} \equiv_{C_{n+1,1}} A'_n$, $A_{n+1} \downarrow_L^{\text{SCF}} C_{n+1,0} C_{n+1,1}$ and $S(\mathcal{G}(A_{n+1})) = S$. Note that this implies $A_{n+1} \downarrow_L^{\text{SCF}} C_{n+2,0}$ because $C_{n+2,0} = \text{acl}(C_{n+1,0} C_{n+1,1})$ and $S(\mathcal{G}(A_{n+1})) \downarrow_{S(\mathcal{G}(L))}^K S(\mathcal{G}(C_{n+2,0}))$ by the choice of S so A_{n+1} and $C_{n+2,0}$ are weakly independent over L . This completes the induction.

Let $p(X; B) = \text{tp}(A/B)$. Observe that, by construction, $A_n \models \bigcup_{i < 2^n} p(X; B_i)$ so $\bigcup_{i < \omega} p(X; B_i)$ is consistent by compactness. As the sequence $\langle B_i : i < \omega \rangle$ is an arbitrary L -finitely satisfiable Morley sequence over L , this shows $A \downarrow_L^K B$. $\square$

Remark 5.10. The above result first appeared in our dissertation [Ram18, Theorem 7.2.6] though there was an error, pointed out to us by Daniel Hoffmann and Jung-Uk Lee: we used that $a \downarrow_M^K b$ if and only if $\text{acl}(aM) \downarrow_M^K \text{acl}(bM)$ in our proof of NSOP_1 , though we only know this equivalence *assuming* that the theory is NSOP_1 . The fix is to first prove NSOP_1 by invoking the equivalence of NSOP_1 with the weak independence theorem, Fact 4.35, and then, having done that, using the same proof to identify Kim-independence.

Fact 5.11. [FJ08, Section 28.10] If K is a field, viewed as a structure in the language of rings, with absolute Galois group G , then $\Gamma(S(G))$ is interpretable in K .

Now we come to the main theorem:

Theorem 5.12. *Given a graph Γ , let K_Γ be a PAC field with absolute Galois group $\widetilde{G}_\Gamma$ (which exists by Fact 5.1). View K_Γ as a structure in the language of rings. Then, for all $n \geq 1$, the field K_Γ has NSOP_n theory if and only if $\text{Th}(\Gamma)$ is NSOP_n .*

Proof. Note that, by Fact 5.11, if $\text{Th}(\Gamma)$ has SOP_n , then $\text{Th}(K_\Gamma)$ also has SOP_n . On the other hand, for $n = 1$, if $\text{Th}(\Gamma)$ is NSOP_1 , then $T_\Gamma = \text{Th}(S(\widetilde{G}_\Gamma))$ is NSOP_1 by Theorem 4.36, and thus $\text{Th}(K_\Gamma)$ is NSOP_1 by Proposition 5.8. By [Mut26], the $n = 2$ case is equivalent to the $n = 1$ case. For $n \geq 3$, we know that if $\text{Th}(\Gamma)$ is NSOP_n , then T_Γ is NSOP_n by Theorem 4.37, which entails $\text{Th}(K_\Gamma)$ is NSOP_n by Fact 5.3. $\square$

Corollary 5.13. *The SOP_n hierarchy is strict in fields. That is, for all $n \geq 3$, there is a (pure) field F such that $\text{Th}(F)$ is SOP_n and $NSOP_{n+1}$.*

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